Municipal Waste Management in Israel

From Cleanup to System Design

A Civic Handbook

Bela Nikitina / Re-Haifa Lab, 2026

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This civic handbook explains how municipal waste systems in Israel actually work – and how they can be changed rather than merely cleaned up.

Using examples from Haifa and other Israeli contexts, it connects everyday waste problems with the hidden systems behind them: contracts, tariffs, infrastructure, data, governance and reform. Its central argument is that good waste management begins before waste is created – through repair, reuse, food-sharing, composting, bulky-item transport, second-hand exchange and local circularity hubs.

The book follows the route of waste only after prevention has failed, offering tools for civic audits, municipal dialogue, and practical local action. It invites readers to move from the role of “free janitors” to that of systemic civic designers.

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Introduction

From Cleanup to System Design

> *"Most discussions of waste begin with a question about behaviour: why don't people sort their rubbish, why do they dump things by the bin, why don't they compost. This book begins with a different question: why is the system built in a way that makes the right behaviour inconvenient, expensive, or simply impossible – and what can be done about that."*

Why This Book – and Why Now

Israel entered 2026 with an officially confirmed shortage of landfill capacity. According to the State Comptroller, there will not be enough sites even by the end of 2026. In parallel, two legislative processes are under way – a broad reform of sector regulation and a narrower memorandum amending the Packaging Law – either of which could change the system's basic parameters for the next 15–20 years. And at the same time, according to the Ministry of Environmental Protection's own data, only 6% of household packaging waste reaches dedicated containers.

This is a rare combination: a physical infrastructure crisis, an unfinished institutional reform, and a documented failure of the existing system – all at once. That is precisely why this book is being written now, rather than a few years from now once the dust has settled. The moment when decisions have not yet been made is the moment when civic knowledge matters most.

This book was written in Haifa, drawing on the experience of a Russian-speaking environmental activist community (the Re-Haifa Lab group) working at the intersection of practice – early rescue of discarded items, circular-economy pilots, organics projects – and analysis: how the system is structured, who runs it, and why it produces the results it produces. This is not an academic treatise or a manifesto. It is an attempt to translate a complex, fragmented system into language that supports meaningful participation in it.

The central argument of this book is that municipal waste management should begin before waste appears – before a useful object, edible food, or organic material becomes part of the waste stream.

In Israel, there is a missing layer between everyday household life and the formal waste collection system: local prevention infrastructure. Neighbourhood circularity hubs can bring together repair, reuse, second-hand exchange, food-sharing, composting, bulky-item transport, creative upcycling, and practical education. Such hubs do not replace municipal waste services; they complement them at the point where the standard system usually starts too late.

In this perspective, separation into material fractions remains important, but it is no longer the first or central act. It is the last gate – what happens after prevention, repair, reuse, food redistribution, and local organics work have not been enough. A good waste-management system should not only process waste more efficiently; it should create the conditions under which fewer things become waste at all.

Who This Book Is For

The book is written for four audiences at once.

1. The activist and resident who want to understand why their efforts to sort waste achieve less than they should – and what can be changed through concrete action at the neighbourhood or city level. Part VII (Chapters 25–27) is written directly for you: data tools, an audit methodology, a seven-day action plan.

2. NGOs and environmental organisations that work professionally on waste issues or are looking for an entry point to partner with a municipality. The book describes not only how the system works, but where its built-in conflicts of interest sit – understanding that makes negotiation more precise.

3. Municipal officials and local council representatives who make practical decisions every day about contracts, bins, and complaints. The chapters on financial structure (19–21) and on how incentives are built into contractor contracts (21) are written for a reader who already knows, from the inside, what a tonne of waste costs and why the sanitation budget never quite adds up.

4. Academic readers and reviewers assessing the evidence base. This book applies an explicit evidence scale, described in Chapter 8, under which every figure in every section on material streams is assigned to one of several categories: ✓✓Confirmed / ✓Likely / ~ Reconstructed / ? Unknown. No figure is used without stating its source and evidence level. That is the only way to operate responsibly under the fairly high level of uncertainty currently present in Israel's waste sector.

How the Book Is Organised

The seven parts are organised as a sequence of progressively more detailed analysis. Each part assumes an understanding of the one before it – but each can also be read on its own.

Handbook Structure

Part I. Why the Policy Is Changing (Chapters 1–3)
The legal framework, the map of actors, and the infrastructure crisis that make waste policy in Israel a live political issue rather than a technical background service. The part begins with the legal definition of waste and municipal responsibility, then moves to the institutional actor map and to the urgent question of landfill capacity, WtE, infrastructure lock-in, and why civic engagement matters right now – literally, not rhetorically.

Part II. How Waste Moves Through the City (Chapters 4–7)
The physical route of waste from the kitchen, the yard and the bin site through trucks, transfer stations, sorting facilities, RDF, landfills and energy recovery. This part makes visible the "invisible middle" of the system and shows how collection routes, housing types, transfer infrastructure and geography shape both service quality and environmental burden – including who lives next to landfills, not by choice.

Part III. What Actually Happens to Recyclables (Chapters 8–13)
The evidence scale as a tool for separating documented facts from optimistic declarations. Six material streams are examined in detail – from paper and cardboard, where a relatively complete downstream chain exists, to flexible, multilayer and small packaging, which often enters the system already structurally difficult or impossible to recycle.

Part IV. Packaging Policy and Extended Producer Responsibility (Chapters 14–18)
How TAMIR was designed, why it fell short, and what the Israeli case reveals about the limits of packaging-focused Extended Producer Responsibility. This part examines the deposit system, the failure to integrate parallel collection streams, the 2026 reform, the MoEP memorandum of November 2025, and the European PPWR as a possible but not automatically transferable model.

Part V. Money, Contracts, and Incentives (Chapters 19–21)
The financial architecture behind municipal waste services: gate fees, landfill levies, hidden subsidies, the "pay-twice" paradox, and the gap between who is legally responsible and who actually pays. This part explains why contracts often reward tonnes collected rather than residual waste reduced – and how KPIs, PAYT principles, data requirements and contract design could shift incentives from managing volume to reducing it.

Part VI. Waste Prevention and Local Circularity Infrastructure (Chapters 22–24)
Why waste prevention cannot be reduced to individual behaviour or occasional volunteer initiatives. Organics, food, clothing, furniture, toys and household items are treated here not simply as "waste streams," but as materials that can be intercepted before they enter the waste system. The central idea is the neighbourhood circularity hub: a missing municipal layer between household life and formal waste collection, where repair, reuse, exchange, food-sharing, composting, bulky-item logistics, education and impact measurement are brought together into one working system.

Part VII. Law, Oversight, and Civic Action (Chapters 25–27)
The practical civic toolkit: what residents, NGOs and local groups have the right to ask, how to audit a waste stream without being experts, and how to turn observations into arguments. The book ends with a one-week action framework for working with municipalities, schools and NGOs, and for taking part in the reform process not as "free janitors," but as informed civic actors capable of engaging with data, contracts and policy choices.

Three Principles That Run Through the Book

First Principle: Data Matters More Than Declarations

Israel's official recycling figures sound convincing. TAMIR reports 84% recycling – relative to a declared base that covers roughly 50% of the actual packaging stream. The MoEP promoted separate collection for years – until its own 2025 scan showed that 6% of household packaging reaches the right bin.

This book applies the evidence scale to every claim – official and independent alike. A fact with a ✓✓ source is an argument. A claim with no source is a declaration. The difference is not a technicality.

Second Principle: Incentives Explain Behaviour

When a municipality does not expand separate collection, that is not always apathy or corruption. It is often a rational response to the "pay-twice" paradox: moving from direct landfilling to sorting increases costs, because the system recovers less than 10% of the material and the rest goes to landfill anyway. When TAMIR favours the commercial stream over the household stream, that is not arbitrary – it is a price signal: 135 NIS/tonne versus 642 NIS/tonne.

Understanding the built-in incentives is not an excuse for poor results. It is a tool for engaging with the system: if you know why an actor does what it does, you can propose changing the structure, not just its motivation.

Third Principle: Systemic Change Starts from Below and Above at Once

This book does not ask readers to choose between civic action at the neighbourhood level and participation in the national legislative process. Both levels matter, and they reinforce each other. A circularity hub that collects data and publishes an annual report becomes an argument in a conversation with the municipality. A municipality that has that argument becomes stronger in its dialogue with the Union of Local Authorities. The Union of Local Authorities, representing an aggregated position, influences the reform.

This is not naive optimism – it is a description of a mechanism that works in countries with more developed waste-management systems. In 2026, Israel is at the point where that mechanism is only beginning to take shape.

A Note on the Evidence Base

This book was written during a period of active change. The State Comptroller's report (July 2025), the MoEP scan (August 2025), the Packaging Law memorandum (November 2025), and the Ministry of Finance's reform decision (December 2025) were all published within the twelve months before this book went to print.

That has two consequences. First, the book describes the system at a moment of maximum relevance, drawing on the most current official data available. Second, some of the processes it describes – above all the 2026 reform (Chapter 16) – remain unresolved. Where the outcome is uncertain, the book says so plainly and describes scenarios rather than predicting a result.

> For readers picking up this book a year or two from now: data on landfill capacity, tariffs, and the legislative status of the reform can change quickly. The core analytical frameworks and methodology are more durable. The evidence scale, the principle that "incentives explain behaviour," and the seven-step civic audit cycle all work regardless of whether the reform is ultimately adopted.

> **What a Reader Can Take Away From This Book – in Practice**
>
> → An understanding of the legal architecture: who is legally responsible for what, and why both sides of almost any waste dispute are often right at the same time.
>
> → A map of the actors with their built-in conflicts of interest – a practical tool for knowing who to address a specific question to.
>
> → The evidence scale – a way of telling a documented fact apart from a likely assumption and

from “unknown.”

→ An understanding of why the system’s financial incentives systematically fail to align with its stated environmental goals.

→ A civic audit methodology – what to request, how to observe, how to turn it into an argument.

→ A concrete action plan at different levels of engagement – from an individual resident to a municipal official.

The book begins with how the law is structured. It ends with a question about what one person, one neighbourhood, one organisation can do inside a system that seems too large and too slow to change. The answer is not simple – but it exists, and it is concrete.

Chapter 1. What Counts as Waste: Definitions, Law, and Responsibility

"Before asking who is to blame for a system that does not work, we need to understand something simpler: who is legally required to build it in the first place."

This book does not begin with numbers or criticism. It begins with definitions. That is not a technicality. Most waste disputes in Israel – between residents and municipalities, between municipalities and the state, between NGOs and regulators – are in fact disputes about the boundaries of responsibility: who is obliged to remove what, and at whose expense.

To take part in that conversation meaningfully, we first need to clarify three things: what legally counts as waste in Israel, who bears primary responsibility for its collection and removal, and how that responsibility has been distributed across law in a historically fragmented way.

1. What counts as "waste" under Israeli law

The basic law governing waste in Israel is the Cleanliness Law of 1984. Article 7 defines the principal categories on which the rest of the system is built.

Main waste categories under the Cleanliness Law

- **Mixed waste (psolaʾ meʿuravt):** solid waste from any source – household, commercial, industrial, or agricultural – containing mixed organic and inorganic components such as food residues, plastic packaging, garden waste, and the like, excluding hazardous materials.
- **Construction waste (psolaʾ binyan):** waste generated by construction, renovation, and demolition.
- **Bulky waste (psolaʾ gushit):** furniture, appliances, and similar items.
- **Special waste streams:** regulated by separate laws, including packaging (), electronics and batteries (Electronic Equipment Law, 2012), and beverage containers with a deposit system (Deposit Law, 1999).Packaging Law, 2011

The structure matters. Israel does not have a single comprehensive waste law. Instead, it relies on a foundational cleanliness law plus a series of separate laws for specific waste streams, adopted at different times in response to different problems. That is not an accident. It is a historical pattern, and we will return to it at the end of this chapter. It also explains directly why the 2026 reform in Chapter 16 proposes a single framework law.

2. Who is responsible for removal: law and precedent

Article 13 of the Cleanliness Law assigns the local authority the duty to organize waste removal within its jurisdiction. This is reinforced by Article 146(8) of the Local Councils Order of 1950, which places responsibility for removing nuisances – including waste – on the local authority.

Under the hood: the legal precedent

The Israeli Supreme Court considered a dispute over how waste-removal responsibility is divided between the state and a local council. Justice Alex Stein, with the support of Justice Yosef Elron, rejected the claim against the state but upheld it against the local council.

The Court held that responsibility for organizing waste removal lies exclusively with the local council, on the basis of Article 146(8) of the Local Councils Order and Article 13 of the Cleanliness Law.

The state's role, as accepted by the Court, is regulatory: enforcement, supervision, and funding support – but not the operational execution of waste removal.

The local council, for its part, did not deny its powers in sanitary matters, but argued that it could not finance the elimination of waste problems and that the budget allocated to it by the state was insufficient.

This is not an abstract legal dispute. It directly explains the hidden subsidy structure discussed in Chapter 14: the municipality is legally obligated to provide waste removal, but the funding it receives from the EPR system and from the state systematically does not cover actual costs.

This precedent is the key to understanding why so many waste disputes in Israel ultimately turn into municipal-budget disputes rather than state-policy disputes. Legally, you may complain to the mayor – because the mayor is formally responsible. But the mayor will rightly point out that state funding is insufficient. Both sides are right at the same time – and that is a structural problem, not a personal conflict.

3. Additional laws: a patchwork, not a system

After the basic Cleanliness Law of 1984, Israel adopted separate laws for specific waste streams – each time in response to a concrete problem, not as part of an overall plan.

Year	Law	What it regulates
1984	Cleanliness Law	Basic responsibility for removal; prohibition of littering; enforcement powers
1993	Collection and Removal of Waste for Recycling Law	Regulates the collection of recyclable waste; complements, rather than replaces, the Cleanliness Law
1999	Beverage Container Deposit Law	Deposit system (discussed in detail in Chapter 15)
2011	Packaging Law	EPR mechanism through TAMIR (discussed in Chapters 14–15)
2012	Electronic Equipment and Batteries Law	Disposal of electronics and batteries
2026 (in process)	Construction Waste Amendment	Regulation of construction waste (~7.3 million tons/year) for the first time

The last line of the table is not a historical note but a live development. In June 2026, the Knesset Interior and Environmental Protection Committee approved an amendment to the Cleanliness Law specifically regulating construction waste – a stream that, more than forty years after the main law, still lacked its own dedicated regulatory regime.

What the construction waste example shows

Construction waste amounts to about 7.3 million tons a year – a substantial stream that remained without a special legal framework for decades.

The State Comptroller documented as early as 2021 that the absence of regulation led to widespread illegal dumping of construction debris in open areas and along roads.

The law was incorporated into a broader package – the Ministry of Finance's 2023 Economic Arrangements Law – the same legislative instrument now being used for the much larger reform described in Chapter 16.

This illustrates a systemic pattern: Israeli waste legislation has historically developed reactively, stream by stream, rather than as a unified system. The 2026 reform is designed precisely to interrupt that pattern.

4. Who else is involved: fragmented enforcement

Responsibility for enforcement is distributed across several bodies that are not always coordinated with one another.

Who enforces the Cleanliness Law

- **Israel Police:** the source of most fines (63%) for individual littering processed by the Ministry of Environmental Protection.
- **Green Police (HaMishteret HaYeruka):** a specialized unit within the Ministry of Environmental Protection.
- **Municipal inspectors:** operate within their own municipality's jurisdiction; fines are processed independently by each municipality.
- **Rangers of the Nature and Parks Authority:** operate in nature reserves and national parks.
- **"Neʾamnei Nekiʾut" volunteers:** civic volunteers who help document violations.

Crucially, no single body centrally collects data on all violations and fines. Municipal inspectors' fines are processed separately by each municipality, which means that a national picture of enforcement simply does not exist.

Conclusion

The main conclusions of this chapter are simple, but they matter.

- Israel does not have a single waste law. It has a foundational Cleanliness Law of 1984 plus a series of separate laws for specific waste streams, adopted reactively over the course of four decades.
- Legal responsibility for waste removal lies with the local council, under Article 13 of the Cleanliness Law and Article 146(8) of the Local Councils Order, as confirmed by Supreme Court precedent.

- The state plays a regulatory, not operational, role. That means municipalities are physically responsible for removal, even when funding is insufficient.
- Enforcement is fragmented across the police, the Green Police, municipal inspectors, and rangers, with no unified data system.
- The 2026 construction waste amendment is a live example of how the system continues to evolve stream by stream rather than as a whole. That leads directly to the question Chapter 16 addresses in detail: what if the problem is not another patch, but the absence of a true framework law?

The next chapter continues this line of inquiry, but shifts from the law itself to the actual actors: who physically manages waste flows in Israel today, and what conflicts of interest are built into that structure.

References

Cleanliness Law, 1984 (Hebrew: חוק שמירת הניקיון, התשמ"ד–1984). Article 2 prohibits littering in public space; Article 7 defines mixed waste, construction waste, and special waste categories; Article 13 obligates local authorities to arrange waste removal within their jurisdiction.

Cleanliness Law, 1984. Article 13; Local Councils Order, 1950 (Hebrew: צו המועצות המקומיות, תשי"א–1950). Article 146(8) assigns responsibility for removal of nuisances within a local authority's jurisdiction to that authority; Supreme Court precedent on division of waste-removal responsibility between the state and local councils.

Israel Hayom, "7 Million Tons of Waste a Year: The New Law Designed to Improve Israel's Environmental Quality," June 2026. The Knesset Interior and Environmental Protection Committee approved an amendment to the Cleanliness Law regulating construction waste (~7.3 million tons/year), folded into the Ministry of Finance's 2023 Economic Arrangements Law. State Comptroller materials on construction waste regulation.

Research Institute for the Knesset, "Enforcement of the Prohibition on Personal Littering in Public Space." Documents the fragmented enforcement structure: Israel Police account for 63% of fines processed by MoEP; the Green Police, Nature and Parks Authority rangers, and municipal inspectors each operate independently, without a centralized data system.

Chapter 2. Who Manages Waste in Israel: Actors and Powers

"The system has no single owner. That is not a bug – it is the result of a historical architecture in which every actor is responsible for a piece of the puzzle, and almost no one is responsible for the whole."

Chapter 1 showed that legal responsibility for waste removal lies with municipalities, while the state plays a regulatory role. In practice, however, the system involves a wide range of actors, each with its own powers, interests, and constraints. Understanding this map is essential if you want to know whom to approach with a specific question – and why the answer so often sounds like "that is not our department."

1. The actor map

Who is who in Israel's waste governance system

- **Ministry of Environmental Protection (MoEP)** – the national regulator. It sets strategy (the National Plan 2030), licenses facilities, administers the Cleanliness Fund, and exercises partial oversight over EPR systems.
- **Local authorities** – more than 250 municipalities, local councils, and regional councils. They are legally responsible for organizing waste collection and removal within their territories, as established in Chapter 1. Their resources, scale, and service quality vary enormously.
- **TAMIR (Packaging Producer Responsibility Corporation)** – the recognized operator of the packaging EPR system under the Packaging Law. Discussed in detail in Chapters 14–15.
- **Deposit system operators (ELA, Asofta)** – manage the return of beverage containers. Discussed in Chapter 15.
- **Private collection and hauling contractors** – work under contract with municipalities; there are many small and medium-sized companies in this sector.
- **Operators of sorting facilities and landfills** – licensed by MoEP; some are private companies, others are municipal associations (iʿugei arim).
- **Israel Land Authority (RMI)** – controls the land needed to expand landfills; coordination with it is a critical bottleneck (Chapter 6).
- **Ministry of Finance** – not formally a sectoral ministry, but in practice one of the most influential actors through the Economic Arrangements Law and its control over the Cleanliness Fund.
- **State Comptroller** – does not manage the system, but conducts periodic audits that become the main source of publicly available data on its condition.

2. Where conflict is built in

This map is not just a list. It contains structural conflicts of interest that explain much of what looks like "malfunction" but is in fact the logical result of how incentives are distributed.

Conflict	What it means
MoEP as regulator and as defender of its own system	MoEP sets recycling targets and also oversees TAMIR – the operator that repeatedly falls short of those targets (Chapter 14). This creates an internal incentive to soften the public framing of the problem.
Ministry of Finance as budget controller and reform sponsor	The Ministry of Finance both determines how much money municipalities receive for waste services and promotes a reform that promises them savings (Chapter 16). That is not a neutral position; the ministry has a direct fiscal interest in the reform's outcome.
Municipality as responsible party and underfunded implementer	Municipalities are legally responsible for removal (Chapter 1), but the funding they receive may not cover actual costs because of the hidden subsidy structure (Chapter 14). This creates an incentive to provide only the minimum acceptable level of service, not the optimal one.
RMI as land controller and detached observer of the crisis	The Israel Land Authority controls the land needed to expand landfills, yet it does not bear direct responsibility for the consequences of the capacity crisis (Chapter 6). That creates an asymmetry between the institution that can approve land use and the one that answers for the fallout.

Under the hood: why this matters for activists

When you write a request or a complaint, it matters which of these actors your question is actually directed to – and what built-in interest shapes their response.

A request about the tonnage of collected packaging is a question for TAMIR (Chapter 25), but interpreting that data requires recognizing that TAMIR has an interest in presenting an optimistic picture of its own performance.

A request about the sanitation budget is a question for the municipality, but the reply will often point to insufficient state funding – and that is not just an excuse, but a real structural problem, confirmed by legal precedent in Chapter 1.

Understanding this map of conflicts of interest is not a reason to trust no one. It is a practical tool for knowing which questions to ask each actor.

3. State infrastructure power: how major decisions are made

Beyond the everyday governance of waste, there is another, rarer, but crucial mechanism: national infrastructure planning. This is the channel through which decisions on the largest facilities – such as waste-to-energy plants – are made.

The decision to build the Neot Hovav facility was not taken through the ordinary municipal or regional planning process, but through a special mechanism: Government Decision No. 1895 (October 2022), within the National Infrastructure Plan.

Why this matters: national planning bypasses local procedures

National infrastructure plans (TATLs) allow the government to approve major facilities more quickly than through standard local and regional planning procedures.

This accelerates projects that might otherwise be delayed for years in local approval processes – which helps explain why WtE projects move forward faster than, for example, the expansion of conventional landfills (Chapter 6).

But it also means that local civic participation in such decisions is structurally weaker than in ordinary municipal development. The points of influence described in Chapter 3 exist precisely because the national process – though faster – still formally includes stages of public consultation that cannot be skipped altogether.

Conclusion

The main conclusions of this chapter are straightforward.

- There is no single "owner" of the system – there are around a dozen actors with different, and sometimes conflicting, powers and interests.
- Structural conflicts of interest are not an accident but a built-in feature of the architecture: the regulator both supervises and protects its own system, the Ministry of Finance both allocates budgets and promotes reform, and municipalities both carry responsibility and remain underfunded.
- The largest infrastructure decisions, such as WtE facilities, are made through a special national planning mechanism that speeds implementation but narrows the space for local civic participation.
- Understanding this map is a practical tool: it tells you whom to address with a specific question and what kind of answer to expect.

Now that the legal framework (Chapter 1) and the actor map (Chapter 2) are in place, we turn to the sharpest example of how this system makes decisions under pressure: the construction of WtE plants at a moment when alternatives are, quite literally, running out.

References

Government of Israel, Decision No. 1895, October 2022. National Infrastructure Plan TATL 107 – Neot Hovav Waste-to-Energy Facility. Approved the first major WtE project under national infrastructure planning powers, bypassing standard local planning timelines.

Chapter 3. WtE, Infrastructure Lock-In, and Why Citizens Matter Now

"By the end of 2026, there will be no remaining landfill capacity in Israel. That is not a scenario or an activist warning – it is the wording of the State Comptroller. It is under these conditions that decisions are being made about plants that will operate for the next 25–30 years."

How to read this chapter

In Chapter 1, we established who is legally responsible for waste management. In Chapter 2, we mapped the actors involved in the system and the conflicts of interest built into it. This chapter applies both of those insights to the most urgent and fastest-moving issue in Israeli waste policy: the construction of waste-to-energy plants, or WtE.

1. Crisis, not choice

At the end of 2025, Israel selected the winner of the tender for its first major WtE plant. The Shapir-Blugene-Dekel consortium won the right to build the Neot Hovav facility in the Negev: 300,000 tons a year of capacity and roughly 50 MW of output.

Two more plants of similar or larger scale are currently in pre-tender planning, each sized at 650,000 tons a year. All of this is part of the National Strategic Plan 2030, which sets a target of turning 23% of Israel's municipal solid waste into energy.

Before July 2025, this chapter could have been written in the following terms: "Israel is considering WtE as one of several tools in a long-term strategy for managing a growing waste stream." That would have been correct, but not precise enough. The reality is sharper.

According to the State Comptroller's follow-up audit, remaining landfill capacity at the start of 2025 stood at 7.67 million cubic metres – not enough to absorb even the waste volume expected by the end of 2026.

What this changes

WtE in Israel is not a strategic choice for the distant future. It is one of the few available responses to a crisis with a specific and near-term deadline. That means any long-term infrastructure decision – whether WtE, accelerated landfill expansion, or a major push in recycling capacity – is being made not in conditions of calm planning, but under intense time pressure.

In November 2025, the Manufacturers Association of Israel petitioned the Supreme Court, describing the situation as approaching "a point of no return." That phrase came not from an environmental NGO but from industrial lobbyists – a sign of how broad the consensus has become about the severity of the crisis.

The WtE contracts discussed below are signed for 25–30 years, but the decisions about them are being taken under a clock measured in months. That combination of long-term commitment and short-term pressure is the central risk factor this chapter examines.

2. Why Israel needs WtE

Before discussing the risks, we need to answer the question honestly: why did this idea emerge at all – and why is it moving so quickly now?

Israel is a small country, densely populated, and short on land. Even now, 83% of municipal solid waste goes to landfill, compared with an average of 35% across Europe and the OECD. Most of the country's large landfills are in the south, far from the main waste-generating areas in the north and center. That means long transport routes, concentrated burdens on southern regions (discussed in Chapter 7), and now – physical exhaustion of available capacity.

What WtE actually solves in the 2026 crisis

- It reduces the volume of waste going to landfill, cutting weight by about 70% and volume by up to 90% – directly addressing the shortage identified by the Comptroller.
- It generates electricity from residual waste that cannot be materially recycled.
- It reduces methane emissions from landfills.
- It allows waste to be treated closer to where it is generated, reducing transport routes – especially important under the logistics stress described in Chapter 6, including the 2023 wartime stress test.
- It is a technically familiar, already proven solution, which shortens implementation time compared with less established alternatives – a major advantage under compressed timelines.

These are real arguments, and in a capacity crisis they carry more weight. The question is no longer "Is it technologically useful?" but "What can actually be built quickly enough?" That shift is the key difference between the present moment and what could still have been written in 2024.

3. What WtE does not solve

The urgency of the crisis does not remove the technology's structural limits. If anything, the faster the decisions are made, the more important it is to understand in advance what those decisions will not do. Local prevention infrastructure is therefore not a soft alternative to "real" waste infrastructure. It is one of the few ways to reduce the future feedstock of landfills, RDF and WtE before long-term capacity commitments are locked in. Repair, reuse, food-sharing, organics diversion and bulky-item rescue operate upstream of the end-of-pipe system. If they remain absent, the country does not simply manage waste badly; it designs a future in which more residual waste must continue to exist in order to feed the infrastructure already built.

What WtE does not solve

- It does not reduce the amount of waste produced – and given the failure of the EPR system (Chapter 14: only 6% of household packaging reaches the correct container), that means the feedstock for WtE will include a substantial share of material that should never have been there in the first place under a functioning separate-collection system.
- It does not return materials to circulation.
- It does not solve the problem of hazardous ash – it requires special landfills, meaning the disposal problem is transformed rather than eliminated.
- It does not replace separate collection – on the contrary, when the input stream is not clean enough, WtE efficiency declines.

- It does not make a country independent of recycling infrastructure. Countries with high WtE rates, such as Sweden, Denmark, and the Netherlands, also have high recycling rates. Israel, with a recycling rate of 23.4%, is moving toward WtE without that parallel foundation.

This creates a specifically Israeli risk: building WtE capacity without comparable progress in recycling means the country risks repeating a path that several European states are now reconsidering – excessive dependence on incineration alongside an underdeveloped system of sorting and material recovery.

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State Comptroller of Israel. Follow-up Audit: Municipal Solid Waste Disposal. July 2025. Key finding: remaining landfill capacity (7.67 million cubic metres at the start of 2025) will not suffice for waste volumes expected by the end of 2026.

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4. Infrastructure Lock-In: How Crisis Makes It More, Not Less, Dangerous

Lock-in is a concrete economic mechanism built into any large infrastructure project backed by a long-term contract. In stable planning conditions, regulators have time to design safeguards against it. In a crisis, that time shrinks.

How lock-in happens

1. A plant costing hundreds of millions of shekels is built, and financing depends on securing a return on investment.
2. The contract guarantees a minimum waste flow for 25–30 years; without that guarantee, private financing would not be possible.
3. If the actual flow falls below the guaranteed minimum, municipalities pay a penalty, or capacity charge.
4. This creates a systemic disincentive for waste prevention: the more successfully waste is reduced, the closer a municipality comes to the penalty zone.
5. Twenty-five years later, the question of whether to build a new plant returns under conditions where alternative infrastructure was never developed, because there was no incentive to develop it.

Under the hood: why crisis increases, rather than reduces, lock-in risk

Under calm planning conditions, regulators can spend time refining contractual safeguards: KPIs, protections for recycling priority, and transparent monitoring systems.

When industrial lobbyists speak of a “point of no return” and the Supreme Court is hearing an urgent petition, political pressure shifts toward “build faster” rather than “build with stronger safeguards.”

This is the classic dilemma of emergency regulation: decisions made under time pressure are less likely to include long-term protective mechanisms, because bargaining power shifts toward the contractor that can build quickly. Zero Waste Europe and the UK DEFRA both document this as a general pattern; in Israel’s 2025–2026 case, the risk is intensified by a specific crisis with a clear date attached to it.

5. Ash and emissions

Urgency does not remove the need to ask technical questions – it only reduces the time available to ask them.

Two types of ash

- **Bottom ash** – about 20–30% of the incoming waste stream by weight; after treatment, part of it can be used in construction.
- **Fly ash** – about 3–5% of the stream; contains heavy metals, dioxins, and furans; must be disposed of in hazardous-waste landfills.

Key questions about ash

- How many tons of fly ash will be generated each year, and does Israel already have certified disposal capacity for it, or will that also have to be built in haste?
- Who is responsible for monitoring the ash landfill over the next 50 years?
- Will the accelerated contract contain the same emissions-monitoring requirements that would have been included under slower planning, or has urgency lowered the standard?

6. Why citizens matter now

Under ordinary circumstances, the phrase “now is the best time for civic participation” sounds rhetorical. Here, it has a literal, dated meaning.

Under the hood: where civic influence still exists

At the time this chapter is being written, the tender stage for the two remaining major projects – Jerusalem and the southern region – has not yet been finalized. That means the key contractual terms, including whether to include a take-or-pay clause, what monitoring requirements to impose, and how to protect the priority of recycling, are still subject to revision.

The faster the process moves under crisis pressure, the more important it becomes for civil society organizations to submit specific, ready-to-insert demands – not broad positions for or against WtE, on which there may soon be no time left to deliberate.

At the same time, the 2026 reform discussed in Chapter 16 may create a national regulator with authority to oversee exactly these kinds of contracts – but that institutional protection does not yet exist. Until it does, direct participation in the discussion of specific tenders is the only real lever.

7. Questions to put to the system now

Financial model and contractual obligations

1. Does the contract contain a take-or-pay clause – an obligation to pay for capacity even when the actual waste flow is lower?

2. What is the minimum guaranteed waste volume in the contract, and how does it align with prevention and recycling targets?

3. If accelerated construction timelines require accelerated contract approval, which protective clauses may be simplified or omitted under time pressure, and who oversees that process?

Ash, emissions, and recycling priority

4. Where will fly ash be disposed of, and is certified capacity already available – or does that too still need to be built on a compressed schedule?

5. Which body will carry out independent emissions monitoring?

6. Does the contract include a mechanism preventing materials for which functioning recycling technology already exists from being sent to WtE?

7. How has the failure of the EPR system – with only 6% of household packaging reaching the correct container, as discussed in Chapter 8 – been factored into projections of the WtE feedstock for the next 25 years?

Conclusion

The main conclusions of this chapter are clear.

- In 2026, WtE in Israel is not a matter of calm strategic choice but an urgent response to a crisis with an officially confirmed date for the exhaustion of capacity.

- That urgency does not remove the technology's structural limits: WtE does not reduce waste generation and does not return materials to circulation.

- Infrastructure lock-in becomes more, not less, dangerous under crisis conditions: decisions made under time pressure are less likely to include long-term safeguards.

- Questions about ash, emissions, and recycling priority remain relevant regardless of urgency – but urgency shortens the time available to debate them.

- Citizens matter now in the most literal, dated sense: the tender documents for the remaining projects are being shaped right now, alongside discussion of the 2026 reform in Chapter 16, which may eventually create institutional protection but has not yet done so.

In the next parts of the book, we will see this same choice – between speed and caution, between urgent response and long-term planning – recur again and again: in the EPR system (Chapters 14–15), in the 2026 reform itself (Chapter 16), and in the ways citizens can build alternative infrastructure at neighborhood level without waiting for the system at the top to resolve its own crisis (Chapter 23).

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Chapter 4. First Stage: The Kitchen, the Yard, the Bin Site

"Waste management does not begin at the landfill or the sorting facility. It begins at arm's length from the sink – at the moment you decide which bin to put an empty package into."

Chapters 1–3 set out the law, the actors, and the system's sharpest political question: WtE. Chapters 6–8 will take us to the end of the chain – landfills, spatial injustice, and the evidentiary base. This chapter and the next are the bridge between them. They are about what happens physically, on the ground, before material ever enters a larger statistical system.

And it is precisely here, at this very first step, that the main system failure occurs, according to the MoEP's 2025 national scan: only about 6% of household packaging waste reaches a dedicated recycling container. Before we ask what is wrong with sorting or recycling, we need to understand what happens a few metres from every Israeli kitchen.

1. The anatomy of the first mile

The "first mile" is the path from the point where waste is generated – the kitchen, bathroom, or yard – to the bin site. On this short stretch, a large part of the material's fate is decided, before any municipal infrastructure or TAMIR process comes into play.

Three decisions made on the first mile

- **Decision 1: Which bin does it go in?**
  This depends on how many bins are physically available, how clearly they are marked, and whether residents actually know what belongs where.

- **Decision 2: How clean is the material when it is discarded?**
  Contaminated packaging is one of the main causes of sorting rejects, discussed in Chapter 10. That decision is made in the kitchen, not at the sorting facility.

- **Decision 3: Is it worth using the dedicated bin at all, or is it easier to throw everything into the general waste bin?**
  This decision is shaped by convenience, distance to the bin, and habit – factors that belong entirely to the first mile, not to later stages.

Each of these decisions is made millions of times a day across the country. The MoEP's 2025 scan is, in effect, the aggregate outcome of those decisions: if only 6% of packaging ends up in the correct container, that means the vast majority of these micro-decisions are going against separate collection.

2. Bin types and what belongs in them

Israel's household separation system is built around several colour-coded bins, each governed by a different set of rules.

Container	What belongs inside	Who regulates it	Common mistake
Orange	Plastic and metal packaging, beverage	TAMIR /	Glass, which breaks, injures sorters, and

Container	What belongs inside	Who regulates it	Common mistake
	cartons	Packaging Law	disrupts the line
Blue	Paper and cardboard	Municipality / 1993 Law	Dirty or wet paper; large corrugated cardboard not flattened
Purple / special collection points	Glass packaging	TAMIR	Confused with the orange bin; not all neighbourhoods have nearby access points
Grey / green (mixed)	Everything else	Municipality	About 60% of recyclable materials end up here, according to the MoEP 2025 scan

Under the hood: why glass is a separate story

Glass is physically hazardous for a standard sorting line: it breaks during transport and sorting, shards can injure workers, and it can damage machinery designed for plastic and metal.

That is why glass never goes into the orange bin. It is collected through separate dedicated points. But those points are much less dense than orange bins, which creates an additional convenience barrier.

This connects directly to the problem examined in Chapter 15: the recycling rate for non-deposit glass in Israel is only 4–8%, among the lowest material-specific rates. The first mile is part of the explanation: if the collection point is inconvenient, the material is more likely to end up in the mixed bin.

3. When the standard bin does not fit: dense urban neighbourhoods

The standard model of a “bin in the yard” does not work for every type of housing. In dense urban neighbourhoods, especially older buildings without a separate waste room, there is simply no physical space for a full set of colour-coded bins.

Tel Aviv-Yafo Municipality acknowledges this directly: in areas where orange bins cannot be placed on private property, alternatives are offered – public recycling points, solar bins, and separate collection through special bags. After a successful pilot in several neighbourhoods, separate collection in orange bags was expanded: residents of buildings without yard space are given special packaging bags, which they place on the street once a week on a fixed day.

Why this matters structurally, not just technically

This means that access to separate collection is uneven not only between municipalities, but also within the same city – between different housing types.

Residents of newer buildings with spacious waste rooms have fundamentally better access to the separation system than residents of older housing in the city centre.

That adds another layer of inequality on top of the geographic one discussed in Chapter 7: inequality by housing type. It is rarely discussed separately, but it directly affects the very 6% identified in the MoEP's 2025 scan.

4. Bin capacity and collection frequency: the economics of the first mile

The State Comptroller found an unexpected pattern: the amount of container volume per resident correlates with the cost of collection for the municipality.

The logic is straightforward. The larger the total bin volume per resident, the more equipment, fuel, and labour hours are needed per unit of waste actually collected – especially when bins are not filled to capacity. In denser neighbourhoods with a smaller container volume per resident, by contrast, collection costs per ton can be lower. Rahat is a useful example: because of its density, roughly half of its housing units are served through a different collection system.

In all of the municipalities examined, mixed household waste is collected two to three times a week. Collection frequency for recyclable streams – orange and blue bins – is often lower, sometimes only once every one or two weeks. That creates an additional incentive for residents to throw material into the more frequently emptied mixed bin if the orange bin is already full.

5. Technological monitoring of the first mile: what becomes visible

The Eastern Negev cluster of municipalities is implementing an RFID-based monitoring system that allows the municipality to track in real time which bins were collected, identify areas that are not being serviced, and compare contractor reports with the actual volumes arriving at the transfer station.

Under the hood: why a municipality needs this system

- It helps monitor the contractor: payment only for containers actually collected and identified as belonging to the municipality, rather than for containers added illegally.
- It allows early detection of unserved neighbourhoods, before residents' complaints accumulate.
- It makes it possible to compare daily collection volumes with transfer-station or end-facility data, so the municipality can check whether the contractor's reported volumes match what physically arrived for sorting or disposal.

This is an example of how the first mile is gradually becoming measurable – and measurability, as shown in Chapter 25, is a necessary condition for meaningful civic oversight.

Conclusion

The main conclusions of this chapter are clear.

- The first mile is the point where most of the decisions that determine the fate of material are made, before any major infrastructure enters the picture.
- According to the MoEP's 2025 scan, only 6% of household packaging reaches a dedicated recycling container, while 60% of the mixed-bin contents are materials that should never have been there.

- Access to separate collection is uneven not only geographically, as Chapter 7 will show, but also by housing type: older, dense residential stock is systematically served worse by the standard courtyard-bin model.
- Bin volume and collection frequency directly shape the economics of the first mile for municipalities – and the behaviour of residents, who are more likely to choose the mixed bin when the orange bin is full.
- Technological monitoring (RFID, digital recordkeeping) is gradually making the first mile measurable, which is crucial for building meaningful civic oversight (Chapter 25).

The next chapter continues the material beyond the first mile – to where the truck goes after the bin site: the invisible middle of collection, transfer, and sorting.

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Chapter 5. Collection, Transfer, and Sorting: The Invisible Middle

> *"Between the bin outside your home and the headline about 'X% recycling' in the news lies an entire invisible infrastructure: trucks, transfer stations, dozens of private contractors. Most residents never see it – and that is precisely why it has remained outside public scrutiny for years."*

Chapter 4 ended at the bin collection point. This chapter follows the material further along – into territory most residents never look at: the network of transfer stations, private contractors, and sorting complexes that physically connects the first mile to the end-of-pipe zone (Chapter 6) or to recycling (Chapters 9–13).

This middle stretch is invisible for a reason. It is structured so that an ordinary resident never encounters it, and even the municipality sees only part of the picture – whatever part the contractor chooses to provide.

1. What a Transfer Station Is

The Cleanliness Law defines a transfer station (tachanat ma'avar) as a fixed facility where waste is moved from one container (including a vehicle) into another, or sorted into components for recycling or reuse.[1]

> **Why a Transfer Station Is Needed**
>
> Small collection trucks (built for narrow city streets) cannot physically make the trip all the way to a distant landfill – it simply isn't cost-effective. A transfer station lets the contents be reloaded onto high-capacity haulage vehicles.
>
> Volume reduction happens at the station (compaction, shredding), which lowers the cost of onward transport.
>
> Some stations combine the transfer function with primary sorting – pulling out large recyclable fractions before sending the remainder onward.
>
> A transfer station is, in effect, the bottleneck through which an entire district's waste stream physically passes. That is precisely why transfer station data (tonnage by day, by source) is one of the most valuable targets for a civic data request (Chapter 25).

2. Who Physically Hauls the Waste: A Fragmented Market of Contractors

Contrary to what one might expect, Israel has no single state operator for waste collection and haulage. The market consists of dozens of private contractors, each working under contract with one or several municipalities, or directly with TAMIR.

TAMIR's list of authorised contractors includes dozens of companies – ranging from major players who own their own transfer stations and vehicle fleets, to highly specialised operators who handle a single waste stream in a single region.[2]

> **◆Under the Hood: A Typical Contractor Structure – An Industry Example**
>
> *Vehicle fleet: RAMSA trucks (with a hydraulic hook for swapping containers) for routine collection; crane trucks for direct haulage from industrial sites; compactors for reducing volume right at the point of collection.*
>
> *Its own transfer station: large contractors typically own or lease a transfer station, where material is sorted and prepared for onward transport.*
>
> *Licences: a company must hold a valid business licence under Article 5.1(b) of the Business Licensing Order, and, to handle packaging, authorisation from TAMIR.*
>
> *Geographic specialisation: most contractors operate in a specific region (for example, "covers the whole North" or "Eilat and the Arava") – which means it is practically impossible for a resident to find out who physically hauls their waste without submitting a direct request to the municipality.*

3. What Happens at the Transfer Station: Primary Sorting

Many transfer stations do more than just reload waste – they also carry out primary sorting: pulling out large, easily identifiable fractions before the remainder is sent onward.

What Gets Separated at This Stage	Where It Goes
Large scrap metal	Directly to metal-recycling plants – high market value makes this profitable even without a formal EPR system.
Large cardboard (if not collected separately)	To a cardboard sorting facility, where there is market demand.
Garden waste / gezem (branches, leaves)	Shredded and sent for composting or for producing wood-chip mulch.
Hazardous waste spotted visually	Removed and sent for specialised disposal – though detection at this stage is limited to visual inspection.
The remainder (mixed, unsorted)	Transported to landfill, to RDF, or to a specialised sorting complex (for orange-bin content – see Chapter 10).

It is worth understanding that this primary sorting at a transfer station is technically different from the detailed fraction-by-fraction sorting carried out at TAMIR's specialised complexes (Rishon LeZion, Afula, Eilat – Chapter 10). A transfer station works more coarsely and faster; detailed sorting by polymer type happens only downstream, at specialised facilities.

4. New Capacity: What's Being Built Right Now

The invisible middle is not a static system. Against the backdrop of the end-of-pipe crisis (Chapter 6) and the failures at the first-mile level (the MoEP scan, Chapter 4), serious investment is now flowing into it.

> **◆Link to the 2026 Reform (→ Chapter 16)**
>
> The new flagship TAMIR-KMM sorting complex in Rishon LeZion (Chapter 10) is designed with the stated goal of tripling the volume and pace at which material from orange bins nationwide is processed – a direct response to the insufficient throughput of the invisible middle.
>
> The Cleanliness Fund approved 900 million shekels for building sorting capacity and a further 144 million for additional projects in December 2025 – direct funding to expand precisely the link in the chain this chapter is about.
>
> The Ministry of Environmental Protection has been advancing 32 initiatives to build new sorting and recycling facilities since the Comptroller's previous audit – an attempt to build up the capacity of the invisible middle systematically, rather than patching individual stations one at a time.
>
> If the 2026 reform (Chapter 16) results in a national regulator, licensing and standards for this fragmented market of contractors and transfer stations could become considerably more unified than they are today.

5. Why This Middle Stretch Stays Invisible to Residents – and What to Do About It

A resident sees the bin collection point (the first mile) and, sometimes, headline statistics on overall recycling rates. Everything that happens between those two points remains opaque almost by default.

> **◆Structural Reasons for the Opacity**
>
> Market fragmentation: dozens of independent contractors mean there is no single point of contact for data on routes and volumes.
>
> Commercial confidentiality: routes, tonnage, and the financial terms of contracts between a municipality and a contractor are often shielded by a commercial-confidentiality claim in response to Freedom of Information requests (see Chapter 25).
>
> No standardised public reporting: unlike the first mile (where the bin collection point is at least visible), a transfer station is physically closed to outside observation.
>
> Technological monitoring systems (RFID, digital logging – see Chapter 4) do exist, but in most cases the data they generate remains an internal municipal tool rather than something published.

◆Questions That Make the Invisible Middle Visible

1. Which company physically hauls waste from my bin collection point, and where is its transfer station located?

2. Does that station carry out primary sorting – and which fractions are separated out before onward transport?

3. How many tonnes per day/week pass through that station, and where does the remainder go after primary sorting?

4. Does my municipality use a technological haulage-monitoring system (such as RFID) – and is that data published?

5. Does the contract between the municipality and the contractor include a minimum-share requirement for material sent to recycling rather than direct landfilling?

Chapter Takeaways

Key Takeaways

→ Between the bin collection point and the endpoint (recycling, landfill, RDF) lies infrastructure invisible to most residents: transfer stations and a network of dozens of fragmented private contractors.

→ Transfer stations often carry out coarse primary sorting – separating out metal, large cardboard, garden waste – before the remainder moves on.

→ Significant investment is now flowing into this invisible middle: the new flagship TAMIR-KMM complex, 900+144 million shekels from the Cleanliness Fund, 32 MoEP initiatives – a direct response to the end-of-pipe crisis (Chapter 6) and first-mile failures (Chapter 4).

→ The opacity of this link is structural: market fragmentation, commercial-confidentiality claims, and the absence of standardised public reporting.

→ Technological monitoring exists, but mostly remains an internal municipal tool – which makes a civic data request (Chapter 25) the only reliable way to see this part of the system.

Part II of the book has now been covered in full: from the first mile (Chapter 4), through the invisible middle (this chapter), to the end-of-pipe zone (Chapter 6) and its geographic distribution (Chapter 7). Chapter 8 introduces the tool needed to discuss all of this honestly – the evidence scale, used throughout the chapters that follow on specific material streams.

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Notes

1. Cleanliness Law (Israel), 1984. Art. 7 defines a "transfer station" (tachanat ma'avar) as a fixed facility where waste is transferred between collection vehicles/containers during its removal and disposal, including sorting into components for recycling or reuse purposes.

2. TAMIR Authorized Collection Contractors list (tmir.org.il/Contractors). Documents dozens of regional contractors, each typically operating their own transfer station and serving a specific geographic area – illustrating the fragmented, contractor-based structure of Israel's collection and transfer layer.

Chapter 6. Landfills, RDF, and Energy Recovery: The End-of-Pipe Zone

> *"By the end of 2026, Israel will have nowhere left to bury its waste. This is not a metaphor, and it is not an activist's forecast. It is the State Comptroller's own formulation."*

Whatever happens at the earlier stages – however much material is diverted to recycling, however much is prevented in the first place – a waste management system always has a last line: the place where everything that found no other use ends up. In Israel, that place is physically running out.

This chapter is about the end-of-pipe zone: landfills, RDF, and energy recovery. And the first thing that needs to be said plainly is this: Israel entered 2026 with an officially confirmed shortage of landfill capacity.[1]

1. The Capacity Crisis: What the Numbers Say

In July 2025, the State Comptroller published the results of a follow-up audit (מעקב ביקורת) of the state of Israel's landfills. The conclusion is stated in the most concrete terms possible: remaining landfill capacity at the start of 2025 stood at 7.67 million cubic metres – and this will **not be enough** even for the volume of waste expected by the end of 2026.[2]

> **The Crisis in Numbers**
>
> 83% – the share of Israel's municipal solid waste that goes to landfill (compared with an OECD/European average of roughly 35%).
>
> 68% – the share of all the country's waste that is landfilled in the south – often after a long and polluting journey from north to south.
>
> 23.4% – Israel's official recycling rate, against roughly 60% for the OECD average.
>
> 18% → 26% – the growth in the recycling rate over a decade of rising landfill levies. Minimal progress despite a substantial increase in fees.
>
> 7.67 million m³ – remaining landfill capacity at the start of 2025. Not even enough for 2026.

Industry sounded the alarm first – in November 2025, the Manufacturers Association of Israel filed a petition with the High Court of Justice, warning that the country is "approaching the point of no return."[3] The petition documents describe trucks standing in queues for hours outside southern landfills, sorting facilities filled beyond capacity, and drivers dumping waste by the roadside – sometimes with fires breaking out.

2. Why Capacity Isn't Growing

The shortage did not appear overnight. The Comptroller warned about it as far back as 2021. The problem is that between diagnosis and solution lies a long chain of administrative and political obstacles.

Why Landfill Expansion Can't Keep Pace with Waste Growth

Planning bureaucracy. Expanding or opening a landfill requires approval from the Israel Land Authority (Rami), the Planning Administration, and local committees – each stage taking months or years.

Site-specific technical constraints. One southern landfill, for instance, required special lining requirements because of oil-shale deposits in the ground – an engineering problem, not a bureaucratic one.

Aviation safety. Landfills attract birds, which pose a hazard to nearby air bases – another technical barrier to expanding certain sites.

Political reluctance. Expanding a landfill is an unpopular decision for local authorities anywhere in the world; Israel is no exception.

Unfinished institutional reform. A 2024 government decision called for an inter-ministerial team to draft a framework law – the team never actually convened (see Chapter 16).

◆Under the Hood: The Evron Landfill Case – A Microcosm of the National Crisis

The Evron landfill in the north became a vivid illustration of the crisis. Once its capacity was exhausted, municipalities that had previously hauled waste there were forced to find alternatives – which for many meant higher transport and processing costs.

In December 2025, the Cleanliness Fund approved 900 million shekels to fund sorting facilities and 144 million shekels for additional projects – a response, in part, to the situation surrounding Evron.

A decision to raise the height of the Evron landfill by 5 metres starting in 2026 will provide an additional million tonnes of capacity – enough for roughly three more years. But the Ministry of Environmental Protection stresses that this extension will be permitted only for sorted mixed waste – not for direct landfilling.

Karmiel Mayor Moshe Koninsky, at a heated session of the Knesset Interior Committee, described how the closure of Evron directly drove up waste disposal costs for his city.

3. RDF and Energy Recovery: What's Already Working

Against the backdrop of the landfill crisis, interest is growing in end-of-pipe alternatives – not only WtE (covered in detail in Chapter 3), but also a less visible mechanism already in operation: RDF (refuse-derived fuel).

According to the Knesset Research Institute,[4] organic waste makes up about 45% of Israel's municipal solid waste, and another roughly 45% is residual material left after sorting – material that is not technically recyclable, but can be used for energy recovery.

The Hierarchy of End-of-Pipe Methods: What's Worse, What's Less Harmful

Direct landfilling of unsorted waste – the most harmful method. Methane, land use, fire risk, no value recovery at all.

Landfilling after sorting – better, but still landfilling. Used for the residual fraction after recyclable materials have been extracted.

Anaerobic digestion of organics – the least harmful of the end-of-pipe methods. Produces biogas (energy) and digestate (potential fertiliser), substantially reducing methane emissions compared with landfilling organic waste.

RDF / energy recovery – fuel derived from waste, used mainly in cement kilns. Reduces the volume sent to landfill but is not material recycling.

WtE (waste-to-energy incineration with electricity generation) – covered in detail in Chapter 3. Substantially reduces waste volume and weight, but creates infrastructural lock-in and requires management of hazardous ash.

The cost of new end-of-pipe methods is higher than direct landfilling, which is why the Ministry of Environmental Protection uses the Cleanliness Fund to subsidise the price difference – so that sorting and recycling can remain competitive with landfilling.[5]

4. What Happened During the War: A Stress Test for the System

October 2023 became an unplanned stress test for the entire end-of-pipe system – and its lessons matter for understanding structural fragility.

◆Wartime Stress Test: What the Crisis Revealed

Most sorting and recycling facilities in Israel are not protected against rocket fire. As a result, in the first week of the war most waste was sent directly to landfill, bypassing sorting entirely.

Some recycling facilities sustained direct damage from intercepted-rocket debris – including fires at major sites.

Waste-hauling trucks were redirected to military and humanitarian needs, creating a transport shortage and a sharp rise in prices among the remaining contractors.

Southern regional councils recorded a tripling of waste volumes due to the influx of military personnel and evacuated civilians – with no corresponding increase in infrastructure capacity.

This episode showed plainly that Israel's end-of-pipe system operates at the limit even in peacetime. Any shock – military, logistical, or weather-related – immediately exposes its fragility.

5. Reform and the End-of-Pipe Zone

◆Link to the 2026 Reform (→ Chapter 16)

The landfill capacity crisis is one of the three main drivers of the 2026 reform (see Chapter 16). The Ministry of Finance's proposal explicitly links the creation of a national regulator to the need to accelerate planning for alternative end-of-pipe capacity.

In parallel, the Ministry of Environmental Protection is advancing 32 initiatives to build sorting and recycling facilities – but, according to the Comptroller, even full implementation would leave timing uncertain.

The key question for civil society: will the new framework law entrench the priority of the 9R hierarchy (prevention and recycling ranked above energy recovery) – or will the regulator's economic logic be driven purely by capacity-sufficiency considerations?

Chapter Takeaways

Key Takeaways

→ Israel entered 2026 with an officially confirmed landfill capacity shortage – the State Comptroller's data, not an activist estimate.

→ 83% of waste goes to landfill (versus 35% for the OECD/Europe average); 68% of all the country's waste is landfilled in the south.

→ Israel's recycling rate is 23.4%, against roughly 60% for the OECD. A decade of rising landfill levies produced only modest recycling growth (18% → 26%).

→ Capacity expansion is blocked not only by political factors, but by technical and bureaucratic barriers as well.

→ The wartime crisis of October 2023 was a stress test that exposed the structural fragility of the entire end-of-pipe system.

→ The capacity crisis is one of three key drivers of the 2026 reform, covered in Chapter 16.

The end-of-pipe zone is the last step in the chain, but it is not the only factor that determines who bears this system's burden. The next chapter looks at the geographic distribution of that burden – and at who lives next to landfills, not by choice.

Chapter 7. The Geography of Waste and Environmental Justice in Israel

> *"A landfill does not appear by accident. It appears wherever it is politically easiest to put one. And this pattern is not an exception for Israel – it is part of a broader international pattern known as environmental injustice."*

The previous chapter showed that 68% of all of Israel's waste is landfilled in the south of the country. That figure is not simply a logistical fact. It raises a question rarely heard in technical discussions of waste management: who lives near the places where society's most unpleasant, dirtiest, most unwanted material ends up?

This chapter applies the environmental justice framework[6] to the geography of waste in Israel. This approach originated in the United States, in the context of the civil rights and environmental movements, and is used today around the world as an analytical tool for examining who bears an environmental burden and who does not.

1. The Baseline Pattern: The South as the End-of-Pipe Destination

A fact already established in Chapter 6 – that 68% of all Israel's waste is landfilled in the south[7] – forms a fundamental geographic asymmetry. The north and centre of the country, where most of the population and economic activity is concentrated, generate the bulk of the waste; the south – the Negev – bears the main burden of disposing of it.

This asymmetry has a direct economic dimension. The Knesset Research Institute notes that landfilling costs for northern municipalities are lower than for southern municipalities,[8] which seems paradoxical at first glance – given that it is the south that receives the main flow of waste from across the country. The logic is explained by tariff and logistics structures, but the fact itself underscores that living close to disposal sites does not necessarily bring any economic benefit.

2. The Bedouin Communities of the Negev: A Documented Case

Within the southern part of the country there is a further, sharper asymmetry – affecting the Bedouin communities of the Negev (in Arabic, the Naqab).

> ◆**Context: The Status of Bedouin Settlements in the Negev**
>
> The Bedouin population of the Negev numbers, by various estimates, around 305,000 – roughly 3.5% of Israel's population. Some Bedouin communities are recognised by the state and organised into permanent municipal entities (Rahat, Tel Sheva, Segev Shalom, Ar'arat an-Naqab, Kuseife, Lakiya, Hura); others live in settlements with partial or no official recognition, which directly affects access to organised municipal infrastructure, including waste collection.
>
> This is a complex and contested political issue, tied to the region's history of land use, and it lies beyond the scope of this book. Here we confine ourselves to documented facts about its intersection with the waste issue.

Academic research conducted in an unrecognised Bedouin village in the Negev documented that about 40% of the Negev's Bedouin population lives in settlements with no organised solid-waste collection infrastructure.[9] The study found that, in the absence of municipal collection, residents – predominantly women, who traditionally manage household waste – resort to backyard burning, creating a direct risk of exposure to corrosive, toxic, and flammable substances.

◆**Under the Hood: What Exactly Is Documented in the Academic Sources**

Meallem, Garb, and Cwikel (2010, Archives of Environmental and Occupational Health) conducted a multimethod study in the village of Umm Batin, finding that modern consumer lifestyles generate both household and construction waste that is dumped in the immediate vicinity of households in the absence of organised collection.

The State Comptroller, in a 2022 audit of the Western Negev cluster of municipalities, explicitly included among the audit's five main themes "gaps in waste collection between Jewish and non-Jewish municipalities and measures to narrow these gaps." This is the formulation of an official state body, not an activist assessment – and it directly acknowledges the existence of a measurable gap in municipal service.

IWGIA (the International Work Group for Indigenous Affairs) documents, in a 2025 review, that land-use status directly affects Negev Bedouin communities' access to organised municipal services, including waste collection.

The academic article by Bahar (2015, Current Anthropology) analyses how discourse about "trash and disorder" has historically been used to naturalise segregationist planning approaches in the Negev – a methodological observation about language and perception worth bearing in mind when reading official documents and public debate.

Beyond the question of organised collection, the geographic location of the landfills and illegal dumps themselves in the Negev has also been documented as disproportionately close to a number of Bedouin population centres – a pattern noted in several academic sources studying this territory.[10]

3. Not Unique to Israel: The International Context

The pattern in which waste disposal and processing sites are disproportionately located near marginalised, low-income, or minority ethnic communities has been documented in dozens of countries, and it forms the foundation of the very concept of environmental justice as an academic and activist field.

Country / Region	Documented Pattern
United States (classic environmental-justice studies, 1980s–90s)	Landfills and hazardous industries were disproportionately sited in African-American and Latino neighbourhoods – the foundation of the environmental justice movement.
India	Dumps and waste-processing facilities are concentrated near Dalit settlements (the

	former “untouchable” caste) and low-income areas.
Brazil	Favelas and peripheral urban districts disproportionately border landfill and incineration sites.
Kenya / Ghana (e-waste exports)	Developed countries export electronic waste to countries with weaker regulation, shifting the environmental burden onto low-income local populations.
Israel (this chapter)	The south of the country (the Negev) receives 68% of all waste; within the Negev, a documented gap in municipal service quality exists between Jewish and unrecognised/partially recognised Bedouin settlements.

This comparative context matters: the problem raised in this chapter is not a uniquely Israeli anomaly, but a local expression of a pattern studied and contested in many countries worldwide. That does not make it any less significant – on the contrary, it means there is a substantial body of knowledge on how such gaps can be measured and narrowed.

4. What Could Change: International Practices for Narrowing the Gap

In countries where environmental justice has become part of official policy, a number of specific tools are in use. They do not resolve political disputes over land use, but they create measurable mechanisms for narrowing the service gap.

Tools for Narrowing the Environmental Gap: International Experience

Environmental burden mapping. The United States (EPA EJScreen) and a number of EU member states use mapping tools that overlay data on the siting of hazardous facilities with demographic data – making gaps visible and quantifiable.

Mandatory impact assessment for vulnerable communities. Some jurisdictions require a separate assessment of any new landfill or processing facility’s impact specifically on nearby marginalised communities – not just a standard environmental review.

Targeted infrastructure funding in underserved areas. For example, targeted grants for organised waste collection in areas with a historical service gap.

Transparent reporting on service gaps. Regular publication of data on municipal service quality, broken down by geography and demographics – a tool that in itself creates public pressure to narrow the gap.

◆Link to the 2026 Reform (→ Chapter 16)

The State Comptroller (Western Negev cluster audit, 2022) recommended that the municipalities already audited work specifically to narrow the service gaps between Jewish and non-Jewish municipalities. If the 2026 reform (Chapter 16) results in a national regulator with a public-reporting duty, this could become an institutional mechanism for systematically tracking such gaps, rather than relying on one-off audits.

5. What This Means for Activists and NGOs

Chapter 25 of this book describes civic data-request tools in detail. Applied to the theme of environmental justice, these tools take on particular significance.

◆Questions for Civic Monitoring of Environmental Justice

1. Does my region publish data on waste-collection quality broken down by locality?

2. Is there a documented gap in the frequency and quality of collection between different municipalities or districts in my region?

3. What measures has the State Comptroller recommended for my region – and have they been implemented?

4. When a new end-of-pipe facility (a landfill, sorting complex, or WtE plant) is sited – was an impact assessment carried out specifically for the nearest communities?

5. Are there organisations already working on this issue in my region with whom I could coordinate?

Chapter Takeaways

Key Takeaways

→ 68% of all of Israel's waste is landfilled in the south of the country – a geographic asymmetry between where waste is produced and where it is disposed of.

→ Within the southern region there is a further, officially documented gap: about 40% of the Negev's Bedouin population lacks organised solid-waste collection infrastructure.

→ This is documented not only by academic research but by an official State Comptroller audit, which explicitly included the gap between Jewish and non-Jewish municipalities among its audit objectives.

→ This pattern is not a unique Israeli anomaly, but a local expression of the internationally studied phenomenon of environmental justice, documented in dozens of countries.

→ Specific international tools exist for measuring and narrowing such gaps: burden mapping, targeted impact assessment, targeted funding, and transparent reporting.

→ The 2026 reform potentially creates an institutional opportunity for systematically tracking such gaps through the duties of the new regulator.

Part II of the book ends here. We have followed waste from the first mile – the kitchen and the bin collection point – through the invisible middle of sorting and transfer, to the end-of-pipe zone and its geographic distribution. Part III, opening with the evidence scale in Chapter 8, moves to a more detailed look at what actually happens to the materials the system does try to bring back into circulation.

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Chapter 8. “Collected for Recycling” ≠ “Recycled”: The Evidentiary Scale

“In August 2025, the Ministry of Environmental Protection itself published data that confirmed what activists had been saying for years: the system does not work. This chapter is about how to read such data – and why official recognition of failure changes the terms of the discussion.”

When an official report says that “65% of packaging is recycled,” what does that actually mean? Was the material really turned into new material? Or was it collected, sent to sorting – and then the trail disappears?

Until August 2025, this question could only be answered through assumptions, reconstructions, and indirect calculations. Then Israel’s Ministry of Environmental Protection published the results of a national scan of waste streams – and it turned out that the assumptions had not been pessimistic, but optimistic.

According to that scan, only about 6% of household packaging waste reaches dedicated separate-collection containers. And about 60% of the contents of the ordinary mixed (green) bin consist of recyclable materials – plastic, cardboard, and the like – that should have ended up in the orange bin but did not.

This is not an activist estimate. It is not a journalistic investigation. It is official data published by the regulator itself. And it changes the way every recycling statistic in this book – and in Israel more generally – should be read.

This chapter introduces the tool used in all later chapters on waste flows (Chapters 9–13): the evidentiary scale. Its purpose is to clearly distinguish between what is known with documentary precision, what can be reasonably inferred, and what we simply do not know. After the MoEP 2025 data, this task became even more important: the gap between official rhetoric and official data is now more visible than ever.

1. Why “collected” is not the same as “recycled”

Imagine the following scene. You carefully rinse a yogurt cup and place it in the orange container. The truck comes and picks it up. That’s it?

Not at all. That moment is only the beginning of a chain of at least six steps, and at each of them part of the material drops out of the system.

Step	What happens	What is lost or hidden
1. Collected	The container is emptied by the truck; the material is in the vehicle.	Often the only number in official reports. Says nothing about what happens next.
2. Delivered to sorting	The material arrives at the sorting facility.	Some collected material may go directly to landfill or RDF without sorting.
3. Sorted	The material passes through the line and is separated into fractions.	This is where the first “tails” appear – rejected, contaminated, or unsortable material.

Step	What happens	What is lost or hidden
4. Prepared for recycling	The fraction is baled, washed, shredded – ready for sale to a recycler.	More material is lost here: technically difficult fractions, commercially unattractive materials.
5. Actually recycled	The recycler accepts the material and turns it into new feedstock or a product.	This is the only step that closes the loop. Less reaches this point than is commonly assumed.
6. Used in manufacturing	Secondary feedstock enters a new product.	Export does not mean recycling: material may become waste again abroad.

In practice, this means that the figure "X tons collected" and the figure "X tons recycled" are fundamentally different things. But in 2026, the problem in Israel is deeper than confusing those two figures. The MoEP data show that even the first step – collection – is working critically poorly: only 6% of packaging even reaches the dedicated container.

This means that for most of the packaging stream in Israel, the discussion about downstream recycling is secondary. The material often does not even begin the chain correctly: it remains in the mixed stream and goes straight to landfill or RDF, bypassing sorting altogether.

2. The MoEP 2025 data: what they actually show

It is worth pausing on these data, because they are the factual foundation for everything that follows in the book.

National MoEP scan, August 2025: key figures

6% – the share of household packaging waste that reaches dedicated separate-collection containers (orange, blue, purple bins).

60% – the share of the contents of the ordinary mixed (green) bin that consists of recyclable materials: plastic, cardboard, and similar items. In other words, material that could technically have been recycled but physically ended up in the wrong stream.

Source: direct national scanning conducted by the regulator itself – the Ministry of Environmental Protection.

Practical meaning: the overwhelming majority of recyclable material in Israel never even reaches the first step of the recycling chain – the correct container.

Under the hood: Why this is stronger than earlier estimates

Before this scan, estimates of recycling levels in Israel were built on TAMIR reports about the "declared base" – that is, the amount of packaging declared by companies contracted with TAMIR. This produced figures such as "84% recycling" (for 2021), which sounded encouraging.

But the declared base is not the full packaging stream in the country. It is estimated to cover only about 50% of the real volume. That means even "84% of the declared base" is, at best, about 42% of the full stream.

The MoEP 2025 data remove the need for such recalculation: they are a direct measurement of what physically happens at the household level, not a calculation through reporting by system participants. The figure 6% is not "recycling from the declared base"; it is the share of material physically reaching the correct container.

A further independent confirmation came from inside the system itself: TAMIR CEO Alon Tal publicly confirmed in September 2025 that only about half of businesses comply with the law, which is a structural confirmation of the same failure from the producers' side rather than the household side.

3. The evidentiary scale: how to read this book

To speak honestly about what happens to materials after the sorting container, we need a common language. In this book, all statements about waste flows are marked on a four-level evidentiary scale.

This is not a quality label for the source – it is a label for how complete our picture is. Sometimes the picture is incomplete not because the researcher did a poor job, but because the data are not publicly available.

Level	**Meaning**	**Typical source**	**How to mark it in the text**
✓✓Confirmed	Official data or object-level documentation	MoEP scan, TAMIR report, Comptroller report, licensing registry	The source exists – give the citation and state the figures
✓Probable	Several independent sources point in the same direction, but the figures are not directly verified	Journalistic investigation + operator data + tender documents	Name all sources and indicate the range of estimates
~ Reconstruction	No direct data; the figure is calculated indirectly from adjacent indicators	Indirect estimate from tonnage, capacity, routes	Show the method and provide a confidence range
? Unknown	No public data available; a formal request will not quickly solve it	Sorting-facility fraction data, downstream fate of a specific plastic	State "unknown" and indicate what request might clarify it

Why does this matter? Because an activist who confuses "probable" with "confirmed" risks being rebutted by an official and losing the audience's trust. And an official who publicly claims "confirmed" where the truth is actually "reconstruction" creates the illusion of a system that does not exist.

After the MoEP 2025 data, this scale works differently than before: now we have a case where the ✓✓Confirmed level is reached not despite the system, but because the regulator itself published an uncomfortable truth about it.

4. Seven questions to ask about any stream

The evidentiary scale is useful only if we ask the right questions. Here is the minimum set for any stream – whether plastic from the orange bin, paper from the blue bin, or organic waste from the brown bin.

Activist's checklist

1. Does the material even reach the correct container – or, as the MoEP data show, does it remain in the mixed stream?

2. Where does the truck go? (sorting facility, landfill, RDF plant)

3. What happens at the sorting facility to this specific material?

4. What is the rejection rate – the percentage of material the sorting facility rejects as unsuitable?

5. Who is the downstream recycler? Does it have documented capacity and actual output products?

6. Is part of the material exported? If so, to which country and under what classification?

7. Is there a public report that confirms all of the above – or are we working with a reconstruction?

In Chapters 9–13 we will go through this list for each stream. Where there are answers, we will give them with a level of evidence. Where there are no answers, we will state "unknown" and indicate what kind of request could change that.

5. Where material goes: four destinies

After the sorting facility, material can take one of four paths. It is important to distinguish them, because official statistics often collapse them into one category of "not landfill." But in light of the MoEP 2025 data, it is worth remembering that for 94% of household packaging the question "what happens after sorting" does not arise at all – the material never reaches sorting, because it never reaches the correct container.

Four possible destinies of material after sorting

1. Real recycling (material recycling). The material becomes new feedstock or a product. This is the only path that closes the loop.

2. Export. The material is shipped abroad. That does not automatically mean recycling: some exported material is reclassified abroad as waste and ends up in local landfills.

3. RDF (refuse-derived fuel). The material is turned into fuel and burned – usually in cement kilns.

Four possible destinies of material after sorting

This is energy recovery, not material recycling.

4. Landfill as "tails." Even material that passed sorting may partly become contaminated or unsortable residue.

When a report says "60% of packaging is recycled," we do not know the ratio among these four paths. And now we know something additional: only a small share of the material even reaches this stage. That is why the figure alone says nothing about the quality of the system – and after the MoEP 2025 data, it says even less than it seemed to before.

6. Three traps of official statistics – now confirmed, not hypothetical

Before 2025, this book could have said: "official statistics may overstate real recycling rates because of several methodological traps." Now the wording must be different: these traps are documented, acknowledged by the regulator, and no longer a matter of debate.

Trap 1: counting from the declared base

TAMIR reports not from the full packaging stream in the country, but from the volume declared by system participants. According to the MoEP, more than 40% of packaging producers and importers have not signed contracts with TAMIR at all. TAMIR CEO Alon Tal confirmed this publicly in 2025: "Only about half of businesses comply with the law."

Trap 2: counting collection or export as recycling

In some public communications, the moment of recycling is treated as the handover to a recycler or the crossing of a border in the context of export. That does not match either the Basel Convention or Eurostat methodology: exported material may end up in a foreign landfill.

Trap 3: failing to account for material that never entered the system

The main trap confirmed by the MoEP 2025 data

The first two traps concern how material that has already entered the system is counted. But the 2025 scan revealed something more fundamental: 94% of household packaging does not reach dedicated containers at all.

That means any "X% recycled" statistic – even if mathematically correct for what was collected – describes only a tiny part of the total picture.

Analogy: if only 6 out of 100 letters reach the recipient, and among those 6, 84% are "delivered successfully," that is not evidence of a good postal system.

7. How to use the scale: practical guide

Below is the working algorithm. It repeats, in modified form, in each of the chapters on waste streams.

Algorithm for working with the evidentiary scale

Algorithm for working with the evidentiary scale

Step 1. Define the statement precisely.

Not “plastic is recycled,” but “HDPE bottles from the orange bin are sorted in Rishon LeZion and transferred to recycler X.”

Step 2. Find the primary source.

Official MoEP scan / TAMIR report / Comptroller report → ✓✓Confirmed

Several independent secondary sources → ✓Probable

Indirect calculation from adjacent data → ~ Reconstruction

No data → ? Unknown

Step 3. Indicate the level in the text. Do not hide it in a footnote. Write directly: “According to the MoEP 2025 scan (✓✓), only 6% of packaging…” or “Exact data on the downstream fate of this plastic are unavailable in the public domain (?)”.

Step 4. If the level is ?, indicate what request might change that. Such requests are listed in Appendix D, and the methodology for formulating them is explained in Chapter 25.

Chapter conclusions

- “Collected for recycling” and “recycled” are different events, separated by a chain of six steps. After 2025, it is known that for 94% of household packaging in Israel, the first step of that chain does not even begin.
- The MoEP national scan (August 2025) is the most important event in the evidentiary base on waste in Israel in the last decade: the regulator itself confirmed failure at the household level.
- The three traps of official statistics – declared base, counting collection/export as recycling, and excluding material that never entered the system – are no longer hypotheses. They are confirmed by regulator data and by public statements from the TAMIR CEO.
- The evidentiary scale (✓✓/ ✓/ ~ / ?) remains a working tool for an honest discussion of data – but now we have a rare case of ✓✓Confirmed for the most pessimistic scenario.
- This chapter is the foundation for understanding why a full sectoral reform is being discussed in 2026 (Chapter 16): the official recognition of a 6% failure rate is one of the main arguments for systemic change.

In the next chapter, we will apply this tool to the first two streams – paper and cardboard from the blue bin, and packaging glass and deposit materials from the deposit system. These are the "cleanest" streams in the system – a good place to see how the scale works under the best evidentiary conditions.

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Ynet / Alon Tal (CEO of TAMIR). "Producers and Importers – Now Is the Time for Shared Responsibility." September 2025. Direct quote from TAMIR's own CEO: "Only about half of businesses comply with the law." Confirms from inside the system what MoEP's scan confirmed from outside.

Chapter 9. Paper, Cardboard, and Deposit Packaging: The Cleanest Streams

> *"If this book has one example of what recycling should look like, it is paper and deposit packaging. Not because the system is perfect, but because here, for once, we actually have the data to check it."*

1. The Blue Bin: Where Paper and Cardboard Go

Israel's leading paper recycler is Infinya (formerly Hadera Paper, and before that also known as Amnir).[1] This is one of the rare cases in this book where a precise figure can be given with a high level of confidence: the company recycles over 400,000 tonnes of paper waste a year into paper rolls for the cardboard packaging used by Israel's packaging industry.[2]

> **Infinya: What We Know About the Company Directly**
>
> Structure: part of the Infinya Group (formerly Hadera Paper), controlled by the FIMI fund. Brings together three brands under one name: Infinya Recycling (collection and sorting), Infinya Paper (paper production), Infinya Packaging (packaging production).
>
> Network scale: more than 15,000 collection points nationwide – from office waste bins to containers for large industrial clients. Around 400 employees.
>
> Two production facilities: in the Modi'in-Maccabim-Re'ut industrial zone and in Hadera.
>
> Evidence level: √√Confirmed – the data is published by the company itself, which for this book is a rare case of full downstream-chain transparency.

> **◆Under the Hood: The Full Cycle – From Blue Bin to New Paper Roll**
>
> *Collection: compactor trucks visit collection points (office waste bins, shopping-mall containers, blue bins outside residential buildings) on a fixed schedule.*
>
> *On-site sorting: paper is sorted by type (white office paper, mixed cardboard, newspaper) and cleared of contaminants (staples, plastic sleeves, tape).*
>
> *Baling: the material is baled into bundles weighing around 700 kg.*
>
> *Recycling: the bales are transferred to the Hadera plant, where they are made into new paper and cardboard.*
>
> *Film: plastic that the company collects alongside paper (such as wrapping film) is passed to a separate subsidiary – Amnir Plastic – for separate recycling.*
>
> *This chain is a rare case in the book where every step is confirmed without needing reconstruction or assumptions.*

2. Why Paper Recycles Better Than Plastic: Three Reasons

The contrast between this chapter and Chapter 11 is not accidental. There are three structural reasons why paper is recycled more successfully than most plastics as a matter of course – not only in Israel, but globally.

Reason	Paper and Cardboard	Rigid Plastic
Material uniformity	Despite different grades, paper is technologically similar enough that a recycler can mix different sources.	HDPE, PP, PET are different polymers requiring separate processing and different equipment.
Market demand	Steady demand from the cardboard packaging industry – secondary paper stock is price-competitive.	Depends on oil prices: when oil is cheap, virgin plastic can be cheaper than recycled.
Identifiability	Paper is easy to tell visually from other materials at any stage.	NIR sorting is needed to separate polymers; black plastic is not technically identifiable.

This is consistent with European statistics: the EU's average paper and cardboard recycling rate has consistently exceeded 80%[3] – one of the highest rates of any material category.

◆A Physical Constraint Rarely Discussed

Paper fibre cannot be recycled indefinitely. With each recycling cycle, the fibres shorten and paper quality degrades. Typically, material can go through 5–7 recycling cycles before the fibre becomes too short to make new paper.

This means that even a perfectly functioning paper-recycling system requires a constant inflow of virgin fibre – either from forests, or through more efficient management of the cycle itself.

This is not a reason to deprioritise paper recycling – it is simply an honest reminder that "recycles well" does not mean "solves the problem completely and forever."

3. Deposit Packaging: The Same Principle, a Different Mechanism

Chapter 15 examines the deposit system and its structural conflict with TAMIR in detail. Here is a short technical summary, useful for understanding why the deposit stream belongs among the same "cleanest" streams as paper.

Green PET Recycling is Israel's leading recycler of deposit PET, with a plant in the Kidmat Galil industrial zone with a capacity of around 3,000 tonnes a month, covering roughly half of Israel's PET waste stream. Its output is certified for food contact (FDA, EFSA, FSSC 22000) – 75% goes to the Israeli market, 25% is exported.[4]

Why the Deposit Stream Is Just as Reliable as the Paper Stream

A financial incentive at the point of entry: the consumer gets money back for returning the container – this creates a clean, uniform stream with no need for complex sorting of mixed materials.

A direct line from consumer to recycler: the bottle travels through an RVM machine or a shop straight to the recycler, bypassing the mixed-bin stage and the contamination risk that comes with it.

Genuine closed-loop recycling (bottle-to-bottle): unlike most plastic streams, deposit PET can become a new food-grade bottle – the most complete form of recycling available for plastic.

4. Even Here – the Upstream Constraint Remains

It would be dishonest to end this chapter on an optimistic note without noting that even the book's most reliable streams are not entirely free of the upstream problem examined in Chapter 11. The MoEP's 2025 scan records an overall figure: only about 6% of household packaging waste reaches dedicated containers at all.[5]

This means that even if the downstream chain for paper and deposit packaging works almost perfectly – as this chapter shows – that only covers the share of paper and containers that physically made it to the blue bin or was returned through the deposit system. Paper thrown into the mixed bin meets the same fate as any other material in the general waste stream: landfill or RDF, not recycling.

The good news is that for paper and deposit packaging, the incentive to reach the right stream is stronger than for most plastic: a deposit is direct cash, and large cardboard is inconvenient to keep at home, which encourages people to get rid of it properly. This probably accounts, in part, for why these two streams show the best downstream results – but it does not remove the need to work on upstream accessibility (Chapter 4) for them too.

Chapter Takeaways

Key Takeaways

→ Paper/cardboard and deposit packaging are the two streams with the most reliable evidence base in the entire book, because the key operators (Infinya, Green PET Recycling) publish operational data directly.

→ Infinya recycles over 400,000 tonnes of paper a year through a network of more than 15,000 collection points – a rare case of ✓✓evidence level for a complete downstream chain.

→ Paper recycles more successfully than plastic for three structural reasons: material uniformity, stable market demand, ease of identification.

→ Deposit packaging achieves the same reliability through a different mechanism – a direct

> financial incentive at the point of entry that bypasses the mixed-collection stage.
>
> → Even these best-performing streams are not free of the upstream constraint (Chapters 4 and 11): whatever does not reach the right bin meets the same fate as the rest of the mixed stream.

Notes

1. Infinya Recycling Ltd. (formerly Amnir Recycling Industries). Operates two facilities (Modi'in-Maccabim-Re'ut and Hadera). Over 15,000 collection points nationwide. Approximately 400 employees. Wholly owned by Infinya Group (formerly Hadera Paper), controlled by FIMI Opportunity Funds. Source: hamichlol.org.il (Hebrew encyclopedia, citing company data); infinya.co.il. Evidence level: ✓✓Confirmed – operational details published by the company.

2. Infinya Group. Company website. "Converts over 400,000 tonnes of paper waste per year into paper rolls for cardboard, used by the packaging industry in Israel." infinya.co.il. This is the single most well-documented downstream recycling claim in this handbook, with specific tonnage published directly by the operating company.

3. Eurostat. Paper and Cardboard Recycling Statistics. Luxembourg: European Commission. EU average paper/cardboard recycling rate consistently above 80% across multiple reporting years – among the highest of any major material category, reflecting both strong commercial demand and decades of established infrastructure.

4. Deposit Law on Beverage Containers (Israel), 1999, as amended 2010, 2021. Operators: ELA (Agudat A.L.H.) and Asofta. Green PET Recycling Ltd.: plant at Kidmat Galil Industrial Zone, established 2019, processing capacity ~3,000 t/month (36,000 t/year), ~50% of Israeli PET waste, food-contact certified rPET (FDA, EFSA, FSSC 22000). 75% local industry, 25% exported. Evidence level: ✓✓Confirmed.

5. Ministry of Environmental Protection (Israel). National Waste Scan, August 2025. Key finding relevant to this chapter: even for the comparatively well-functioning paper and deposit streams, only ~6% of total household packaging waste reaches dedicated containers – paper/cardboard collection at the household level, while better documented downstream than plastic, is not exempt from the upstream collection gap.

Chapter 10. The Orange Bin: Anatomy of the TAMIR Stream

"Every time you put something in the orange bin, a journey begins. Where exactly it leads depends on what you put in, how clean it was, and what is happening at the other end of the chain – the end you have no access to."

Note: connection to the 2026 reform (→ Chapter 16)
This chapter describes the orange bin and TAMIR – a system that in 2026 faces a radical reform. The draft Economic Arrangements Law proposes abolishing the Packaging Law and liquidating TAMIR as the monopoly operator. This does not make the chapter obsolete: understanding how the current

system works – including its failures – is necessary for evaluating any new architecture. More in Chapter 16.

The orange bin is the most recognisable element of Israel's source-separation system. Most residents know it is for packaging. Many assume the packaging "gets recycled." But between those two points – closing the bin lid and a new product appearing – lies a chain that is largely invisible to the person who set it in motion.

This chapter opens that chain, to the extent that publicly available data permits. Where data is absent, we say so plainly.

1. What Belongs in the Orange Bin – and What Does Not

The orange bin accepts packaging waste. Not all waste – only packaging. This distinction is fundamental: it is precisely packaging that falls under the Packaging Law and is financed through TAMIR.

What goes in the orange bin – the simple rule:

YES – plastic packaging: bottles, flasks, containers, crates.
YES – metal packaging: tin and aluminium cans, lids.
YES – beverage cartons: Tetra Pak, Combibloc and equivalents.
YES – packaging film and bags (technically acceptable, though sorted less well).

NO – glass (→ purple bin).
NO – paper and cardboard (→ blue bin).
NO – food waste – it contaminates the entire batch.
NO – deposit bottles marked "חייב בפיקדון" (→ return to shop).
NO – clothing, electronics, hazardous waste.

Practical rule: if the packaging carries a material marking (1–7 in a triangle) or is clearly packaging – orange bin. If in doubt, the mixed bin is preferable to contaminating the batch.

An important nuance: beverage bottles marked "חייב בפיקדון" do not belong in the orange bin. They enter the deposit circuit (the *pikkadon* system) and must be returned to shops or via RVM machines. The difference lies in where the material ends up and how valuable it is downstream – this stream conflict is described in detail in Chapter 15.

2. From Bin to Facility: The First Steps

Dedicated collection – not a mixed truck

The orange bin is collected separately from mixed waste – this is fundamental.[1]

A dedicated truck goes only to orange bins. This means the material does not mix with food waste, construction debris, or other streams at the collection stage. That is precisely why the quality of what you put in matters: one contaminated load can degrade the quality of the entire truck.

Three sorting facilities

Almost all of Israel's orange bin contents go to one of TAMIR's three contracted sorting facilities: Rishon LeZion, Afula, and a smaller facility in Eilat.[2]

Facility	Location	Region served	Operator and status
Rishon LeZion (K.M.M / KMM)	Central District – Gan Raveh Regional Council	Gush Dan, centre of the country	KMM (ק.מ.מ מפעלי מחזור). Contract with TAMIR renewed for 5 years. New large facility under construction on Dan Cities Association land – expected to be the most advanced in the country. ✓Confirmed
Afula	Jezreel Valley, north	Northern region	Operator unconfirmed in open sources. ~ Reconstructed: serves northern region
Eilat	Southern Negev	South, Arava	Small volume. Operator: partial data from TAMIR tender documents. ~ Reconstructed

This means the entire country depends on three physical facilities. Any breakdown, overload, or quality problem at one facility affects an entire region.

◆Under the Hood: The New Flagship Facility – What Is Being Built in Rishon

In May 2024 TAMIR announced an agreement with KMM (ק.מ.מ מפעלי מחזור) to build a new large sorting facility on Dan Cities Association land in the central district. Investment: tens of millions of shekels. Technological level: comparable to European standards.

As part of the agreement KMM also invested in upgrading the current Rishon LeZion facility – to ensure continuity until the new one opens.

The new facility will include a public visitor centre – a step TAMIR positions as increasing transparency for citizens.

Evidence level: ✓✓Confirmed (public TAMIR announcement, May 2024).

Key question in the context of the 2026 reform: if TAMIR is liquidated as the monopoly operator – what happens to the investment in the new facility and to the contract with KMM?

3. What Happens at the Sorting Facility

The sorting process is a sequence of technological and manual operations. Their goal is to divide the mixed packaging stream into homogeneous fractions suitable for transfer to recyclers.

Eight sorting stages (per TAMIR data):

1. **Unloading.** The truck empties its contents into a pile. This is the first point where the actual composition of the stream is visible.

2. **Manual pre-sorting.** Workers remove large contaminants: food bags, non-sortable items, hazardous materials.

3. **Conveyor and primary separation.** Material passes onto conveyor belts. Bags and film are separated by air separators.

4. **NIR optical sorting.** Infrared sensors identify polymer type by spectral reflection. Air jets direct PET, HDPE, PP into separate hoppers. Problem: black and dark plastic absorbs infrared and is not detected.
5. **Magnetic separation.** Magnets extract steel packaging. Eddy currents – aluminium.
6. **Manual quality control.** Workers on the line remove incorrectly placed items. This is the last barrier before baling.
7. **Baling.** Each fraction is pressed into standard bales of 700–1,000 kg. Bale quality depends on fraction purity.
8. **Transfer to recyclers.** Bales are sold or transferred to recyclers – most of which, according to TAMIR, operate in Israel.

Per TAMIR data, four of the eight fractions are sorted technologically, without human involvement. The remainder require manual intervention. This means "contamination" of the stream (food residues, mixed materials) affects the speed and quality of sorting.

4. Fraction Map: What Goes Where

Here is a map of what actually happens to the different materials from the orange bin – with evidence levels.

Fraction	What is accepted	Sorting method	Where it goes (evidence level)
Rigid plastic (HDPE, PP)	Cleaning product bottles, shampoo bottles, non-deposit bottles	NIR optics, manual inspection	Local recyclers, partial export. ~ Reconstructed
PET plastic (non-deposit)	Trays, non-standard bottles	NIR optics	Partial export. Lower value than deposit PET. ? Partially unknown
Flexible plastic	Bags, film	Air separators	Predominantly RDF or landfill. ✓ Probable
Metal – aluminium	Aluminium cans and packaging (non-deposit)	Magnets, eddy currents	Recycled locally. ✓Confirmed (TAMIR data)
Metal – steel	Tin cans, lids	Magnets	Recycled locally. ✓Confirmed
Beverage cartons (Tetra Pak)	Juice and milk cartons	Manual sorting	Specialist recycler, partial export. ~ Reconstructed
Dirty / mixed residual	"Tails" after sorting	—	RDF or landfill. ✓Probable

Note the "Where it goes" column: most downstream data are either reconstructed (~ Reconstructed) or unknown (? Unknown). This is not accidental incompleteness – it is the systemic opacity described in Chapter 8.

5. The Problem of "Tails": What Does Not Become Secondary Raw Material

Even after passing through the sorting facility, part of the material does not become recyclable feedstock. These are the so-called "tails" (residual fraction).

Where "tails" come from:

Food contamination. A bottle with sauce residue, a mouldy container – these can spoil an entire batch. A single item does not kill the fraction, but systematic contamination lowers quality grade and bale value.

Black and dark plastic. The NIR sensor cannot see black plastic. It ends up in the "unidentified" stream – most often RDF or landfill. This is not a question of resident behaviour – it is a technical limitation.

Multilayer and laminated packaging. Sachets, stand-up pouches, metallised wrappers – NIR gives a mixed signal, so such material is not reliably sorted.

Small items (<3–5 cm). Lids, small containers fall through conveyor gaps or miss the air stream.

Materials with no market demand. Even technically sorted plastic of certain types (e.g. PS polystyrene) has low or zero market demand. No recycler wants it – it goes to RDF.

The share of "tails" in the TAMIR system is not published in the public domain. Evidence level: ? Unknown.

◆Under the Hood: The Tariff Signal on Stream Quality

TAMIR's 2023 tariff data (Infospot) shows a gap between the commercial and household streams. For example, processing PP plastic: 135 NIS/tonne (commercial) vs 642 NIS/tonne (household). A gap of almost five times.

This tariff signal says directly: material from households is significantly more expensive to process. The reasons: higher contamination levels, smaller volumes per collection point, more complex logistics.

This confirms the structural argument of Chapters 14–15: TAMIR systematically favours the commercial stream because it is economically more rational within the existing model.

6. Downstream: What We Know and What We Don't

TAMIR publicly states that "most recyclers are in Israel." This is difficult to verify: there is no systematic public data on which specific operator accepts which fraction, in what volume, and what it produces.

Per TAMIR's description (website, public materials), after sorting the bales go to recycling:[3]

- Plastic bags → film for agriculture and insulation.

- Cleaning product bottles and containers → plant pots and technical parts.
- Metal food cans → smelting, steel rods for construction.

This is a plausible description, consistent with what is known about HDPE, PP, and steel chains from other sources (see Chapter 11). But specific recyclers with confirmed capacity data and products are not named.

Note: connection to the 2026 reform (→ Chapter 16)
One of the key arguments for the 2026 reform: the creation of a national waste regulator would introduce a duty of annual public reporting on the downstream fate of all streams. If the reform passes in a version that includes this element, data that currently does not exist will become mandatory to publish. This would fundamentally change the possibilities for civic oversight.

7. What Gets in the Way of Sorting: Practical Advice

A number of sorting errors systematically reduce stream quality. This is not shaming – these are practical facts about what works and what does not.

Five rules that genuinely affect quality:

1. Rinse, but don't scrub.
Remove visible food residue – that is sufficient. Thorough washing with soap is unnecessary and wastes water.

2. Don't put things in closed bags.
The sorting line cannot detect what is inside a bag. Contents go in the orange bin; the bag goes there too, but empty and open – or into a separate film collection point.

3. Caps on bottles.
In several European countries the recommendation is to screw caps on before discarding – they don't get lost on the conveyor. TAMIR does not make this recommendation explicitly, but a cap on a bottle does not impede sorting.

4. Small items.
Items smaller than 3–5 cm will most likely not be sorted – they fall through the gaps. Small caps and tiny containers can be collected in a slightly larger container and discarded together.

5. Black plastic – not in the orange bin.
The NIR system cannot see it. It will end up in the tails. There is no good solution for black plastic in the current system – it is an architectural problem, not yours.

Chapter Summary

Key conclusions:

→ The orange bin is not a general waste receptacle: it accepts packaging and only packaging. Glass goes in the purple bin, paper in the blue bin, deposit packaging back to the shop.
→ Three sorting facilities serve the entire country: Rishon LeZion (centre/south), Afula (north), Eilat (far south). A new flagship facility with KMM is under construction in Rishon LeZion.
→ Eight fractions are sorted – four technologically. NIR optics, magnets, air separators + manual

inspection.
→ Downstream data are not published systematically. The evidence level for most fractions: ~ Reconstructed or ? Unknown.
→ "Tails" exist and go to RDF or landfill. Their share is not publicly reported.
→ The commercial stream is cheaper to process than the household stream – which is why TAMIR systematically favours it. This is a tariff signal, not an accident.
→ The 2026 reform would potentially change the entire architecture. If adopted – TAMIR as monopoly operator disappears, and mandatory downstream reporting may become part of the new law.

◆Orange Bin Checklist: Questions for the Activist

1. Which of the three sorting facilities does the orange bin from my neighbourhood go to?
2. Who is the operator of that facility? Does it publish a public report on rejection rates?
3. Where do the "tails" go – RDF or landfill? Which landfill?
4. Which specific recyclers accept plastic and metal from that facility?
5. Does TAMIR publish downstream data for my city or region?

The next chapter examines specific plastic streams – HDPE, PP and PET – with a more detailed analysis of downstream chains and known recyclers in Israel.

Sources for Chapter 10

Israeli sources

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Legal sources

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Footnotes

1. TAMIR FAQ (tmir.org.il/faq): "Packaging collected in orange bins is collected separately by a dedicated truck and transferred to one of three sorting facilities located in Rishon LeZion, Afula and Eilat." Eight material fractions are sorted; four are sorted technologically without human contact. After sorting, each material is pressed into bales and transferred to recycling plants, most of which are in Israel. ↩

2. Sources disagree on the exact facility count. TAMIR's own public FAQ (tmir.org.il/faq) states packaging is sent to "2 sorting stations located in Rishon LeZion and Afula." However, Infospot—a specialist industry tracker already cited elsewhere in this book—reports TAMIR currently operates three dedicated sorting facilities via contractors: Rishon LeZion, Afula, and a smaller facility in Eilat (infospot.co.il). Mako (December 2024) also lists three, matching Infospot. A fourth, much larger facility at Hiriya broke ground in June 2026 and is expected to roughly triple national capacity once operational (Calcalist, May 2024; Times of Israel). Given the industry-press consensus, this book follows the three-facility count, while flagging TAMIR's own page as an outlier that may simply be unupdated.

3. Mako. "Israel's Recycling Day: The New Lives of Packaging." December 2024. Describes the full journey of orange bin content: dedicated truck → sorting facility (Rishon, Afula or Eilat) → optical sorting by material type → baling → transfer to recycling plants "most of which are in Israel." Note: "most of which are in Israel" is a claim that requires downstream verification – no public data on specific recyclers, volumes and rejection rates is available. ↩

Chapter 11. Rigid Plastic (HDPE, PP) and PET: Different Fates for Similar Bottles

"A shampoo bottle and a water bottle look the same. Both plastic, both in the orange bin. But their subsequent fate is fundamentally different. And the gap between them begins not at the sorting facility, but at the filling point."

When people think about plastic in the orange bin, they usually picture something unified. Plastic is plastic. You put it in, it gets recycled.

In reality, the word "plastic" covers several fundamentally different polymers with different chemistry, different prices on the secondary market, different technical requirements for sorting and recycling – and very different chances of actually closing the loop.

Israel generates approximately 250,000 tonnes of household plastic waste per year – about 18% by weight and 30% by volume of all household waste.[1] Meanwhile, plastic recycling in the country stands at an estimated 6–7% of total volume – significantly below the OECD average of 39%.[2]

This chapter examines three streams that are often confused: rigid plastic HDPE (high-density polyethylene) and PP (polypropylene) from the orange bin, and PET – in two entirely different variants: deposit and non-deposit. For each – an evidence scale and an honest picture of what happens after the bin.

1. Plastic Map: What's in the Orange Bin

The orange bin accepts packaging. Among the packaging – several types of plastic with fundamentally different characteristics. Here is a brief map.

Type	Examples of packaging	Sortability from orange bin	Downstream in Israel
HDPE (high-density polyethylene)	Shampoo, conditioner, dish soap, bleach, liquid soap bottles; jerrycans	✓Good – rigid, large, easily identified by NIR	Mechanical recycling locally; products: pipes, pots, technical parts. ~ Reconstructed
PP (polypropylene)	Bottle caps, some food containers, crates, buckets	✓Good for large items; smaller parts – less so	Mechanical recycling partially; export. ? Partially unknown
PET (non-deposit)	Trays, blisters, some non-deposit bottles	~ Medium – depends on cleanliness and colour	More complex than deposit PET; not always suitable for food contact. ~ Reconstructed
LDPE (low-density polyethylene)	Bags, film, soft packaging	✗ Poor – clogs sorting lines, adheres to equipment	Mostly RDF or landfill. ✓ Probable

Type	Examples of packaging	Sortability from orange bin	Downstream in Israel
PS (polystyrene)	Disposable cups, trays, packaging "peanuts"	✗ Very poor – lightweight, flies off conveyor	Almost never recycled locally. ✓ Probable
Multilayer packaging	Sachets, stand-up pouches, some Tetra equivalents	✗ Impossible with standard NIR	RDF or landfill. ✓Confirmed (no market demand)

This table is not a verdict but a navigator. Below we examine HDPE, PP and PET in detail.

2. HDPE and PP: Rigid Plastic That Is Actually Recycled

What they are

HDPE is high-density polyethylene. You can feel it by hand: hard, slightly matte, does not flex. Typical examples in an Israeli home: shampoo, hair conditioner, dish soap, bleach, liquid soap bottles. Most 5-litre jerrycans of cleaning products are also HDPE.

PP is polypropylene. Slightly more flexible, often marked "5" in a triangle. The caps on most bottles – PP. Fruit crates at the market – PP. Some food containers – also PP.

Both polymers are thermoplastics: they melt when heated and solidify when cooled. This makes them mechanically recyclable – in theory.

The recycling chain

Here is the path from orange bin to secondary granulate – assuming everything goes correctly.

Stage	What happens	Where material is lost
1. Sorting	NIR scanner identifies polymer by reflectance spectrum. Air jet directs item into the appropriate hopper. Magnets remove metal.	Contaminated items, dark plastic (absorbs NIR), multilayer – go to "tails" or mixed stream.
2. Manual sorting	Workers on the conveyor remove incorrectly sorted items and contaminants.	Line capacity is limited; at high throughput some rejects pass through.
3. Baling	Sorted HDPE or PP is pressed into standard bales (~700–1,000 kg) for transport.	Quality control at this stage is critical: mixed bales reduce recycling value.
4. Shredding	Bales are shredded into flakes of 10–20 mm.	Losses minimal, but dust and fine particles drop out.

Stage	What happens	Where material is lost
5. Washing	Hot water with detergent removes food residues, labels, adhesive.	Dirty water becomes industrial effluent. Heavy contamination rejects the entire batch.
6. Drying and granulation	Flakes are melted in an extruder and pushed through a die into granules – the finished secondary raw material (REPRO).	Polymer degrades slightly with each heat cycle. Colour contamination lowers granulate grade.
7. Sale / application	Granulate is sold to manufacturers. For non-food applications (pipes, pots, crates) – suitable; for food contact – not without additional purification.	Granulate price competes with virgin plastic. When oil is cheap, recycling becomes less profitable.

◆Under the Hood: NIR Sorting Technology – How It Works

NIR (near-infrared) is the standard for sorting at modern MRFs (material recovery facilities). A sensor above the conveyor illuminates the item with infrared light and analyses the reflectance spectrum. Each polymer has a unique spectral "fingerprint" – HDPE, PP, PET, LDPE, PS are identified reliably.

An air jet fires on the sensor's signal and directs the item into the correct hopper.

Problems: (1) Dark and especially black plastic absorbs infrared and gives no reflectance – the sensor "cannot see" it. A solution exists (hyperspectral or RGB cameras), but is not universally installed. (2) Dirty or wet plastic: food residues alter the spectrum. (3) Very small items (<3 cm) fall through gaps. (4) Multilayer packaging gives a mixed signal.

Israeli context: TAMIR's sorting facilities in Rishon LeZion, Afula and Eilat are equipped with NIR lines, however detailed data on their technical specifications and rejection rates is not published in the public domain. Evidence level for sorting quality: ? Unknown.

Who Recycles in Israel: What Is Known

This is the most difficult section from an evidence standpoint. Several recyclers of rigid plastic exist in Israel, but public data on their capacities, actual throughput and downstream products is extremely limited.

Known and probable HDPE/PP recyclers in Israel (data as of 2025)

✓Probable – K.B. Recycling Industries Ltd. (Beit Shean). A family-owned company (the Crystal family), operating for over 20 years. Has a truck fleet, two sorting facilities, and a plastic recycling plant. Produces REPRO from LDPE, HDPE and PP. Trades in plastic, metal, cardboard and textiles. Capacity data and downstream products not published in open sources.

✓Probable – KMM (ק.מ.מ מפעלי מחזור). One of the oldest players in Israeli recycling. Collects plastic and directs it to shredding/processing. Per the company's own data, capacity is 100,000 tonnes of plastic per year; plastic without local demand is exported. Downstream end-products not described in open sources.

? Unknown – SuperPipe Israel. The name appeared in several references, but documentary confirmation was not found during research for this chapter. It may refer to a manufacturer of pipes from secondary HDPE, but no public data exists. Readers with direct knowledge of this company are invited to send clarifications.

✓Probable – Keter Group. Israel's largest manufacturer of home and garden products. Uses over 40% secondary plastic in its products. Runs pilots for collecting used plastic garden furniture and household items from residents (in partnership with municipalities). This is a post-consumer stream, not from the orange bin – but an important secondary market actor.

~ Reconstructed – Export. According to OECD (2023), Israel imports 80% of primary resins and exports most of its collected plastic. This means that a significant portion of HDPE/PP sorted from the orange bin goes abroad – to Turkey, EU countries, Asia. Data on export destinations and downstream fate is not published systematically.

◆Under the Hood: Why Data Is So Scarce – the Structural Reason

Israel has historically been a net exporter of collected plastic waste rather than building local recycling chains. The OECD 2023 report states explicitly: the country imports 80% of primary resins and exports most of its collected plastic waste. This means there are fewer incentives for public disclosure of downstream data: the main transactions occur on international markets, not within Israeli public reporting.

A requirement to publish downstream data (where sorted material goes, what product results) does not exist in Israeli legislation. TAMIR publishes aggregate recycling figures against the declared base, but not data by specific fraction and recycler.

This is a systemic gap, not an accidental omission. And that is precisely why an activist who asks "where does HDPE from the orange bin in Haifa go?" gets no answer – the answer simply does not exist in the public domain.

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3. Two Types of PET: Why Deposit and Non-Deposit Are Different Worlds

PET (polyethylene terephthalate) is the most familiar plastic to recyclers. Transparent bottles for water, carbonated drinks, juice – nearly all PET. Marking: "1" in a triangle.

But in Israel PET exists in two parallel streams that are technically identical but economically and systemically entirely different. Confusing them means failing to understand how the system works.

Deposit PET vs non-deposit PET – the fundamental difference:

DEPOSIT PET (via the pikkadon / פיקדון system)

→ These are beverage bottles up to 5 litres, covered by the deposit law.

→ The consumer pays a deposit at purchase and recovers it by returning the bottle to a shop or RVM (reverse vending machine).

→ The financial incentive creates a clean, homogeneous, uncontaminated stream.

→ Operators: ELA (אגודת א.ל.ה.) – historically the main one, and Asofta – which appeared later. Both operate through shop networks and RVM machines.

→ Main recycler in Israel: Green PET Recycling (plant at Kidmat Galil, Galilee). Capacity ~3,000 t/month, covers ~50% of Israeli PET waste. Produces rPET certified for food contact (FDA + EFSA).

→ This is called bottle-to-bottle recycling – the most complete loop closure available for plastic.

NON-DEPOSIT PET (from the orange bin)

→ These are trays, blisters, some non-standard bottles, non-beverage packaging.

→ Enters the orange bin alongside HDPE, PP and other polymers.

→ Stream quality is lower: mixed, often contaminated, various colours.

→ Not suitable for bottle-to-bottle without additional purification.

→ Part is sorted and exported, part goes to RDF or landfill.

→ Downstream data: ? Unknown in any systematic form.

This distinction is the main reason why the expansion of the deposit system (extended to larger bottles in 2021) did not solve the plastic problem in the orange bin but changed its composition: the most valuable PET moved into the deposit circuit, while the less valuable and less clean PET remained in the TAMIR system.

◆Under the Hood: Green PET Recycling – The First Real Case of Bottle-to-Bottle in Israel

Green PET Recycling was founded in 2019 with the participation of Yafora-Tavori (one of Israel's largest beverage producers) and CEO Zohar Levi.

The plant at Kidmat Galil uses a line from Austria's Starlinger – a leader in PET recycling. SSP (Solid-State Polymerization) technology produces rPET approved for food contact.

This is an example of vertical integration: a beverage producer participates in recycling bottles from its own drinks. This model reduces dependence on virgin PET and ensures stable demand for secondary raw material.

Evidence level: ✓✓Confirmed – the company publishes detailed operational data. 75% of output goes to the Israeli market, 25% is exported.

4. The Problem Starts Earlier: Upstream Matters More Than It Seems

Until this point, this chapter has focused on the downstream question: what happens to plastic AFTER it has entered the orange bin – who sorts it, who recycles it, where unsuitable material goes. This is an important question. But data from the Ministry of Environmental Protection's national scan, published in August 2025, shows that for most rigid plastic in Israel, the problem arises significantly earlier – at a stage that precedes sorting altogether.

According to this scan, only around 6% of household packaging waste in Israel reaches dedicated source-separation containers – that is, the very orange bin whose fate is analysed in this chapter. The remaining 94% ends up in the mixed stream, where around 60% of the contents are materials that could technically be recycled, including a significant share of HDPE and PP.

This reframes the entire chapter. The question "where does an HDPE bottle go after the sorting facility" is important, but it applies only to a small fraction of all the rigid plastic actually produced and discarded by Israeli households. For the vast majority of shampoo bottles, cleaning product flasks and food containers, the relevant question is different: why did they never reach the orange bin in the first place?

◆Upstream vs downstream: two different plastic problems

Downstream problem (where this section ends and the rest of the chapter concentrates): what happens to plastic AFTER sorting – who recycles it, what is the rejection rate, where does the export go. This problem is about the efficiency of the sorting and recycling system.

Upstream problem (what the MoEP 2025 scan reveals): why 94% of packaging never reaches the source-separation system at all. This problem is about behaviour at the first-mile level (Chapter 4), the density and accessibility of containers, and systemic EPR failures (Chapters 14–15).

Key conclusion: even if the entire downstream chain worked perfectly – no sorting losses, no rejection, 100% recycling of all sorted material – it would resolve the fate of only 6% of the stream. The upstream problem is quantitatively far larger than the downstream problem.

◆Under the Hood: Why This Changes Priorities for the Activist

Chapter 8 introduces the evidence scale and shows: MoEP 2025 data is not a reconstruction but an official direct measurement (✓✓Confirmed). This means the upstream problem is the best-documented part of the entire plastic picture in Israel, in contrast to downstream data where the levels "~ Reconstructed" and "? Unknown" predominate (see the recyclers table above).

Practical conclusion: efforts aimed at improving the upstream indicator (more containers, more convenient placement, clearer marking – Chapter 4) are likely to have greater impact than efforts focused exclusively on improving downstream sorting of specific polymers.

This does not mean downstream questions (where is the recycler, what is the rejection rate) are unimportant – they remain necessary for the full picture and are the subject of legitimate civic requests (Chapter 25). But the priority by volume of impact is clearly shifted towards the beginning of the chain.

5. What Turns Plastic into "Tails"

Even if plastic has correctly entered the orange bin and passed through sorting, it may not reach recycling. There are several points at which material drops out of the cycle.

Five reasons why rigid plastic becomes "tails":

1. Food contamination. A bottle with yogurt or oil residue spoils the batch: only clean plastic goes to recycling. Rule: rinsing matters. Thorough scrubbing is unnecessary; removing visible residue is sufficient.

2. Black and dark plastic. The NIR sensor "cannot see" black plastic. It goes to the mixed stream or to RDF. This is a systemic problem, not a question of resident behaviour.

3. Small items. Caps and small containers under 3–5 cm fall through conveyor gaps or miss the air stream. A practice in several countries: place caps inside bottles and close them before discarding.

4. Multilayer packaging. A container where plastic is bonded to metal or another polymer cannot be unambiguously sorted by NIR. Result – mixed stream or RDF.

5. Absence of market demand. Even sorted and clean PP or HDPE of certain colours and types may find no market – especially when virgin plastic prices fall. In such cases the recycler accepts it at a lower price, reducing chain profitability, or refuses to take it at all.

Per TAMIR data, the overall recycling rate for plastic packaging from the declared base is approximately 30%.[3] What lies behind that figure – which specific plastic, how much of it actually closed the loop, how much went to export or RDF – is not described in the public domain. Evidence level for downstream: ? Unknown.

6. Packaging Design as a Condition for Recycling

One of the main conclusions of this chapter is: recycling begins before the product reaches the bin. Packaging design determines its fate.

UNIDO conducted a pilot project in Israel that illustrates this well.[4] A standard yogurt or salad cup in Israel is a PP cup with a PET sleeve. Because of this combination, the NIR scanner sees a mixed signal. Simply replacing the PET sleeve with a polyolefin sleeve makes the packaging detectable. A marginal increase in production cost, a proportional reduction in environmental cost.

The European PPWR regulation of 2023 introduced "recyclability by design" as a market-entry condition for the first time: by 2030 all packaging must be recyclable. This means producers working with the EU are already redesigning their packaging. In Israel no such requirement exists yet – and this is precisely what creates a situation in which packaging design systematically works against recycling.

◆Under the Hood: What Makes Packaging Recyclable – Technical Criteria

Mono-material: one polymer without mixing. HDPE + HDPE – good. HDPE + PET sleeve – bad for sorting.

NIR detectability: transparent, white and light-coloured polymers – detected well. Black, metallised, opaque dark – poorly or not at all.

Removable labels: glued paper labels detach during washing. Heat-shrink sleeves (shrink sleeves) from a different polymer contaminate the batch.

Size: an item smaller than 3–5 cm will most likely not be sorted.

Content cleanliness: oily, sticky substances – the main cause of HDPE batch contamination.

Chapter Summary

Key conclusions:

→ Plastic in the orange bin covers several fundamentally different polymers. HDPE and PP from the orange bin are genuinely recyclable – provided they are clean, the right size, and market demand exists.
→ Deposit PET and non-deposit PET are different systems. Deposit goes through a clean circuit with high circular potential (bottle-to-bottle at Green PET). Non-deposit from the orange bin – a significantly more complex fate.
→ Several HDPE/PP recyclers exist in Israel, but documented data on capacities and downstream products in the public domain is extremely limited. Evidence level for most downstream claims: ? Unknown.
→ Israel has historically exported most of its collected plastic rather than building local chains. This is a structural deficit, not a temporary failure.
→ Black plastic, multilayer packaging and small items systematically drop out of sorting – regardless of resident behaviour. This is a design problem, not a motivation problem.
→ The key question a citizen can put to TAMIR or the municipality: which specific plastic from the orange bin reaches a real recycler, in what volume, and what product comes out at the other end?

◆Activist Checklist for Plastic

1. What percentage of HDPE/PP collected from the orange bin in your city actually reaches a recycler (rather than going to export or RDF)?
2. Who specifically is the downstream recycler for HDPE/PP collected by TAMIR in your region?
3. What is the rejection rate at the sorting facility for rigid plastic?
4. How has the composition of the plastic stream in the orange bin changed since the deposit system was extended to larger bottles in 2021?
5. Does the sorting tender include a requirement for a minimum share of material actually recycled in Israel (rather than exported)?

The next chapter examines what the system handles considerably worse: flexible, multilayer and small packaging – the part of the stream that in almost all circumstances has likely become RDF or landfill tails.

Sources for Chapter 11

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A Note on the Evidence Base

This chapter contains more "? Unknown" markers than most other chapters in the handbook. This is not a research failure – it is an honest reflection of the actual state of public information about the downstream fate of plastic in Israel. The data that would allow these question marks to be replaced by confirmed facts must be requested from TAMIR, MoEP and sorting operators. Request templates are in Appendix D.

Footnotes

1. UNIDO / SwitchMed. "Increasing the circularity of plastic value chains in Israel." MED TEST III Project, 2022. Available at: circemed.org. Key finding: Israel generates ~250,000 t/year of plastic waste from households; only 6% recycled in 2019. Israel is the second largest per-capita consumer of single-use plastics globally. ↩
2. CDS-Luthi / IP Group. "Recycling Plastic With Plastic." Analysis of Israeli plastic recycling market, 2024. Data: Israel recycles only 7% of plastic waste vs 39% OECD average. Plastic waste = 18% of household waste by weight. Available at: cds-luthi.com ↩
3. TAMIR (תאגיד מחזור יצרנים). Annual Report 2022. Plastic packaging recycling rate from declared base: ~30% for plastics overall. Downstream data by facility and fraction not published systematically. Evidence level for downstream data: ? Unknown. ↩
4. UNIDO / SwitchMed, 2022 (see note 1). Pilot project on yogurt/salad packaging redesign: replacing PP cup + PET sleeve with mono-material polyolefin sleeve makes packaging NIR-detectable. Estimated impact: 2,100 t/year of additional recyclable plastic if adopted broadly by Israeli brands. ↩

Chapter 12. Flexible, Multilayer and Small Packaging: The Problem Stream

"A crisp packet is not plastic. Technically it is five or six layers of different materials bonded together so that they can never be separated. The design that protects the product guarantees the impossibility of recycling."

Chapter 9 showed the best-case scenario: paper and deposit packaging, where the evidence base is strong and the downstream chain is traceable. This chapter is the opposite pole. Flexible, multilayer and small packaging is a material that systematically fails on every criterion examined in the previous chapter: it is non-homogeneous, has no stable market demand, and is physically difficult to identify.

1. What This Category Is

This category covers several different types of packaging that share one characteristic – low or zero recyclability within the existing infrastructure.

Sub-categories of problem packaging:

Flexible film (LDPE) – bags, wrapping film, frozen food bags.

Multilayer packaging – crisp packets, coffee bags, sachets for spices and cosmetics. Technically composed of several layers of different materials (plastic + aluminium foil + another layer of plastic), bonded together to protect against moisture, light and oxygen.

Stand-up pouches (doypack) – packaging for dried fruit, nuts, pet food.

Small packaging – caps, small bottles, single-serving packaging – physically too small for standard sorting.

2. Why This Packaging Systematically Fails

UNIDO, in the framework of its project on plastic circularity in Israel, explicitly identifies flexible packaging as the most technically difficult fraction for processing on standard sorting lines.[1] The reason is the combination of low density, tendency to wrap around equipment, and frequent multi-polymer composition.

Barrier	What it means	Consequence
Low density	Lightweight material is poorly separated by weight and shape on standard equipment	High probability of sorting error
Wrapping around equipment	Film winds around moving parts of conveyors and sorting machines	Equipment downtime; requires specialised air separators
Multilayer structure	Several different polymers bonded into a single object	NIR scanner cannot unambiguously identify the material

Per TAMIR data, only PE-type (polyethylene) "plastic bags" are separately extracted during orange bin sorting – meaning that multilayer packaging and PP-based bags do not technically form a separate marketable stream and by default go into the residual.[2]

3. Where It Actually Goes

Honest picture: downstream for flexible packaging

Clean PE film, separated at sorting: the only sub-category with a real chance of mechanical recycling – becomes granulate for technical non-food applications (bin bags, technical films).

Multilayer and contaminated packaging: based on a combination of sources (Plastics Recyclers Europe, the general TAMIR sorting structure) – goes to RDF (fuel for cement kilns) or landfill. Evidence level for Israel specifically: ✓Probable (no direct confirmation of volumes, but the sorting structure makes any other outcome unlikely).[3]

Small packaging (<3–5 cm): physically falls through sorting equipment regardless of material composition – this applies even to theoretically recyclable small items.

4. Design as the Only Real Solution

Unlike rigid plastic (Chapter 11), where one can talk about improving sorting and downstream capacity, the technological solution on the recycling side for flexible and multilayer packaging is extremely limited. The main lever is changing packaging design before it enters the waste stream.

◆Under the Hood: What Makes Flexible Packaging Recyclable – International Experience

Mono-material redesign: replacing the multilayer structure with packaging made of a single polymer (for example, fully PE instead of PE + aluminium + PP) makes the material technically sortable by standard NIR equipment.

The European PPWR regulation explicitly sets recyclability-by-design requirements for 2030, targeting precisely this packaging category – recognition that downstream investment alone will not solve this problem.[4]

The Ellen MacArthur Foundation identifies flexible and multilayer packaging as one of the key "non-recyclable by design" categories globally – a structural problem, not a specifically Israeli one.[5]

In Israel no such mandatory packaging design requirements yet exist – which makes this category systemically unsolvable at the downstream infrastructure level until regulatory pressure on upstream design appears.

Chapter Summary

Key conclusions:

→ Flexible, multilayer and small packaging systematically fails on density, identifiability and market demand.
→ TAMIR separates only clean PE film into a distinct stream; multilayer and PP-based film do not by default form a marketable stream.[2]
→ The downstream fate for most of this category is RDF or landfill – not material recycling.
→ Unlike rigid plastic, for this category the technological solution on the sorting side is extremely limited – the main lever lies in packaging design, not in recycling.[5]
→ Israel has no regulatory requirements for recyclability-by-design equivalent to the European PPWR – this creates a structural gap precisely for the most difficult packaging category.[4]

The next chapter concludes the review of material streams – glass, metal and beverage cartons – where, as with paper, the technical possibilities of recycling meet economic and logistical barriers.

Chapter 13. Glass, Metal and Beverage Cartons

"Glass can be recycled infinitely, without loss of quality. And that is precisely why its recycling rate in Israel – 4–8% – is the most telling example of the fact that technical recyclability and actual recycling are different things."

This short chapter concludes the review of specific material streams begun in Chapter 9. Three materials – glass, metal, beverage cartons – are grouped here not because they are technologically similar, but because each illustrates its own distinct lesson about what determines recycling success.

1. Glass: The Best Technical Case, the Worst Practical Result

Glass is physically dangerous for TAMIR's standard sorting line: it breaks during transport, shards injure sorting staff and damage equipment calibrated for plastic and metal. For this reason glass never goes in the orange bin – it has separate specialised collection points.[6]

This creates the same first-mile structural problem examined in Chapter 4: glass collection points are physically less accessible than orange bins, which directly reduces the likelihood of correct sorting.[7]

◆Two kinds of glass: deposit and non-deposit

Deposit glass (beverage bottles with a deposit): volume approximately 120,000 tonnes per year. High recycling rate – because the deposit financial incentive (Chapter 15) creates the same effect as for PET: a clean, motivated stream.

Non-deposit glass (jars, sauces, jam – going to the purple bin or specialist points): the volume declared to TAMIR is approximately 30,000 tonnes per year, significantly less than the actual glass packaging flow in the country. Recycling rate: 4–8%.[7]

This is the lowest recycling rate of all materials examined in this book – despite glass being technically the most easily recyclable material of all.[8]

◆Under the Hood: Why Technical Recyclability Doesn't Help

Glass, unlike paper or plastic, can be recycled indefinitely without quality degradation – provided it is correctly sorted by colour. This is not a theoretical advantage but a practically realisable feature of glass recycling technology, confirmed by European data.[8]

The problem is not technical but economic and logistical: glass is heavy (high transport cost per tonne), fragile (requires careful handling), and loses value when colours are mixed.

Without a financial incentive equivalent to a deposit, and without convenient collection infrastructure (like the blue or orange bin), even a technically ideal material can show worse real-world results in the system than far less recyclable materials.

This is direct practical confirmation of the Chapter 4 thesis: infrastructure accessibility predicts behaviour more reliably than material technical recyclability.

2. Metal: The System's Quiet Success

Unlike glass and most plastic, metal is one of the few streams where the economics work in favour of recycling almost automatically, without the need for EPR subsidies or deposit schemes.

Why metal is a simple case:

High secondary market value: aluminium and steel retain commercial value regardless of the state of the EPR system – this makes metal collection profitable even outside formal channels.

Easy identification: magnets separate ferrous metal; eddy currents separate aluminium – without the need for complex optical sorting.

Infinite recyclability: like glass, metal does not lose quality through recycling.

Parallel informal market: in Israel, as in many countries, a well-developed informal scrap metal market exists alongside the official system – TAMIR publishes tariffs for metal from both the commercial and household streams.

TAMIR authorises several specialised metal recycling operators, including companies working exclusively with this stream.[9] This is one of the rare cases where market forces and the EPR system act in the same direction rather than working against each other.

3. Beverage Cartons: A Technically Complex Hybrid

Beverage cartons (Tetra Pak-type packaging for milk and juice) are included in the orange bin as one of seven sorted fractions.[10] But this creates a misleading impression of simplicity: technically this is a composite material – layers of paperboard, polyethylene and aluminium foil, bonded together to protect the contents.

Recycling such material requires a specialised technology (hydrapulping) that separates the paperboard fibre – which can then go into the standard paper recycling stream – from the polyethylene-aluminium layer, which requires separate processing or energy recovery.

◆An Honest Acknowledgement of a Data Gap

Despite beverage cartons being listed among TAMIR's sorted fractions, no systematic public data on which specific operator in Israel processes this material and by what method was found in open sources during research for this chapter.[11]

European data (ACE, Alliance for Beverage Cartons and the Environment) shows that specialised technology exists and is applied in advanced recycling systems – but this does not confirm that equivalent infrastructure operates in Israel.[11]

Evidence level: ? Unknown. This is a specific example of a question worth directing to TAMIR via the tools described in Chapter 25: who recycles beverage cartons collected from orange bins, and what happens to the polyethylene-aluminium layer after the paperboard is separated?

Chapter Summary

Key conclusions:

→ Glass is technically the most recyclable material but shows the worst practical result (4–8% for the non-deposit stream) – due to the absence of a financial incentive and inconvenient collection

infrastructure.[78]
→ Metal is a rare case where market economics and the EPR system act in alignment, without the need for artificial incentives.[9]
→ Beverage cartons are a technically complex composite material for which systematic data on downstream fate in Israel is absent from the public domain.[11]
→ The comparison of three materials confirms the central thesis of this part of the book: recycling success is determined not only by the technical properties of the material, but by the combination of economic incentives, infrastructure convenience, and data transparency.

This concludes the review of specific material streams (Chapters 9–13). Part IV of the book moves to the policy that sets the rules of the game for all these streams simultaneously – beginning with how the EPR mechanism through TAMIR was designed and why it failed.

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Footnotes

1. UNIDO / SwitchMed. "Increasing the circularity of plastic value chains in Israel." MED TEST III Project, 2022. Documents flexible plastic packaging (LDPE film, multilayer pouches, sachets) as the most technically difficult fraction for material recovery facilities to process due to low density, tendency to wrap around machinery, and frequent multi-polymer composition. ↩

2. TAMIR FAQ. Confirms that orange bin sorting separates "plastic bags" (PE type only) as one of the seven material output streams – implying that other flexible plastics (multilayer, PP-based film) are not separated into a marketable fraction and default to the residual/rejected stream. ↩ ↩2

3. Plastics Recyclers Europe. "Flexible Packaging Recycling: Technical and Economic Barriers." Documents that flexible film recycling requires specialised equipment (different from rigid plastic MRF lines) and that contamination tolerance is far lower than for rigid plastics – a small percentage of food residue or multi-material content can render an entire batch non-recyclable. ↩

4. European Commission. Packaging and Packaging Waste Regulation (PPWR), COM(2022) 677 final. Art. 7 sets recyclability-by-design requirements specifically targeting flexible and multilayer packaging by 2030 – recognising that downstream sorting investment alone cannot resolve this material category's structural unrecyclability. ↩ ↩2

5. Ellen MacArthur Foundation. "The New Plastics Economy." 2016. Identifies flexible and multilayer plastic packaging as one of the primary "non-recyclable by design" categories globally, driving the case for redesign-led solutions (mono-material alternatives) over end-of-pipe sorting investment. ↩ ↩2

6. TAMIR FAQ / Wikipedia (Hebrew). "Recycling in Israel." Glass packaging is explicitly excluded from the orange bin: it breaks during transport and sorting, injures sorting staff, and damages equipment calibrated for plastic and metal. Separate collection points are required by design, not by administrative choice. ↩

7. Infospot. "A Decade of the Packaging Law." November 2023. Documents the structural imbalance between deposit glass volume (~120,000 t/year, high recycling rate due to deposit incentive) and non-deposit glass declared to TAMIR (~30,000 t/year, recycling rate estimated at 4–8%) – among the most clearly documented material-level failures in the Israeli system. ↩ ↩2 ↩3

8. Glass Packaging Institute / European Container Glass Federation (FEVE) data: glass is infinitely recyclable without quality loss when properly sorted by colour, unlike paper (fibre degradation) or most plastics (polymer chain degradation). This makes Israel's 4–8% non-deposit glass recycling rate a particularly stark illustration of economic and logistical barriers overriding technical recyclability. ↩ ↩2 ↩3

9. Recycom (ריסייקום). Metal recycling company directory listing via TAMIR authorised contractors. Specialises in metal waste collection, sorting, dismantling and processing for industrial clients across Israel. tmir.org.il/Contractors ↩ ↩2

10. TAMIR FAQ. Beverage cartons (Tetra Pak-type packaging for milk and juice) are included in the orange bin stream as one of the seven sorted material categories, despite being a composite material (paperboard, polyethylene, aluminium foil layers) requiring specialised recycling technology distinct from standard paper recycling. ↩

11. ACE (Alliance for Beverage Cartons and the Environment) European data on beverage carton recycling: specialised "hydrapulping" technology separates paperboard fibre (recyclable in standard paper streams) from the polyethylene/aluminium layer (requires separate processing or energy recovery). No Israel-specific data on beverage carton downstream processing was found in open sources during research for this chapter. Evidence level: ? Unknown. ↩ ↩2 ↩3

Chapter 14. EPR in Practice: How TAMIR Was Designed and Why It Failed

"The architecture of a system determines its outcomes. If a system is designed so that its success is measured by what it reports about itself – that is not a recycling system. That is a reporting-about-recycling system."

Note on this chapter

This chapter was written in 2026 – at the moment when the system described here is proposed for complete abolition. The draft Economic Arrangements Law (חוק ההסדרים) for 2026 proposes cancelling the Packaging Law and liquidating TAMIR as the sole recognised operator. Chapter 16 describes this reform in detail. Here we ask: why did a system that functioned for 15 years produce results that make its abolition politically possible?

In 2011 Israel adopted the Packaging Law – one of the country's first pieces of legislation on extended producer responsibility (EPR). The idea was simple and sensible: those who produce and import packaging should pay for its recycling. Not the state, not municipalities, not taxpayers – producers.

Behind this stood the "polluter pays" principle, enshrined in international environmental law and working successfully in dozens of countries.[1]

Fifteen years passed. The result: approximately 50% of packaging entering Israeli shops remains entirely outside the system. MoEP's own 2025 data shows that only 6% of packaging waste reaches dedicated containers. And yet the system reports successes – because it counts only what has entered it.

This chapter explains how this became possible. Not through malice – but through architectural decisions taken in 2011 that seemed reasonable at the time and revealed their systemic limitations only over time.

1. How the System Was Designed

The Law, TAMIR and the "guf mukar" mechanism

The 2011 Packaging Law obligated all producers and importers of packaged goods in Israel to contract with a "recognised operator" (גוף מוכר, *guf mukar*) – a company licensed by MoEP to fulfil recycling obligations on their behalf.[1]

The sole recognised operator became TAMIR (ת.מ.י.ר – *Tagid Michzur Yatzranim*). Founded by the Manufacturers Association of Israel, TAMIR is a non-profit company: producers and importers pay it *damei tipul* (דמי טיפול, treatment fees), and it organises the collection, sorting and recycling of packaging.[2]

The EPR mechanism through TAMIR – how it worked:

1. Producer/importer → contracts with TAMIR.
2. Producer/importer → declares quarterly the weight of packaging placed on the market.
3. Producer/importer → pays *damei tipul* per kg by material type.
4. TAMIR → pays municipalities for organising source separation (orange bins, collection).

5. TAMIR → pays for sorting at sorting facilities.

6. TAMIR → reports to MoEP on meeting recycling targets.

On paper: a closed chain in which producers finance the recycling of their own packaging.
In practice: the chain is closed only for packaging that entered it.

2. Three Systemic Failures

Failure One: The Declared Base vs the Real Stream

TAMIR reports its recycling performance against the so-called "declared base" (הבסיס המוצהר) – that is, the volume of packaging declared by the companies that have signed contracts with TAMIR.[3]

The problem is that this base represents only part of the actual packaging flow in the country. Per MoEP's estimate, TAMIR covers approximately 50% of all packaging that enters the Israeli market annually.[4]

This means that the "84% recycling" figure (the 2021 result) refers to 50% of the market. Applied to all packaging, this is approximately 42% – already significantly lower. And that is at the most optimistic calculation, not accounting for downstream losses at sorting.

◆Under the Hood: Why the Declared Base Creates Structural Optimism

The denominator is calculated not from "all packaging in Israel" but from "packaging declared by system participants." This creates a paradox: TAMIR can show growing recycling figures simply because new companies join the system – even if the actual volume of recycled material does not change.

An analogous problem is documented by Eurostat for several European countries: "collection" is counted as "recycling," and EPR system coverage is not complete. In the Israeli context the problem is more acute – due to the scale of free riding.

The State Comptroller recorded this methodological trap in 2021, but no systematic recalculation of the figures followed.

Failure Two: Free Riders

Per MoEP data (2025), more than 40% of packaging producers and importers in Israel have still not contracted with TAMIR – meaning they pay no *damei tipul* and declare no packaging.[4]

Only in 2024, 13 years after the law was passed, did TAMIR take legal action against dozens of violators for the first time.[5]

Why did this continue for so long? Because the system had no enforcement mechanism with real teeth. The fine – 150,000 shekels – was insufficient relative to the benefit of non-compliance. And crucially: MoEP had the right to issue fines but did not exercise it systematically.

◆The Economics of Free Riding: Why Non-Compliance Was Profitable

A company not paying *damei tipul* has an operational advantage: its costs are lower than those of law-abiding competitors. Its products end up in the same orange bins that others are paying for – the system collects and recycles the free rider's packaging at no cost to it. A fine of 150,000 NIS against an import volume of hundreds of tonnes per year is a disproportionate sanction. Paradoxically: as TAMIR expanded (more bins, more collection), even non-paying participants received better recycling service for their packaging. This is a classic collective action problem: the benefits of the system accrue to everyone, while the costs are borne only by those who support it.

Failure Three: Households Were Systematically Underserved

A further structural failure: TAMIR collected 75% of packaging from the commercial sector and only 25% from households, despite households accounting for 70% of all packaging placed on the market.[6]

The logic was straightforward from an operational standpoint: collecting packaging from shopping centres and warehouses is cheaper and easier than from 90,000 residential entrances. And the volume from the commercial sector is sufficient to meet the law's statutory targets.

Result: the system formally meets the law's requirements, but does not fulfil the law's purpose. It prevents commercial packaging from being landfilled – and leaves household packaging virtually unchanged.

◆Under the Hood: MoEP 2025 Data – Official Acknowledgement of Failure

In August 2025 MoEP published the results of a national waste stream scan. Key figures: only approximately 6% of household packaging waste reaches dedicated containers. Approximately 60% of the green (mixed) bin content consists of recyclable materials – plastic, cardboard and similar – that should have ended up in the orange bin.

This is not an independent activist assessment. These are official data from the regulator. They mean: 15 years of EPR system operation have not changed household behaviour in any measurable way.

This acknowledgement, made by MoEP itself, is one of the key arguments in favour of the reform being discussed in 2026.

3. The Hidden Municipal Subsidy

There is one more mechanism that is rarely spoken about openly – and that makes this system unfair to local authorities.

TAMIR pays municipalities for organising source separation of packaging. But the amount it pays covers only part of the real costs. The difference is covered by the municipality from *arnona* – that is, from the general tax paid by all residents.

How the hidden subsidy arises:

The cost of collecting, transporting and sorting packaging: real price – X shekels per tonne.

Compensation from TAMIR to the municipality: Y shekels per tonne, where Y < X.

The difference X–Y: covered from the municipal budget, which is financed by *arnona*.

This means: residents of all households – including those whose packaging is not declared in the system – subsidise the recycling of packaging belonging to producers paying *damei tipul*.

This is a redistribution: from the broad taxpayer to producers who are already legally obligated to pay for this themselves.

Per State Comptroller data (2022), municipalities spend on average approximately 7% of their regular budget on the collection and disposal of solid household waste.[7]

The precise size of the "hidden subsidy" – the gap between real costs and TAMIR compensation – has not been published systematically in the public domain. This is itself a characteristic of the system: money moves, but no one is obliged to publicly show exactly how much.

4. Monopoly Status and Its Consequences

TAMIR has been the sole recognised operator (גוף מוכר) since its founding. Competing operators attempted to obtain a licence from 2011 onwards – without success.[8]

The Competition Authority (רשות התחרות) recognised TAMIR as a monopoly and imposed temporary conditions. But the monopoly did not disappear – it was part of the architectural decision: presumably a single operator is simpler to manage and coordinate than several competing ones.

The problem with a monopoly in an EPR system is not so much price (TAMIR is non-profit) as the absence of pressure towards efficiency improvement. If the sole operator meets its statutory targets – that is enough. There is no incentive to perform better, collect more from households, or reduce the rejection rate at sorting facilities.

A class action was filed against TAMIR – a company importing cement claimed that TAMIR had collected 645,000 shekels in retroactive payments as a condition of contracting, exploiting its monopoly position.[9]

5. What the Figures Show

Indicator	What it actually means
"84% recycling" (TAMIR result, 2021)	This is 84% of the declared base – approximately 50% of the real packaging flow. Real coverage: ~42% at best.
~2,800 companies under contract	An estimated ~6,000 producers and importers are outside the system entirely. Coverage: ~31% by number of participants.
75% collection from commercial sector	Households produce 70% of packaging but contribute only 25% of the collected volume. The system serves those who are not paying the *arnona*.
6% of household packaging in dedicated bins (MoEP 2025)	Official regulator data. After 15 years of EPR – zero measurable change in household behaviour.
Retroactive payments for 7–12 years	The incentive to evade was built into the system: permission to pay retroactively at a reduced rate meant it was more advantageous to

Indicator	What it actually means
	wait than to register immediately.

Chapter Summary

Key conclusions:

→ The EPR principle ("polluter pays") is correct. The architecture of its implementation in Israel led to predictable systemic failures.
→ The declared base is the key methodological trap: the system measures success against itself, not against the real packaging flow.
→ Free riders are not a random phenomenon but a predictable result of insufficient enforcement. 40% of producers and importers outside the system is a systemic problem, not an exception.
→ Households were systematically underserved: 25% of collection against 70% contribution to the packaging flow. The law's targets permitted this because they were oriented to tonnage, not source.
→ The hidden municipal subsidy is real: residents pay *arnona*, including to cover the gap between TAMIR compensation and real costs.
→ Monopoly status without competitive pressure reduced incentives to improve efficiency.
→ MoEP's official 2025 data documented the household-level failure. This is not an activist critique. It is the regulator's own acknowledgement.
→ The 2026 reform is a direct consequence of these failures. Chapter 16 describes it in detail.

◆Control Questions for the Municipality and TAMIR

1. How many tonnes of packaging were collected from orange bins in your city last year – and what share does this represent of the estimated packaging flow?

2. What is the gap between TAMIR compensation and the municipality's real costs for organising source separation?

3. Which specific companies in your region have no contract with TAMIR, and what has been done to enforce compliance?

4. What is the rejection rate at the sorting facilities serving your city?

Footnotes

1. Packaging Treatment Regulation Law (Israel), 2011 [Hebrew: חוק להסדרת הטיפול באריזות, התשע"א–2011]. Full text: Nevo Legal Database, nevo.co.il/law_html/law00/75777.htm. The law establishes the framework for EPR for packaging waste in Israel. Art. 5: obligation to contract with a recognised operator (גוף מוכר). Art. 12: criteria for recognition. The law entered into force January 2012; implementation began gradually through 2012–2014. ↩ ↩[2]

2. TAMIR Association (תאגיד מחזור יצרנים). Founded by the Manufacturers Association of Israel (התאחדות התעשיינים). TAMIR was the only recognised operator (גוף מוכר) from 2011 to the present. As of 2024: approximately 2,800 companies under contract. Sources: tmir.org.il; TheMarker, "Instructions for producers/importers under the Packaging Law," 27.08.2024. ↩

3. Infospot. "A Decade of the Packaging Law: About Half of Israel's Packaging Is Recycled." 22.11.2023. Key figure: TAMIR recycled approximately 402,000 tonnes in 2021, representing 84% of the 480,000 tonnes declared to it – but this is only 50% of all packaging estimated to circulate in Israel, since the other 50% is not declared to TAMIR at all. Available at: infospot.co.il/n/*Packaging_Law* ↩

4. Infospot — Recycling Knowledge Center. "Update on Packaging Waste Law: Will Opening the Market to Competition Solve the Problems?" June 2025. Key figure: MoEP estimates that more than 40% of producers and importers of packaging have not contracted with TAMIR, contributing to over 50% of packaging waste being unregistered in the EPR system. Available at: infospot.co.il/n/Packaging_Waste_Law_Update ↩ ↩[2]

5. Infospot. "First Time: TAMIR Sues Importers Who Are Not Under Contract." infospot.co.il/n/tmir. Documents that alongside ~2,800 contracted companies, TAMIR estimates approximately 6,000 producers and importers have not complied with the Packaging Law. TAMIR began litigation for the first time in 2024 – 14 years after the law entered into force. ↩

6. Infospot. "Update on Packaging Waste Law." June 2025 (see note 3). A critical structural problem: TAMIR collects 75% of packaging waste from the commercial stream and only 25% from the household stream, despite the household sector accounting for 70% of all packaging sold. ↩

7. State Comptroller of Israel. Report on Municipal Solid Waste Management, Municipal Government Audit, July 2022. Available at: mevaker.gov.il ↩

8. Israel Competition Authority (רשות התחרות). TAMIR was recognised as a monopoly (מונופולין) under Israeli competition law. Conditions were imposed for a two-year period. Source: gov.il/he/pages/tmir. ↩

9. Infospot. "Class Action Against TAMIR for Retroactive Payment." infospot.co.il/n/Class_action_against_Tamir. A company importing cement filed a class action claiming TAMIR collected 645,000 NIS as a condition of contracting, representing retroactive treatment fees for years without service. ↩

Chapter 15. The Deposit System and TAMIR: Two Streams That Failed to Integrate

"Two systems operating in parallel do not give one plus one. They give one plus one minus friction, minus gaps in coverage, minus materials that ended up in the wrong stream."

Israel's packaging waste management system is built on two parallel mechanisms that emerged independently and were never truly integrated. The first is the deposit system (חוק הפיקדון, *hok ha-pikadon*), in operation since 1999 and based on a financial incentive for the consumer. The second is the EPR system through TAMIR, based on the 2011 Packaging Law and obligating producers to pay for recycling.

On paper they complement each other. In practice they compete for the same materials, create institutional gaps between them – and this structural conflict is one of the factors explaining why both systems together deliver less than their design intended.

This chapter is not about the deposit system not working – it works well precisely within its narrow range. Nor is it about TAMIR being badly designed. It is about why the architectural decision of "two parallel systems" creates systemic gaps that cannot be resolved through operational improvements.

1. The Deposit System: Why Money Works

The 1999 deposit law is simple and elegant: when purchasing a drink, the consumer pays a deposit (*pikadon*, פיקדון), which is returned when the empty container is brought back to a shop or dedicated machine (RVM – reverse vending machine).[1]

The financial incentive creates a fundamentally different dynamic from voluntary sorting. A person returns a bottle not because they "care about the environment" – but because money is at stake. 30 agorot for a small bottle, a similar amount for a large one.

The deposit system in figures:

→ The original law (1999) covered containers up to 1.5 litres in any material (PET, aluminium, glass).
→ 2021: extension to larger containers (1.5–5 litres). Entry into force: 1 December 2021.
→ Operators: ELA (אגודת א.ל.ה.) – historically the primary one, Asofta – which entered the market in 2010 following market opening.
→ Deposit size: minimum 30 agorot; producers may set a higher amount (some containers carry a deposit of 1.20 NIS).
→ Expected additional stream from the 2021 expansion: approximately 750 million large containers per year.
→ Shops with a floor area over 28 sq m are obliged to accept returns even for containers not purchased from them.
→ Green PET Recycling (Kidmat Galil): the main recycler of deposit PET; capacity ~36,000 t/year; produces rPET certified for food contact.

The deposit system is a clear demonstration of the principle that EPR through TAMIR never managed to replicate: the right financial incentive, directed at the consumer, creates a clean, homogeneous, high-value material stream.

2. The Structural Conflict: Who Gets the Valuable Material

This is where the conflict begins – one that is difficult to resolve within the existing architecture.

The deposit system "skims the cream" – it captures the most valuable and cleanest fractions: transparent PET from beverages (precisely the material suitable for bottle-to-bottle recycling), aluminium cans (high secondary market value), and partly glass.

Once these materials have entered the deposit circuit, TAMIR's orange bin receives what remains: more mixed, more contaminated, less valuable plastic. PET trays, non-standard bottles, flexible packaging, HDPE and PP – fractions that are more complex and more expensive to recycle.

Deposit system	TAMIR orange bin
Transparent PET from beverages	PET trays, non-standard bottles
Aluminium cans	Mixed metal, aluminium foil
Deposit glass (partly)	Remaining glass → purple bin
Clean, homogeneous stream	Mixed, heterogeneous stream
Financial incentive for return	Voluntary sorting
High secondary raw material value	Lower secondary raw material value
Downstream: bottle-to-bottle possible	Downstream: predominantly non-food applications

After the deposit was extended to large bottles in December 2021,[1] this structural gap became even more pronounced: the most valuable PET – from large bottles of water, juice, and soft drinks – moved into the deposit circuit. The orange bin was left with packaging of lower recyclability potential.

3. The Glass Problem: Where the Gaps Are Most Visible

Glass is a clear example of the structural conflict.[2]

The deposit system covers glass beverage bottles carrying a deposit mark – approximately 120,000 tonnes per year. This is clean, homogeneous glass that is genuinely recycled.

But outside deposit glass lies an enormous volume of glass packaging: jars, tinned goods, sauces, jams – material that goes into TAMIR's purple bin. Here the volumes are entirely different: only approximately 30,000 tonnes are declared to TAMIR, while the real volume of glass packaging in the country is significantly higher.

What happens to this glass? Israel's recycling rate for non-deposit glass is 4–8% by various estimates. This is one of the lowest rates among developed countries. The explanation: glass is heavy, fragile, the cost of recycling is high, and collection logistics are expensive. Under current financial incentives there is no reason for TAMIR to invest in quality collection of purple-bin glass.

◆Under the Hood: Glass as a Systemic Symptom

Glass is the best indicator that the EPR system measures compliance with targets rather than actual recycling. TAMIR consistently fails to meet glass targets (the only material showing systematic non-compliance). There are no penalty consequences.

The Interdepartmental Committee on Competition in the Packaging Waste Sector (July 2024) recorded explicitly: not a single new operator has been granted permission to work since 2011, meaning there is no competitive pressure on TAMIR to improve glass collection.

Paradox: glass is one of the most technically recyclable materials (100% recyclable without quality loss). In Israel it is recycled less well than plastic – due to the absence of economic logic within the system.

4. Money: Who Pays for What and Why It Is Opaque

The financial flows between the deposit system, TAMIR and municipalities are structured in a way that creates hidden cross-subsidies – and makes it very difficult to answer the simple question: how much does it cost to recycle a specific fraction?

Financial flows in the two systems:

DEPOSIT SYSTEM:
Consumer pays deposit at purchase.
Operator (ELA/Asofta) receives deposit from the beverage producer/importer.
On container return – operator reimburses the deposit to the consumer.
The difference between deposits collected and deposits reimbursed – operator income, which covers operating costs.
The recycler (Green PET and others) receives clean material and sells secondary raw material.

TAMIR SYSTEM:
Producer/importer pays *damei tipul* to TAMIR.
TAMIR pays municipalities compensation for organising collection.
The municipality covers the gap between compensation and real costs from *arnona*.
Sorting facilities receive payment for sorting.
Recyclers purchase sorted material (at market prices, which depend on oil prices).

WHO BEARS THE RISKS:
Falling oil prices → falling value of secondary plastic → sorting facilities and TAMIR lose income.
Growing volume of non-sortable fractions → growing "tails" → additional landfill costs.
Growing volume of unregistered packaging → more packaging in the system without additional financing.

The integration problem

When the deposit system expanded to large bottles in 2021, it took with it the valuable PET from TAMIR's financial model. That is, TAMIR receives less valuable secondary raw material to sell – while maintaining the same organisational and operational overhead. This worsens its financial position without any compensating mechanism.

No regulatory instrument governed this transition. The two systems develop in parallel, each with its own logic, without a coordination mechanism to manage the situation when one system affects the revenue base of the other.

5. What Real Integration Would Have Required

Integration of the two systems does not mean merging them. It means having a coordination mechanism that addresses three tasks.

Three integration tasks that were not resolved:

Task 1. Stream separation without overlap.
A clear definition: which packaging falls under the deposit, and which under EPR. Currently some types of container can enter either circuit depending on consumer behaviour. This creates accounting confusion and double-counting.

Task 2. Financial rebalancing when system boundaries change.
When the deposit expands, taking valuable material from the TAMIR stream, a compensating mechanism should exist. Currently there is none.

Task 3. Unified downstream reporting.
Currently each system reports its own figures separately. A citizen who wants to understand the real recycling rate for packaging in Israel cannot add together the figures from the deposit system and TAMIR – they are calculated using different methodologies, against different bases.

◆Under the Hood: International Context – How Other Countries Address This

Germany (DPG + Grüner Punkt): the deposit system and EPR operate in parallel, but with a clear legislative boundary between streams and unified reporting to the regulator. Producers know in advance which system will handle their packaging.

The Netherlands (Nedvang): EPR and the deposit system are coordinated through a single body that maintains a mass balance across all streams. An annual report with downstream data for all fractions is published.

The general OECD principle: integration of EPR and DRS (deposit-refund systems) requires a formal coordination body with a mandate to set rules for both systems. In Israel no such body exists – directly linked to the absence of a single waste regulator.

Chapter Summary

Key conclusions:

→ The deposit system works – within its narrow range. The financial incentive creates a clean, homogeneous, high-value stream. This is not accidental, it is an architectural decision.
→ The 2021 extension of the deposit to large bottles was ecologically the right step, but it created unforeseen financial pressure on TAMIR.
→ The structural conflict is real: the deposit "skims the cream," leaving TAMIR with more complex fractions under the same operational requirements.
→ Glass is a clear indicator of systemic failure: a technically well-recyclable material with a 4–8% real recycling rate due to the absence of economic incentives.
→ Hidden cross-subsidies and opaque financial flows make it impossible to give a precise answer to

the question: how much does recycling a specific fraction actually cost?
→ The absence of a coordination body is the root of the problem. Without it, every change in one system unpredictably affects the other.
→ The 2026 reform is, among other things, a response to precisely this structural gap. The proposal for a single regulator and unified reporting is a direct answer to the problem described in this chapter.

◆Questions for the Citizen and Activist

1. What share of the packaging I put in the orange bin belongs to the deposit stream? (Answer: almost none – deposit containers must be returned to shops.)
2. Where do the fines for violations of the Packaging Law go – to the Cleanliness Fund or to TAMIR's budget?
3. Does my municipality publish data on the gap between TAMIR compensation and real costs for source separation?
4. When will the system have a single public report that consolidates the streams of both systems with downstream data?

The next chapter describes how the entire system – with its failures and structural contradictions – reached the point where its continuation became politically impossible. The 2026 reform is not an improvement of the existing architecture. It is an attempt to replace it with something new.

Sources for Chapters 14–15

Israeli legislation

- Packaging Treatment Regulation Law 2011 ,(חוק להסדרת הטיפול באריזות). nevo.co.il/law_html/law00/75777.htm
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Footnotes

1. Deposit Law on Beverage Containers (Israel), 1999, as amended 2010, 2021 [Hebrew: חוק הפיקדון על מכלי משקה, התשנ"ט–1999]. Extension to large containers (>1.5L up to 5L): entered into force 1 December 2021. Source: MoEP FAQ, gov.il/he/departments/faq/bottle_deposit_expansion_faq. ↩ ↩[2]
2. For the structural conflict between the deposit system and TAMIR: the deposit system captures high-value PET bottles and aluminium, leaving TAMIR with the more difficult and less valuable packaging mix. As the deposit was extended to large bottles (2021), TAMIR's stream became even more difficult. This is documented in the Interdepartmental Committee report (July 2024) and in Infospot, "A Decade of the Packaging Law," 2023: deposit glass volume in Israel is ~120,000 tonnes/year, while glass packaging declared to TAMIR is only ~30,000 tonnes/year – illustrating the structural imbalance. ↩

Chapter 16. The 2026 Reform: Israel's Waste Sector at a Crossroads

"A reform that has been debated for two years and removed from the bill twice is not an administrative formality. It is an acknowledgement that the existing system has exhausted itself."

Status at time of publication – June 2026

This chapter describes a process in motion. The reform of Israel's waste sector has been under discussion since 2024, was included in the Economic Arrangements Law bill twice and removed twice. The main parameters of the system – who will be the regulator, which laws will be repealed, how financing will change – remain the subject of negotiations between the Knesset, the Ministry of Finance, the Ministry of Environmental Protection and local authorities. No outcome is guaranteed. The chapter honestly describes this uncertainty – and explains why civic society participation matters most precisely now.

The previous two chapters described how Israel's EPR system was designed and why it failed to fulfil the objectives for which it was created. This chapter is about what happens next.

In 2026 three things are happening simultaneously in Israel that make reform inevitable, complex and politically unstable.

Three simultaneous processes shaping the reform:

CRISIS: landfill sites are running out by end of 2026.
Per the State Comptroller's follow-up audit of July 2025, remaining capacity is insufficient for the volume of waste expected by end of 2026. This is a physical constraint, not a forecast.

FAILURE: the regulator's own data confirmed that the system does not work.
MoEP published the results of its scan in August 2025: 6% of household packaging in dedicated bins, 60% of the mixed bin – recyclable materials. 15 years of EPR – and zero measurable change in household behaviour.

PRESSURE: the Ministry of Finance took the initiative into its own hands.
After sector reform stalled for years, the Ministry of Finance included it in the Economic Arrangements Law bill (חוק ההסדרים) – an instrument traditionally used to carry out structural reforms simultaneously with budget adoption.

1. What Is Proposed: the Substance of the Ministry of Finance Reform

In November 2025 the Ministry of Finance published a draft legislative proposal within the framework of the 2026 חוק ההסדרים.[1]

The substance of the proposal – a transition from a fragmented system of separate laws to a single framework law, following the model of other regulated industries: water, electricity, gas.

Five key elements of the Ministry of Finance proposal:

1. A single framework waste law.
One law covering all types of waste. An end to fragmentation: separate laws on packaging, deposit, electronic waste will be repealed after the first implementing regulations of the new law are published.

2. A national waste regulator (רשות פסולת ארצית).
A new body within MoEP. Powers: setting tariffs, licensing operators, long-term planning, service standards, annual public report on the state of the sector.

3. Abolition of the landfill levy (היטל הטמנה) and the Cleanliness Fund (קרן הניקיון).
Replacement: direct service payments in the inter-municipal segment on the "full-cost principle." Rationale: the current levy distorts incentives and has become effectively additional Ministry of Finance revenue rather than an environmental policy instrument.

4. A registry of all sector operators.
Licensing of carriers, sorting facilities, landfill operators. An instrument against organised crime, which has penetrated the sector.

5. Projected savings.
The Ministry of Finance forecasts savings of more than 450–580 million shekels per year for local authorities in the first year of implementation.

◆Under the Hood: What "Closed Market" (משק סגור) Means in the Ministry of Finance Model

The Ministry of Finance uses the formula "closed market based on the full-cost principle" (משק סגור המבוסס על עיקרון העלות). This means: each service in the chain – collection, sorting, processing, landfilling – receives a regulated tariff covering real cost. The gap between real cost and what the municipality or producer pays disappears.

This "hidden subsidy" is described in Chapter 14: currently the gap is covered from arnona, that is, from all residents' pockets. The Ministry of Finance reform aims to make these flows explicit.

The water market analogy: water tariffs are set by a regulator (the Water Authority), cover real costs and are publicly justified. The Ministry of Finance wants the same for waste.

Criticism: the water analogy is incomplete – water does not have the same number of competing streams, material types and downstream markets. Waste management is significantly more technically complex.

2. Three Positions: Who Says What and Why

The reform has generated sharp disagreements between institutions. Here are the three main positions – not as opinions, but as positions with internal logic.

THE MINISTRY OF FINANCE

Central argument: the current system generates inefficiency, opacity and attracts organised crime. A single regulator with public tariffs will fix these problems structurally, not operationally.

What is proposed: framework law + regulator + abolition of the landfill levy and Cleanliness Fund + operator registry.

Main concrete argument: approximately 75% of carcinogenic-substance emissions into Israel's air come from illegal waste burning.

Weakness of the position: the proposal is oriented towards economic efficiency and fighting crime; environmental goals (reducing landfilling, increasing recycling) are named but not enshrined in specific mandatory indicators.

THE MINISTRY OF ENVIRONMENTAL PROTECTION (MoEP)

Central argument: the Ministry of Finance reform is a "machtaf" (מחטף) – a grab. It ignores MoEP's environmental strategy and creates a regulator without environmental goals.

What is proposed: several alternatives, all providing for the retention of environmental KPIs: reducing landfilling, increasing recycling, raising the share of household collection. Agrees with creating a regulator – but with a mandatory environmental mandate.

Main argument: without hard environmental goals at the foundation of the law, the regulator will optimise service costs, not environmental impact reduction. These are different objectives.

Weakness of the position: MoEP also bears responsibility for the current crisis – for 15 years MoEP oversaw the EPR system that, by the ministry's own data, does not work. The "machtaf" critique is perceived by many as protection of departmental interests.

NGOs AND ENVIRONMENTAL ORGANISATIONS (position of Adam Teva VeDin)

Central argument: both proposals – the Ministry of Finance's and MoEP's – do not address the real problem. A single waste law is needed, but one built around the 9R hierarchy: prevention first, then reuse, then recycling – and only at the end energy recovery.

What is said: the Ministry of Finance proposal is "superficial and amorphous," sets no measurable environmental targets. At the same time the current system (which MoEP is defending) is also failing.

Main argument: a law that regulates tariffs will not solve the problem if it contains no mandatory targets for reducing waste generation. Most European countries with successful recycling rates have legally enshrined goals – not just regulators.

Weakness of the position: analytically strong, but politically difficult to implement under current conditions of crisis demanding rapid solutions.

3. The Reform Timeline: Why This Is So Difficult

Understanding the current moment is impossible without understanding how many times the reform has already started and not concluded.[2]

Date	Event
2020	MoEP publishes National Strategic Plan 2030: targets for reducing landfilling, increasing recycling, WtE.
2021	State Comptroller documents the landfill capacity crisis. First recommendations.
October 2024	Government decision: create an inter-ministerial team to develop a framework waste law. The team never convened.

Date	Event
September 2024	Ministry of Finance publishes first version of reform – the sector responds with sharp rejection.
October 2024	Reform tone "softened" following industry pressure.
November 2025	Ministry of Finance includes reform in the draft 2026 חוק ההסדרים.[1]
November 2025	MoEP calls the proposal a "machtaf." Ministry of Finance removes reform from the bill.
December 2025	Reform returns – as a private member's bill (הצעת חוק פרטית).
February 2026	Public consultations – comment deadline for citizens and organisations.
March 2026	Knesset cuts חוק ההסדרים by half. Status of waste reform – unclear.
December 2026	Local authorities call on Ministry of Finance to advance the reform.
June 2026	Reform continues to be discussed. No final decision.

This is not a story of bureaucratic delays. It is a story of real conflicts of interest: sector operators, local authorities, packaging producers, environmental NGOs – each has its own position, and each has internal logic.[2]

4. What Happens Under Different Scenarios

Since the outcome of the reform is unknown, we examine three scenarios – not as forecasts but as a map of consequences. Each scenario affects residents, NGOs and municipalities differently.

Scenario A: The Ministry of Finance reform is adopted in roughly the proposed form

What changes: a national regulator is created, the landfill levy and Cleanliness Fund are abolished, regulated tariffs are introduced, TAMIR loses its monopoly, all existing EPR laws are repealed.

Scenario A: possible consequences

For municipalities: tariffs become transparent and regulated. Projected savings of 450–580 million NIS/year – if the regulator actually sets tariffs below current market rates. Risk: transition period without clear rules.

For NGOs and activists: a regulator appears with an obligation to publish an annual report. This is potentially better data access than currently available. Risk: if the regulator focuses on economic efficiency without environmental KPIs – recycling may not improve.

Scenario A: possible consequences

For recyclers and operators: open market and licensing. New players can enter the EPR system. Risk: competition benefits the sector if the regulator ensures fair rules.

For residents: no direct immediate effect on waste separation. Long-term – if tariffs fall, arnona may (theoretically) decrease.

Scenario B: Reform is blocked again, crisis deepens

What it means: the system stays as is. But this is not the status quo – it is deterioration. Landfill sites continue to be exhausted, municipalities face capacity deficits, costs rise.

◆Scenario B: The Cost of Inaction

Per State Comptroller data 2025: remaining capacity is insufficient for the volume of waste expected by end of 2026. This is a physical fact, not a forecast. Without new end-of-pipe solutions (WtE or new landfills), municipalities will face a choice: pay a many-times-higher price for waste disposal at remaining sites, or allow waste accumulation. Illegal burning and illegal sites are already a documented reality. Under a deficit of official capacity, pressure towards illegal solutions will grow. For citizens: rising arnona costs without improvement in service quality.

Scenario C: Compromise – limited reform with environmental KPIs

What it means: a regulator is created, but with a mixed mandate – economic efficiency AND environmental goals. Some existing laws are retained (possibly the deposit law) or reformed rather than repealed. The landfill levy is reformed, but not abolished.

This is the most likely practical outcome of political negotiations. But it requires all three sides – Ministry of Finance, MoEP, NGOs – to find common ground. History shows this is not easy.

5. What Citizens Can Do Right Now

One of the key arguments of this book: reforms under conditions of crisis are adopted more quickly – and worse – if civil society is silent. The best reforms happen when citizens, NGOs and local communities formulate specific demands before legislative texts are finally agreed.[1]

Below are specific leverage points that exist right now.

Civic leverage points in the reform process:

Public consultations.

Both ministries – Ministry of Finance and MoEP – have held public consultations. A further round is possible with any new legislative proposal. A comment from a citizen or NGO is an official document that the regulator is obliged to consider.

The Knesset Interior and Environmental Protection Committee.

This is the responsible committee reviewing waste legislation. Public hearings are an open process. Organisations can request the opportunity to present.

Local authorities.

Municipalities have their own organisations (Union of Local Authorities – ועד ראשי הרשויות

המקומיות). It was they who publicly called on the Ministry of Finance in December 2026 to advance the reform. Local activists can influence their municipality's position in this dispute.

Data requests.
The reform creates new statutory transparency obligations (annual public report from the regulator). Before the law is adopted – use the tools from Chapter 25: requests for data on current tariffs, costs, gaps.

Position on specific elements – not "for" or "against" the reform in general.
It is more effective not to advocate a general position but to formulate specific demands: "Any law must contain clause X" or "Without mandatory environmental KPI Y we oppose this." Specific demands are harder to ignore than general support or opposition.

◆Questions to Ask During the Reform Process

1. Does the law include measurable, mandatory targets for reducing landfilling and increasing recycling – with specific deadlines?
2. Will the new regulator be obliged to publish an annual report with downstream data for all streams?
3. What will happen to current municipal investments in source-separation infrastructure during the transition period?
4. Who will sit on the regulator's board – and how will representation of environmental organisations and local communities be ensured?
5. How does the law protect residents' right to information about service costs and recycling quality in their area?
6. What will happen to the deposit system? Will it be repealed, retained or reformed?
7. How will the law protect against lock-in when WtE capacity is expanded – so that the regulator's economic incentives do not conflict with waste prevention goals?

6. The Reform and This Book: How to Read Everything Written Above

The reader may already be asking: if the EPR system is proposed for abolition and the new system is not yet defined – why read Chapters 14 and 15 about TAMIR and the deposit system?

The answer is that the failure of the current system is not a story about specific institutions. It is a story about design principles. The declared base instead of the real flow. The absence of enforcement for free riders. The bias towards commercial collection. The absence of a coordination body. Opaque financial flows.

All these problems will not disappear automatically with the creation of a new regulator. They will disappear only if the new law directly addresses them. That is precisely why knowledge of how the old system failed is not a historical excursion. It is a toolkit for evaluating the new one.

◆Under the Hood: How to Evaluate Any New Legislative Text on Waste

Question 1. Against what is recycling measured – against the declared base or against the full packaging flow? If the former – the problem remains.

Question 2. What is the enforcement mechanism for non-compliant producers? Size of fine, timelines, public disclosure of violations.

Question 3. Are there mandatory targets for the household sector – not only by tonnage but by source?

Question 4. Does the regulator publish downstream data – where sorted material actually goes, what the rejection rate is, what happens to exports?

Question 5. How does the law relate to the 9R hierarchy – does prevention and reuse have priority over recycling, which has priority over energy recovery?

Question 6. Who verifies compliance and how? A state body with real powers or a self-reporting mechanism?

Chapter Summary

Key conclusions:

→ The 2026 reform is not an administrative reorganisation. It is an attempt to respond to a physical crisis (no space for landfilling), a systemic failure (EPR does not work), and political pressure (the Ministry of Finance took the initiative).
→ Three positions (Ministry of Finance, MoEP, NGOs) have different priorities: economic efficiency vs. environmental goals vs. a comprehensive systemic approach. None of them is absolutely right.
→ The reform was removed from the bill twice. This is not a signal of its failure – it is a signal of the reality of the conflict of interests that must be resolved legislatively.
→ Three scenarios (adopted, blocked, compromise) have different consequences for citizens, municipalities and recyclers. The most likely is a compromise variant with unclear details.
→ Civic participation now is not rhetoric. Specific demands on specific legislative elements are harder to ignore than general support or opposition.
→ Knowledge of how the old system failed is a toolkit for evaluating the new one. The declared base, free riders, bias towards commercial collection – these problems must be sought in the text of any new law.
→ Uncertainty is not a reason to disengage. On the contrary: the moment when texts are not yet finalised is the moment of maximum influence.

The subsequent parts of the book – on packaging design, PPWR and money in the system – are written with an understanding of this turning point. Regardless of how the 2026 reform concludes, the principles described in them will remain relevant: upstream regulation matters more than downstream recycling, data transparency matters more than polished reports, prevention matters more than optimising landfilling.

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Chapter 17. Packaging Design: Why Some Materials Are Born Recyclable

"Nobody asks the packaging designer whether their product is recyclable. They are asked whether it sells, whether it protects the contents, and what it costs to produce. Recyclability is a decision made many steps before the material reaches anyone's bin."

Chapters 9–13 showed specific material streams – where each succeeds and where it fails. This chapter moves up one level: what determines whether a particular piece of packaging ends up in the "successful" or "problem" material category, before it ever reaches a container?

The answer is design. Decisions made at the product development stage by the manufacturer determine the packaging's fate more powerfully than any subsequent investment in sorting or recycling.

1. The Principle: Recyclability Is Not a Property of a Material, It Is a Property of Design

A common mistake is to think of materials in binary terms: "plastic bad, paper good." Chapters 9–13 already showed this is imprecise: HDPE plastic recycles significantly better than multilayer packaging, while technically perfect glass recycles worse than metal – due to logistics, not chemistry.

The precise formulation: recyclability is determined not by the material as such, but by how that material is combined, coloured, labelled and packaged into a single object.

Designer's decision	Recyclable option	Non-recyclable option	Why
Material composition	Single polymer (mono-material)	Several layers of different materials	NIR scanner cannot distinguish composite material (Chapter 11)
Colour	Transparent or light	Black or dark	Black pigment absorbs infrared light – the sensor "cannot see" the material
Label	Removable paper label	Heat-shrink sleeve from a different polymer	Non-removable label from a foreign material contaminates the batch during recycling
Size	Large, from 5 cm	Small, under 3 cm	Physically falls through sorting equipment (Chapter 4)

UNIDO conducted a specific pilot project in Israel illustrating this in practice: a standard Israeli yogurt or salad cup is a PP cup with a PET sleeve. The combination of two different polymers creates a mixed signal for the NIR scanner. Simply replacing the PET sleeve with a polyolefin sleeve makes the packaging detectable – with a minimal increase in production cost and an estimated impact of 2,100 tonnes of additional recyclable plastic per year if adopted broadly by Israeli brands.[1]

2. Three Design Categories: What Is Born Recyclable and What Is Not

Applying the principle from section 1 to all materials examined in the previous chapters, a simple three-tier categorisation emerges.

Design categories: from born-recyclable to born-unrecyclable:

Category 1: Born recyclable.

HDPE bottles, metal cans, deposit PET, paper and cardboard. Common features: single material, easily identifiable, physically resistant to the sorting process. These are materials for which all the subsequent infrastructure (Chapters 5, 10) works – because design from the outset creates no barriers.

Category 2: Recyclability depends on design details.

PP containers with PET sleeves, packaging with non-removable labels, coloured (but not black) plastic. Technically recyclable, but specific design decisions determine whether they will be in practice.

Category 3: Born unrecyclable.

Multilayer packaging (Chapter 12), black plastic, packaging smaller than 3 cm, glass outside the deposit system (Chapter 13). For these categories no realistic downstream solution exists at the current level of sorting technology – the only path to improvement runs through changing design at the input stage.[2]

3. Who Makes Packaging Design Decisions – and Why It Is Not Their Priority

The decisions examined in sections 1–2 are not made by recycling operators or municipalities – they are made by producers and their packaging suppliers, at a stage that is physically and institutionally separate from the entire waste management system described in this book.

The structural gap between design and consequences:

The producer choosing a PET sleeve for a yogurt cup is optimising for branding (vivid print), cost and product protection – not for NIR detectability.[3]

The costs of non-recyclability (tails, RDF, landfill) are borne not by the producer but by the collection system – by the municipality and ultimately by the taxpayer through arnona (Chapter 14), not by the person who made the design decision.

This is a classic case of externalising costs: the correct EPR principle ("polluter pays") should have corrected this, but as Chapter 14 shows, Israel's implementation of EPR through the declared base and weak enforcement does not create sufficient financial pressure on the design decisions of specific products.

4. What Is Changing in Israel: The November 2025 Memorandum

In November 2025 the Ministry of Environmental Protection published a memorandum of law (a preliminary bill for public comment) on an amendment to the Packaging Law – a document separate from the broader Ministry of Finance reform (Chapter 16), but moving in parallel with it.[4]

The amendment's aim – to reduce the free-rider phenomenon (Chapter 14) and raise recycling targets to European standards.[5]

Key elements of the November 2025 memorandum:

Gradual increase in recycling targets to 2033 in alignment with European standards – the general target will rise from the current 60% to 70% by 2031, with specific sub-targets by material type.[6]

New payments for reusable packaging – producers and importers will be obliged to pay *damei tipul* for reusable packaging, which is currently exempt.

Minimum 50% by weight of recycled waste from the household sector. A direct response to the problem examined in Chapter 14 (75% of TAMIR's collection from the commercial sector while households contribute 70% of the packaging flow) – the amendment directly obliges the operator to expand household coverage rather than optimising in favour of the convenient commercial stream.[5]

Opening the market to competition – until now the only recognised operator was TAMIR. The amendment will allow multiple operators to be recognised simultaneously, under MoEP oversight and with a coordination mechanism between them.[7]

New registry of producers and importers – every producer or importer of packaged goods will be required to register in a new online registry managed by MoEP – a transparency mechanism separate from, but complementary to, the obligation to contract with a recognised operator.[8]

Note: connection to the 2026 reform (→ Chapter 16)
This memorandum is not the same as the broader Ministry of Finance reform from Chapter 16, but they are closely linked. The Ministry of Finance proposes fully abolishing the Packaging Law after the first implementing rules of a new framework law are published; the MoEP memorandum, by contrast, proposes reforming the Packaging Law from within – retaining it, but opening it to competition and tightening requirements. This means Israel at the time of writing has two parallel, potentially competing visions for the future of the EPR system for packaging. Which of them ultimately prevails is an open question at the time of publication. The deadline for public comments on the MoEP memorandum was 10 December 2025 – a specific civic participation point analogous to those described in Chapter 16.

5. Design Requirements as Part of the Memorandum: What Changes for Packaging Specifically

Although the main focus of the November 2025 memorandum is market structure (competition between operators) and financial obligations, the document also signals direction towards alignment with European design standards through the recycling targets themselves.

The comparative table of recycling targets published alongside the memorandum directly shows: Israel's current target (60%) versus the proposed target (70% by 2031) versus the EU target (70% by 2030 under the PPWR framework) – an alignment that is not accidental but deliberate.[6]

This means: raising recycling targets itself creates indirect pressure on producers towards more recyclable design – because a higher target is harder to achieve without improving the detectability and sortability of packaging at the input stage. Direct design requirements (as in PPWR, examined in the next chapter) are not introduced in the current draft of the MoEP memorandum.

Chapter Summary

Key conclusions:

→ Recyclability is not a property of a material – it is the result of specific design decisions: composition, colour, label, size.

→ Three design categories determine packaging fate long before the bin: born recyclable, depends on design details, born unrecyclable.[2]
→ The structural gap between who makes design decisions (the producer) and who bears the costs of non-recyclability (the collection system, the taxpayer) is the root of a problem that EPR should have, but failed to fully, solve (Chapter 14).
→ The MoEP November 2025 memorandum proposes reforming the Packaging Law from within: raising targets, opening the market to competition, mandatory minimum household collection, a new producer registry.[45]
→ This memorandum moves in parallel with, not jointly with, the broader Ministry of Finance reform (Chapter 16) – Israel at time of publication has two competing visions for the future of the system.

The next chapter looks at the source from which both the MoEP memorandum and (in a different form) the Ministry of Finance reform draw inspiration: the European Packaging and Packaging Waste Regulation (PPWR), which came into force in 2025 and offers the most developed model of design regulation in existence.

Chapter 18. What Israel Can Learn from PPWR

"PPWR is not a perfect law. It is a law that passed through the most intensive lobbying process in EU history and came out diluted. But even the diluted version contains mechanisms that in Israel do not exist in any form whatsoever."

Chapter 16 described the 2026 reform as a moment of systemic turning point for the entire Israeli waste sector. Chapter 17 showed that in parallel a narrower reform of the Packaging Law specifically is underway. This chapter concludes Part IV, examining the source that both processes draw on explicitly or implicitly: the European Packaging and Packaging Waste Regulation (PPWR) – Regulation (EU) 2025/40.

This is not a chapter arguing that Israel should copy European legislation word for word. It is a chapter about which specific PPWR mechanisms address problems already documented in this handbook for the Israeli context – and which of them it would be worth considering when drafting the new framework law mentioned in Chapter 16.

1. PPWR: What It Is and Why Form Matters

PPWR was formally adopted by the EU Council on 19 December 2024, published in the Official Journal on 22 January 2025, and entered into force on 11 February 2025. Most of its provisions will begin to apply from 12 August 2026 – after an 18-month transition period.[9]

◆Under the Hood: Regulation vs Directive – Structural Lesson Number One

PPWR replaces the previous Packaging Directive (94/62/EC). This is not a technical detail – it is a structural decision with direct consequences.

A Directive requires each Member State to adopt its own national legislation to achieve common goals – this creates significant divergences between countries in implementation details (for example, in EPR rules and environmental labelling).

A Regulation applies directly across all EU countries without the need for national transposition – a single, consistent set of rules.[10]

Direct parallel with Israel: Chapter 1 showed that Israeli waste legislation has historically developed as a collection of separate laws (Cleanliness Law 1984, Collection Law 1993, Packaging Law 2011, and so on), adopted reactively. The choice between "a new separate law" and "a single comprehensive regulation" is precisely the fork in the road facing the authors of the 2026 reform (Chapter 16).

2. Design Requirements: What Does Not Exist in Israeli Legislation at All

The most direct lesson for Chapter 17: PPWR introduces mandatory recyclability grades (Grade A, B, C – and effectively un-marketable D, E) based on a clear methodology. From 1 January 2030, packaging below Grade C (70% recyclability per the approved methodology) will not be permitted on the EU market.[11]

What PPWR specifically regulates in terms of design:

Recyclability-by-design (Article 6): packaging must be designed so that its components can be separated for recycling, taking into account established collection, sorting and recycling methods.[11]

Minimisation (Article 10): packaging must be of minimum necessary weight and volume; for transport and e-commerce packaging, the empty space ratio must not exceed 50%.[12]

Ban on certain single-use plastics: for example, individual condiment and sauce sachets in food service settings.[12]

Minimum recycled content quotas (Article 7): progressively increasing quotas for the share of recycled material in new plastic packaging – from 10% by 2030 rising to 50% by 2040 for certain categories, verified through mass-balance accounting methodology.[13]

Restriction of hazardous substances (Article 5): ban on PFAS in food-contact packaging above specified thresholds from August 2026; retention of existing heavy-metal restrictions.[14]

This is precisely the level of regulation that, as Chapter 17 showed, is absent from the Israeli MoEP November 2025 memorandum: that document raises recycling targets and opens the market to competition, but introduces no mandatory design criteria analogous to PPWR Article 6.[2]

3. Deposit Systems: PPWR Institutionalises What Israel Already Knows from Practice

PPWR requires EU Member States to establish deposit-return systems (DRS) for single-use plastic and metal beverage containers by 2029 – unless a Member State already achieves a high level of separate collection without such a system.[15]

This directly resonates with the topic of Chapter 15: Israel already has a deposit system since 1999 and knows well both its advantages (a clean, motivated stream) and its structural conflict with the parallel EPR system through TAMIR. PPWR does not directly resolve this conflict for its Member States, but requires explicit institutional coordination between EPR and DRS – precisely the mechanism whose absence Chapter 15 identifies as the root of the problem in the Israeli case.

◆Under the Hood: The Lesson Is About Coordination, Not the Deposit Itself

The lesson here is not that Israel needs a new deposit – it already has one. The lesson is that Israel has no formal coordination mechanism between the deposit system and the EPR system, analogous to what European regulation is beginning to require.

If the 2026 reform (Chapter 16) creates a single national regulator, a natural part of its mandate should be precisely this coordination function – mass balance across all packaging streams, rather than separate, uncoordinated reporting from deposit operators and TAMIR.

4. Extended Producer Responsibility: The Same Principle, A Clearer Mechanism

PPWR confirms and details the EPR principle already present in Israel's Packaging Law since 2011: producers bear responsibility for the entire packaging lifecycle, including financing collection, recycling and disposal.[9]

EPR element	Israel (current system, Chapter 14)	PPWR (EU)
Single or multiple	Single recognised operator (TAMIR) since 2011; November 2025 memorandum proposes opening to	Multiple EPR scheme operators by default, under national oversight

EPR element	Israel (current system, Chapter 14)	PPWR (EU)
operators	competition (Chapter 17)	
Base for calculating targets	Declared base – approximately 50% of real flow (Chapter 8)	Regulation requires direct reporting from producers placing products on the market, via a single registry
Enforcement against free riders	Weak until 2024–2025; first lawsuits only in 2024 (Chapter 14)	Direct obligation to register in producer registry as a condition of market operation
Financing monitoring costs	Not formalised	Producers obliged to finance "costs of labelling waste containers" and "costs of conducting surveys of the composition of collected mixed waste"

The last row of the table is particularly important: PPWR explicitly obliges producers to finance surveys of mixed waste stream composition – meaning precisely the type of national scan that MoEP conducted in August 2025 (Chapter 8), which in the European model becomes not a one-off regulator initiative but a permanently producer-financed obligation.[9]

5. What Should Not Be Copied Word for Word

It would be dishonest to present PPWR as a perfect model without caveats. The regulation has its own weaknesses, which it is worth knowing before proposing it as a template for a new Israeli law.

Limitations and compromises of PPWR to be aware of:

PPWR passed through the most intensive lobbying process in EU history, and the final text is acknowledged as diluted compared with the original 2022 Commission proposal – it is not a technically pure document but the result of political bargaining, just like any reform, including the Israeli one (Chapter 16).[16]

Many key details (the precise methodology for recyclability grades, specific design-for-recycling criteria) are delegated to secondary legislation (delegated acts) that the Commission must adopt only by 1 January 2028 – meaning even in the EU full operational clarity will come several years after formal adoption.[11]

Directly copying the complex multi-tier grading system (A/B/C/D/E) into the Israeli context would require significant institutional capacity for verification and control – a question that should be explicitly addressed when drafting any new law, not assumed automatically resolved.

6. Practical Recommendations for a New Israeli Law

Summarising the lessons of this chapter in a form applicable to the process described in Chapter 16 – whether the Ministry of Finance reform, the MoEP memorandum, or their possible hybrid.

Checklist: what to look for in any new Israeli waste law:

1. Does the law introduce mandatory recyclability-by-design criteria – or does it merely raise recycling target figures without changing what is produced?[11]

2. Is there a formal coordination mechanism between the deposit system and the EPR system for packaging not subject to the deposit?[15]

3. Does the law oblige producers to finance regular (not one-off) surveys of mixed waste stream composition – institutionalising what MoEP did as a one-off exercise in August 2025 (Chapter 8)?[9]

4. Does the law operate as a single regulation with direct application, or as yet another separate law requiring further secondary legislation and interpretation by each municipality in its own way?[10]

5. Is the possibility of multiple operators built in from the outset – or does the system risk again getting a single operator without competitive pressure (Chapter 14)?

6. Is there a realistic transition timeline, acknowledging that even the EU's full operational clarity on complex mechanisms comes years after formal law adoption?[11]

Chapter Summary

Key conclusions:

→ PPWR is not a template for word-for-word copying, but a set of specific institutional solutions that can be compared against the gaps already documented for Israel in Chapters 8, 14, 15 and 17.
→ The main structural lesson: a Regulation with direct application eliminates the fragmentation that Israel's system of separate laws tends towards (Chapter 1) – this is a central question for the form of the future framework law from Chapter 16.[10]
→ Mandatory design criteria (recyclability-by-design) are precisely the link absent from both the current Israeli system and the MoEP November 2025 memorandum (Chapter 17).[11]
→ PPWR requires formal coordination between EPR and deposit systems – the institutional mechanism whose absence Chapter 15 identifies as the root of the structural conflict in Israel.[15]
→ PPWR institutionalises regular producer-financed surveys of waste stream composition – turning the one-off MoEP scan of August 2025 (Chapter 8) into a permanent systemic practice.[9]
→ PPWR is a product of political compromise and intensive lobbying, not a technically pure solution. This is not a reason to ignore it, but a reason not to idealise it.[16]

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10. Khurana & Khurana / KHLaw. "The New EU Packaging and Packaging Waste Regulation – Highlights and Challenges Ahead." Notes the shift from Directive (94/62/EC) to Regulation (2025/40, directly applicable EU-wide) – a structural lesson directly relevant to Israel's fragmented, amendment-by-amendment approach to waste legislation (Chapter 1). ↩ ↩[2] ↩[3]

11. Regulation (EU) 2025/40, Art. 6. Recyclability performance grades A, B, C. From 1 January 2030, packaging must achieve at least grade C to be placed on the EU market. By 1 January 2028 the European Commission must adopt delegated acts setting design-for-recycling criteria and the detailed grading methodology. ↩ ↩[2] ↩[3] ↩[4] ↩[5] ↩[6]

12. Regulation (EU) 2025/40, Art. 10 and recitals on minimisation. Packaging must be designed to use the minimum weight and volume necessary; empty space ratio in e-commerce and transport packaging must not exceed 50%. ↩ ↩[2]

13. Regulation (EU) 2025/40, Art. 7. Minimum recycled content quotas for plastic packaging, phased: e.g. for contact-sensitive plastic packaging, 10% by 2030 rising to 50% by 2040 for certain categories. Verified through mass-balance accounting methodology. ↩

14. Regulation (EU) 2025/40, Art. 5. PFAS restricted in food-contact packaging above specified thresholds from 12 August 2026; existing heavy-metal restrictions retained. ↩

15. EUR-Lex summary of Regulation (EU) 2025/40. Deposit-return systems for single-use plastic or metal beverage containers must be established EU-wide by 2029 unless a Member State already achieves a high separate collection rate without one. ↩ ↩[2] ↩[3]

16. Food Packaging Forum. "European Council adopts final provisions of PPWR." Notes that PPWR has been "the most lobbied political process in the EU's history," with the final text diluted in several respects from the original 2022 Commission proposal. ↩ ↩[2]

Chapter 19. How Much Waste Costs: Mass Balance, Gate Fees and Hidden Costs

"In January 2025, the cost of waste disposal for one cluster of municipalities in the north of the country nearly doubled – from 260 to over 500 shekels per tonne. This is not an abstract statistic. It is a specific figure that a mayor cited at a Knesset committee hearing, holding his city's invoices in his hands."

The previous parts of this book showed how material physically moves through the system – from kitchen to landfill or recycler. This chapter and the next ask a different question: how much does it cost, and who pays.

The answer is more complex than "the taxpayer pays for waste collection." Money in Israel's waste management system moves through several parallel channels, some of which are opaque even to the municipalities themselves – and this opacity is one of the reasons for the crisis examined in Chapter 6.

1. The Anatomy of Cost: Two Segments

The cost of waste management at municipal level divides into two fundamentally different segments.[1]

Intra-municipal segment (פנים-עירוני): the cost of collecting waste from the container site to the transfer station. Depends on distances between collection points, bin type, collection frequency and volume collected (Chapter 4).

Extra-municipal segment (חוץ-עירוני): the cost from the transfer station to the final destination – transport, gate fees at the sorting facility or landfill, including the landfill levy.

This chapter focuses primarily on the extra-municipal segment – it is here that the sharpest and fastest-growing costs are concentrated, directly linked to the capacity crisis from Chapter 6.

2. The Landfill Levy: What It Is and How It Works

Article 15 of the Cleanliness Law establishes the landfill levy (*hitel hatmana*) – a charge per tonne of waste placed in a landfill.[2]

In practice the levy is paid by the landfill operator, but the municipality effectively bears this cost – either directly or through the waste collection contractor, who includes it in their tariff.[2]

The levy was introduced approximately ten years ago with a specific purpose: to create a financial incentive to move away from landfilling in favour of recycling.[3]

Revenues from the levy go to the Cleanliness Fund (*keren ha-nikkayon*), operating within MoEP, and are used, among other things, to develop alternative waste treatment solutions.[3]

3. The "Pay Twice" Paradox: Why Sorting Does Not Reduce Costs

This is perhaps the most counterintuitive and important financial fact in the entire book.

◆The "Pay Twice" Paradox: How It Works

The landfill levy does not reflect the real alternative cost of waste treatment. Every tonne shifted from direct landfilling to sorting *increases*, rather than decreases, the cost for the municipality.

Why: processing at a sorting facility costs money – but extracts less than 10% of recyclable materials. The remaining more than 90% still goes to landfill.

Result: the municipality pays for sorting 100% of the stream – and simultaneously pays the landfill gate fee and landfill levy for over 90% of that same stream, which ends up there anyway after sorting.

This means: the transition from direct landfilling to "sorting + residual landfilling" – ecologically the right step – costs the municipality *more* money under the current tariff structure, not less. This misalignment is one of the sources of municipal financial resistance to more ecological practices.

◆Under the Hood: Why This Matters for Understanding Municipal Motivation

When a municipality appears uninterested in expanding sorting or source separation, this is often not a question of lack of environmental will – it is a rational financial response to a tariff structure that in its current form penalises, rather than rewards, intermediate steps towards recycling.

This directly echoes the hidden subsidy examined in Chapter 14: financial incentives at municipal level systematically diverge from the environmental goals declared at national level.

4. Real Figures: The Crisis in a Concrete Example

Chapter 6 described the capacity crisis in general terms. Here is how it looks in the specific invoices that municipalities receive.[4]

Karmiel Mayor Moshe Koninsky, at a hearing of the Knesset Interior and Environmental Protection Committee, described the direct consequence of the closure of the Evron landfill: from January 2025 the cost of waste disposal for the Western Galilee Cluster of authorities (which includes Karmiel) nearly doubled – from 260 to over 500 shekels per tonne, adding millions of shekels in annual costs per local authority in the region.[4]

Where capacity is specifically insufficient: data from the State Comptroller's follow-up audit, July 2025:[5]

Afe'a landfill (Mishor Rotem, south) – receives approximately 50% of all landfilled waste in the country (2.3 million tonnes/year), but at the start of 2025 had only around 2 million cubic metres of remaining capacity – less than one year's supply at that point.

Hagal landfill (north) – new cell with 1.8 million tonne capacity approved; expected operational from January 2026.

Evron landfill – height extension of approximately 5 metres approved from 2026, adding around 1 million tonnes of capacity – but only for waste that has been sorted, not for direct landfilling.

Longer-term expansions (post-2026): Dudaim/Ganei Hadas near Beer Sheva – planning for 4 million tonnes; Zohar near Arad – planning for 4.8 million tonnes; Afe'a cell #8 – planned for approximately 10 million additional tonnes.

5. The Hidden Cost Item: ILA Royalties

In January 2025 the Union of Local Authorities formally demanded that the Israel Land Authority (ILA / רמ"י) cancel a royalty collection mechanism for land use at landfill sites – arguing that landfill-operating companies had begun passing on this cost (set a few years earlier) to municipalities.[6]

Per the Union's estimate, the cumulative additional burden amounts to approximately 150 million shekels per year nationally – with a direct warning of the risk of "financial collapse" for local authorities.[6]

MoEP acknowledges that it has so far failed to establish direct control over landfill gate-fee pricing and has turned to the Inter-Ministerial Price Committee requesting recommendations.[7]

◆What This Means Structurally

The price of landfilling in Israel is formed without direct regulatory control – a combination of the landfill levy (national instrument, Article 15 of the Cleanliness Law), gate fees (set by the landfill operator), and ILA royalties (a separate channel, recently become contentious) – three independent cost sources that no single body coordinates or regulates as a whole. This is a direct illustration of the absence of a coordination mechanism, described in Chapter 15 in relation to the deposit and EPR – the same structural problem manifesting in the financial sphere.

6. The 2026 Reform: What Is Proposed to Change

Note: connection to the 2026 reform (→ Chapter 16)

The Ministry of Finance, as part of the reform examined in Chapter 16, explicitly proposes abolishing the landfill levy and the Cleanliness Fund, replacing them with direct regulated service payments in the inter-municipal segment based on the full-cost principle – under the management of the proposed national waste regulator.

The logic of the proposal directly addresses the "pay twice" paradox described in this chapter: if each service in the chain (sorting, transport, landfilling) receives a regulated tariff reflecting real cost, the structural penalty for intermediate steps towards recycling theoretically disappears.

MoEP's critique (Chapter 16): abolishing the landfill levy removes the only existing – if imperfect – financial incentive against landfilling, with no guarantee that the new regulator will set equivalently strict environmental priorities in its tariff policy.

Whatever is ultimately decided, the "pay twice" paradox and the unresolved ILA royalty dispute are concrete, measurable problems that any reform must explicitly address – not assume will be automatically solved by the creation of a new body.

Chapter Summary

Key conclusions:

→ The cost of waste management divides into intra-municipal (collection) and extra-municipal (transport, processing, landfilling) segments – the main financial risks are concentrated in the second.[1]

→ The landfill levy – designed as an incentive for recycling – creates the "pay twice" paradox: sorting, which extracts less than 10% of material, increases rather than reduces total municipal costs

under the current tariff structure.[23]
→ Real figures are dramatic: disposal costs in the Western Galilee Cluster nearly doubled in one month after the closure of the Evron landfill.[4]
→ The price of landfilling is formed without unified regulatory control – a combination of levy, gate fee and ILA royalties that no one coordinates as a whole.[67]
→ The 2026 reform explicitly proposes abolishing the landfill levy and replacing it with regulated full-cost tariffs – but critics note this removes the only existing environmental incentive without a guarantee of an adequate replacement.

The next chapter continues this theme from a different angle: not how much the system costs overall, but who specifically pays – arnona, contractors, TAMIR – and how these money flows relate to each other.

Chapter 20. Who Pays: Arnona, Contractors, TAMIR and Residents

"Ask a resident who pays for their waste collection and they will most likely answer: 'I do, through arnona.' This is only partly true – and this incompleteness is precisely why the system is systematically short of money."

Chapter 19 showed how much the system costs and why that cost is growing faster than seems logical. This chapter traces the money further – who specifically pays at each stage, and where the gaps in this chain arise

1. Four Sources of Money in the System

Israel's waste management system is financed from four largely independent sources that are rarely examined together.

Four sources of financing:

1. Arnona (municipal property tax). The municipality's primary revenue source, from which intra-municipal waste collection is financed and the gap between real costs and any external compensation is covered.

2. TAMIR payments to municipalities. Compensation for organising source separation of packaging – partial, as shown in Chapter 14, and not covering full costs.

3. *Damei tipul* from producers and importers. Money that producers pay TAMIR for fulfilling their obligations under the Packaging Law – the source from which TAMIR in turn pays municipalities (point 2).

4. Consumer deposits. A separate, parallel financial circuit for packaging entering the deposit system (Chapter 15) – does not directly intersect with the three previous sources.

2. Arnona: What It Actually Covers

The municipality is legally obliged to organise waste collection and removal (Chapter 1) – and finances this primarily from arnona, a general property tax with no earmarked allocation specifically for waste.[8]

This means: waste expenditure competes for budget with any other municipal need – education, public space, social services. When landfilling costs rise (Chapter 19), this is not an abstract burden on the "waste system" – it is direct pressure on the entire municipal budget.

◆Under the Hood: Why This Creates Inequality Between Municipalities

Wealthy municipalities with a high tax base can absorb rising waste expenditure without visible reduction in the quality of other services.

Municipalities with a low tax base – which often correlates with the geographic inequality examined in Chapter 7 – face a direct choice between sanitation service quality and other basic needs.

The State Comptroller documents that municipalities spend on average approximately 7% of their regular budget on solid household waste management – but this average figure conceals significant variability depending on geography, building density and distance from landfills.

3. TAMIR: Partial Compensation, Not Full Financing

TAMIR acts as a financial intermediary: it receives *damei tipul* from producers and importers under the Packaging Law, and transfers part of these funds to municipalities as compensation for organising source separation.[9]

The size of this compensation is structurally dependent on which stream the material was collected from. Tariff data shows a gap of almost five times between the commercial and household stream for PP plastic – 135 NIS/tonne versus 642 NIS/tonne.[10]

This gap is not an arbitrary number. It reflects a real difference in collection cost (higher cost per tonne for households due to smaller volumes per point and more points), but also explains why TAMIR systematically favours the commercial stream (Chapter 14): it is simultaneously cheaper to service and sufficient to meet statutory tonnage targets.

◆Who Ultimately Covers the Gap

If TAMIR's compensation does not cover the full cost of organising source separation in the household sector, the difference falls on the municipal budget – ultimately on arnona. This is the "hidden subsidy" first described in Chapter 14: packaging producers are legally obliged to finance recycling, but in practice part of this obligation is shifted to the broad taxpayer through arnona. The precise size of this gap is not published systematically in the public domain by either TAMIR or most municipalities – making it a concrete target for a civic data request (Chapter 25).

4. The Deposit System: A Parallel Circuit

Unlike the three previous sources, the deposit system (Chapter 15) is financially almost entirely ring-fenced. The consumer pays the deposit at purchase and recovers it on returning the packaging – the municipal budget is not directly involved in this chain.

This is a structural advantage of the deposit system from a municipal finance perspective: it creates no additional burden on arnona. But precisely for this reason, expanding the deposit (as in 2021) does not relieve the municipality's financial burden in relation to the TAMIR stream – it simply removes some material from that stream without changing the compensation structure for what remains.

5. The Flow Map: Where a Resident's Money Goes

Resident's payment	Where it goes	What it covers
Arnona (the portion relating to sanitation)	Municipal budget	Intra-municipal collection; gap between TAMIR compensation and real costs; administrative expenses
Price of goods (includes *damei tipul* embedded by the producer)	TAMIR via producer/importer	Partial compensation to municipality for source separation of packaging; TAMIR operational costs
Deposit on purchase of a drink	Deposit system operator (ELA/Asofta)	Deposit refund on container return; deposit system operational costs – entirely separate

Resident's payment	Where it goes	What it covers
		from municipal budget

This table is a practical answer to a question that is rarely clearly formulated: a resident pays for the waste system three times – through arnona, through the price of goods (indirectly, through *damei tipul* embedded in the price), and through the deposit on drink purchases. These three payments flow through different channels, finance different parts of the system, and are nowhere aggregated into a single transparent bill for the resident.

6. What Changes Under the 2026 Reform

Note: connection to the 2026 reform (→ Chapter 16)

The Ministry of Finance reform (Chapter 16) proposes creating a single national regulator with powers to set regulated tariffs on the "closed market" principle – potentially bringing today's fragmented flows (arnona, TAMIR compensation, landfill gate fees) into a more transparent and coordinated system.

The parallel MoEP memorandum on the Packaging Law (Chapter 17) directly addresses part of the problem described in this chapter: the requirement that a minimum of 50% by weight of a producer's recycled waste come from the household sector – structural pressure on TAMIR to compensate household collection more fully, rather than optimising towards the commercial stream.

If the Ministry of Finance reform is implemented in full, abolishing the landfill levy (Chapter 19) and the Cleanliness Fund will require a new mechanism for financing alternative solutions (sorting, recycling, WtE) – an open question, the answer to which had not been determined at time of publication.

Regardless of outcome, transparency about who pays how much at each stage is precisely what the reform's proposed mandatory annual public reporting by the new regulator is intended to achieve. This is a direct link to the civic data-request tools examined in Chapter 25.

Chapter Summary

Key conclusions:

→ System financing comes from four largely independent sources: arnona, TAMIR payments to municipalities, *damei tipul* from producers, consumer deposits.
→ Arnona is a non-earmarked municipal tax, meaning waste expenditure directly competes with other municipal needs and creates inequality between wealthy and less wealthy municipalities.[8]
→ TAMIR compensation is structurally lower for the household stream than for the commercial stream – a tariff gap of almost five times that directly explains TAMIR's preference for the cheaper commercial collection (Chapter 14).[10]
→ The deposit system is financially ring-fenced from the municipal budget – a structural advantage, but also the reason why its expansion does not relieve the TAMIR compensation burden for the remaining stream.
→ A resident effectively pays for the system three times – through arnona, through the price of

goods, and through the deposit – with no single transparent bill aggregating these three flows.
→ The 2026 reform and the parallel MoEP memorandum each, in different ways, address the financing fragmentation described in this chapter – the final architecture remains an open question.[9][10]

◆Financial Questions for a Civic Data Request

1. How much of my municipality's arnona is actually spent on waste management – and what share goes to covering the gap between TAMIR compensation and real costs?
2. Has the cost of waste disposal for my municipality changed over the past two years – and if so, by how much and why?
3. Does my municipality pay ILA royalties or additional charges to landfill operators above the established landfill levy?
4. Does TAMIR publish data on the difference in compensation between the commercial and household stream for my region?

This concludes Part V – money, contracts and incentives. The next chapter moves from diagnosing the financial structure to concrete instruments: how contracts between the municipality and contractors can be structured to reward residual waste reduction, rather than simply paying for tonnes collected.

Sources for Chapters 19–20

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3. Knesset Research and Information Center. Estimate of Economic Cost from Waste Burning. Landfill levy revenues to the Cleanliness Fund. fs.knesset.gov.il
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9. Infospot. "A Decade of the Packaging Law." November 2023. TAMIR coverage and payment to municipalities. infospot.co.il

10. Infospot. Treatment fee data 2023–2024. TAMIR tariff differential commercial vs household stream. infospot.co.il/n/_Packaging_Law

Footnotes

1. Dyellin Academic College (see source 1). Two-segment cost structure: intra-municipal and extra-municipal. ↩ ↩[2]

2. Cleanliness Law (Israel), 1984, Art. 15 (see source 2). The levy is technically paid by the landfill operator and passed through to the municipality via contractor tariffs. ↩ ↩[2] ↩[3]

3. Knesset Research and Information Center (see source 3). The levy was designed to create a financial disincentive for landfilling; revenues go to the Cleanliness Fund. ↩ ↩[2] ↩[3]

4. Infospot / Knesset Committee hearing (see source 4). Karmiel Mayor Moshe Koninsky: costs rose from 260 to over 500 NIS/tonne from January 2025 after Evron closure. ↩ ↩[2] ↩[3]

5. State Comptroller of Israel, July 2025 (see source 5). Specific landfill capacity data: Afe'a, Hagal, Evron, and longer-term planned expansions. ↩

6. Haaretz, January 2025 (see source 6). Union of Local Authorities demands cancellation of ILA royalty mechanism; estimated burden ~150 million NIS/year nationally. ↩ ↩[2] ↩[3]

7. Smart Waste (see source 7). MoEP has not established price control over gate fees; referred to Inter-Ministerial Price Committee. ↩ ↩[2]

8. Cleanliness Law Art. 13 / Local Councils Order Art. 146(8) (see source 8). Arnona is a general property tax with no earmarked waste allocation. State Comptroller documents municipalities spend ~7% of regular budget on solid waste. ↩ ↩[2]

9. Infospot, "A Decade of the Packaging Law" (see source 9). TAMIR collected 439,000 tonnes in 2022; 254 of 257 local authorities under contract. ↩ ↩[2]

10. Infospot, treatment fee data 2023–2024 (see source 10). PP plastic: 135 NIS/tonne commercial vs 642 NIS/tonne household. Metal packaging: 115 NIS/tonne commercial vs 229 NIS/tonne household. ↩ ↩[2] ↩[3]

Chapter 21. Contracts, KPIs and PAYT: From Tonnes to Residual Waste Reduced

"'The tender must define the expected outputs and not the processes.' This phrase from the Neaman Institute report sounds obvious. But applied to Israeli waste collection contracts, it describes a gap that explains a great deal."

Chapters 19 and 20 examined the financial structure of the waste system – how much it costs and who pays. This chapter concludes Part V by asking the question that follows logically from the two previous: if money exists in the system (albeit chronically insufficient), how exactly does it "speak" to those making operational decisions? What do existing contracts actually incentivise – and what do they not?

The answer lies in how KPIs (key performance indicators) are structured in contracts between municipalities and contractors – and why in their current form they measure primarily the quantity of waste collected, not the quality of how it is handled.

1. The Problem: Contracts Pay for Tonnes, Not for Outcomes

Most Israeli municipal waste collection contracts are structured as volume-based service contracts: the contractor receives payment for the number of trips, tonnes collected, or a combination of both.[1]

This sounds reasonable – but creates a structural incentive towards quantity rather than quality.

◆What "Pay-per-Tonne" Contracts Incentivise – and What They Do Not

Incentivise: as many trips as possible, trucks as full as possible. The contractor's interest is to collect a large volume.

Do not incentivise: reducing stream contamination – the contractor is indifferent to how clean the contents of the orange bin are. A clean bottle and a dirty plastic bag count equally in the tonnage.

Do not incentivise: reducing total waste volume – the contractor earns less if there is less waste.

Direct consequence: the contractor is neutral or hostile towards programmes that reduce waste generation or improve source-separation quality. Its financial model is built on volume.

The State Comptroller documented in 2022: among the local authorities examined, performance measurement against waste-reduction targets was limited or absent; contracts focused on collection frequency and vehicle availability rather than environmental outcomes.[2]

2. The Alternative: Outcome-Based Contracts

The Neaman Institute, analysing the feasibility of PPP contracts for large waste treatment facilities in Israel, formulated a key principle: "The tender must define the expected outputs and not the processes."[3]

Applied to collection and removal contracts, this means shifting from "how much was collected" to "how much residual waste was generated," "what share of sorted material comes from the total stream," or "what is the quality of the stream entering the sorting facility."

KPI "per tonne"	KPI "per outcome"
Number of trips per collection point	Percentage of correctly sorted material from the collection point's stream
Tonnage of mixed stream collected	Year-on-year reduction in mixed stream (residual waste) tonnage
Equipment availability (% working days)	Rejection rate of orange bin material at the sorting facility
Response time to 106 complaint	Stream contamination (% non-sortable items in dedicated bins)
Technical compliance with routes	Citizens actually using source-separation bins (measured by sampling)

◆Under the Hood: International Context – How Other Countries Do This

Germany and the Netherlands require contracts with explicitly stated targets for improving recycling performance – the contractor faces penalties if performance falls below the threshold.[4]

The UK is transitioning to outcome-based performance contracts in a number of municipalities: a minimum recycling percentage is enshrined in the contract, not just in municipal intentions.[4]

Taipei implemented PAYT (see below) precisely through changing incentives at the consumer level – and achieved a 35% reduction in waste volume and a 2.6-fold increase in recycling.[5]

OECD directly recommends: municipal waste contracts should include targets for reducing residual waste and increasing recycling rates, not only volume and service parameters.[4]

3. PAYT – "Pay As You Throw": The Principle and Its Limitations

What PAYT Is

PAYT (Pay As You Throw) is a system under which residents and organisations pay for waste management in proportion to how much they actually discard: by weight, volume, or number of special bags or containers.[5]

The logic is simple: if you pay for each unit discarded, you receive a direct financial incentive to generate less waste and sort more (since recycling is usually cheaper or free).

Three main PAYT models:

Model 1: Special bags. Residents purchase official numbered bags for mixed waste; sortable materials are deposited in standard containers for free. Popular in South Korea and Taiwan.

Model 2: Chipped containers with scales. Each household has an RFID-tagged container; every time it is emptied, weight or number of lifts is automatically measured. The bill reflects actual use. Used in the Netherlands and parts of Belgium.

Model 3: Partial pricing. A base volume of collection is included in a fixed charge; anything above that is charged additionally. A compromise that reduces the risk of social resistance.

Why PAYT Works – and What It Requires

Taiwan (Taipei) – the best-documented example: the introduction of special bags led to a 35% reduction in waste volume and a 2.6-fold increase in recycling. Key enabling conditions: clear and simple rules, parallel coverage of all source-separation access points, and neutralisation of increased costs for low-income households (who statistically generate less waste).[5]

◆The Main Risk of PAYT: Illegal Dumping

PAYT creates a direct incentive against paying – and therefore potentially against legal disposal. Illegal dumping ("fly-tipping") and waste burning have been documented to increase in a number of municipalities following PAYT introduction without a sufficient monitoring and enforcement system.[5]

The Knesset Research and Information Center's document on the economic costs of waste burning explicitly warns: the rising cost of landfilling (Chapter 19) already creates an incentive for illegal burning. PAYT may intensify this effect if not accompanied by a proportionate enforcement system.[6]

In Israel's context, where waste management enforcement is already fragmented (Chapter 2), introducing PAYT without first strengthening the monitoring and penalty system is potentially counterproductive.

4. PAYT in Israel: An Honest Assessment of Prospects

At the time of writing, no systematic PAYT exists in Israeli municipalities. Arnona is a non-earmarked tax based on floor area, not on volume discarded. This means there is no direct financial incentive for the resident towards waste reduction under the current system.

Can PAYT be introduced in Israel? Technically – yes. Organisationally and politically – it is a more complex question.

Factor	**In favour of PAYT**	**Against / requires resolution**
Legal basis	Municipalities have powers to set charges for municipal services	Requires changing an arrangement traditionally embedded in arnona – politically sensitive
Monitoring infrastructure	RFID systems for containers are already being tested (Chapters 4, 5)	No national household-level waste measurement system
Social justice	PAYT is documented to be fairer for lower-income residents (less waste = less paid)	Requires protective mechanisms for households facing genuine hardship; perception as a new tax
Illegal dumping	Manageable with sufficient monitoring	Current enforcement of illegal burning and dumping is insufficient for PAYT (Chapter 7)

Factor	In favour of PAYT	Against / requires resolution
2026 reform	Creating a regulator potentially simplifies PAYT standardisation	Until the reform is adopted, no national platform for a pilot

Conclusion: PAYT is not an imminent but a potentially achievable instrument for Israel given certain preconditions. The priority for the next 3–5 years is not full-scale PAYT but a more realistic step: transitioning from volume-based KPIs to outcome-based KPIs in existing contracts.

5. What Can Change Right Now: A Realistic Plan

There is no need to wait for the 2026 reform or the introduction of PAYT to improve incentives in the system. A number of changes are possible at the level of an individual municipal contract right now.

Four elements of an improved waste collection contract:

1. A target indicator for residual waste reduction.
Example: "The contractor undertakes to contribute to a 5% year-on-year reduction in residual mixed waste tonnage in the serviced area." This is not 100% within the contractor's control – but its presence in the contract changes the incentive.

2. An incoming stream quality indicator for the orange bin.
Example: "The rejection rate at the sorting facility for this district's stream shall not exceed X% – if exceeded, a payment adjustment is triggered or additional resident education obligations apply."

3. Data as part of the service.
The contractor must provide quarterly data on tonnage by stream, observed violations (what ended up in the wrong bin), and statistics on 106 complaints. Without data, meaningful oversight is impossible.

4. Diversified payment: fixed + variable.
The base portion of payment covers service provision (fixed). The variable portion covers achieving quality targets and residual waste reduction (bonus or penalty). This does not abolish volume pricing but adds a results-based component to it.

5. **A verified prevention indicator.**

A fifth element should be added where prevention infrastructure exists or can be piloted: a verified prevention indicator. The municipality should not measure only how many tonnes were collected, sorted or rejected, but also how many potential tonnes were intercepted before collection. This can include repaired items, redistributed goods, bulky objects moved to reuse rather than disposal, edible food shared, and organics processed locally through composting or digestion. Without such an indicator, municipal waste management remains a system for handling waste after it has appeared. With it, prevention becomes visible, contractable and reportable.

6. How Citizens and NGOs Can Influence Contracts

A contract between a municipality and a waste collection contractor is a public tender. In Israel, tender specifications are publicly published, and adopted contracts are requestable under the Freedom of Information Law (Chapter 25).

This opens specific leverage points for civic organisations.

◆Questions About Contracts: What to Request and What to Demand

1. Does my municipality's current contract include any KPI for reducing residual waste or improving source-separation quality? (Request a copy of the contract under the FOI Law)

2. What is the rejection rate at the sorting facility for the orange bin stream from our district – and is there a penalty in the contract if it is exceeded?

3. Is the contractor required to provide quarterly data on tonnage by stream?

4. When does the current contract end and when is the next tender announced? (The tender moment is the maximum point of civic participation)

5. Has my municipality published the results of the previous contract: did the contractor meet its KPIs?

6. Is a PAYT pilot or partial volume-based payment being considered as part of the next contract?

Note: connection to the 2026 reform (→ Chapter 16)

The 2026 reform (Chapter 16) proposes creating a national regulator setting regulated tariffs on the "closed market" principle. If implemented in full, this means the possibility (and necessity) of standardising KPIs across all municipal contracts through a unified tariff system.

The parallel MoEP memorandum (Chapter 17) explicitly proposes that a minimum of 50% by weight of a producer's recycled packaging waste come from the household sector. This is, in effect, a KPI embedded in the EPR system – one that creates pressure on TAMIR, which creates pressure on municipalities, which creates pressure on contractors towards quality rather than volume alone.

Key conclusion: if waste collection KPIs emerge in Israel, they are more likely to come through EPR reform (top-down) or through requirements of a new regulator, rather than bottom-up through voluntary changes to individual contracts – although the latter is also possible and creates demonstration examples.

Chapter Summary

Key conclusions of Chapter 21:

→ Most Israeli waste collection contracts are volume-oriented rather than outcome-oriented – which structurally does not incentivise waste reduction or improved sorting quality.[12]
→ Key principle: the tender must define expected outputs, not processes. The transition from volume KPIs to outcome-based KPIs is possible at the level of an individual contract without waiting for national reform.[3]
→ PAYT is a systemically proven instrument (Taipei: -35% waste, ×2.6 recycling), but requires

preconditions in Israel: developed monitoring, strengthened enforcement, and protective mechanisms for vulnerable households.[5]
→ Four elements of an improved contract are applicable right now: a residual waste reduction target, a rejection rate KPI, a data-provision obligation, and diversified payment with a variable component.
→ The tender moment is the maximum civic leverage point. Tender specifications are public, contract copies are requestable, participation in public tender consultations is lawful.

Concluding conclusions of Part V: Money, Contracts and Incentives:

→ Israel's waste management financial system is structurally misaligned with environmental goals: the landfill levy penalises intermediate steps towards recycling; TAMIR compensation incentivises commercial rather than household collection; contracts reward volume, not quality.
→ The 2025–2026 crisis is not only a capacity crisis (Chapter 6) but also a financial incentives crisis: the system is designed so that rationally acting actors at every level make choices that collectively reproduce the problem rather than solve it.
→ The 2026 reform potentially addresses all three levels (landfill levy, EPR compensation, regulated tariffs) – but its outcome remains an open question.
→ Citizens and NGOs have concrete leverage points: data requests on costs (Chapter 25), participation in public consultation on reform (Chapter 16), and monitoring of tenders – this is where contracts and KPIs for the next 5–10 years are being set.

Part VI moves from money to alternative practices: organics, goods streams and circularity hubs – tools that exist not to wait for systemic reform, but to begin changing the system from below right now.

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2. State Comptroller of Israel. Municipal Government Audit. 2022 (see source 2). Among examined local authorities, performance measurement against waste-reduction targets was limited or absent; contracts focused on collection frequency and vehicle availability rather than environmental outcomes. ↩ ↩[2]

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Chapter 22. Organics and Food Waste: From a Methane Problem to a Local Resource

"Organics is the largest fraction in the waste bin. It is also the most harmful when landfilled. And the most valuable when handled correctly. The gap between what exists and what is possible is most visible here."

Food and plant waste accounts for approximately 40–45% of all solid household waste in Israel by weight. It is the single largest fraction – more than plastic, cardboard and glass combined. And it is precisely this fraction that the waste management system handles worst of all.

This is not accidental. Organics is a chemically active material. Under anaerobic landfill conditions it decomposes releasing methane – a greenhouse gas approximately 28 times more potent than CO_2. It is estimated that proper treatment of Israel's organic waste could reduce national greenhouse gas emissions by approximately 5.5%.[1]

This chapter describes the problem, the available technologies, and where workable solutions are beginning to emerge in Israel.

1. The Problem: Organics in the Mixed Bin

The vast majority of Israeli households discard organics in the mixed (green) bin. This organic material makes up a significant portion of the 60% of mixed bin content that, per MoEP's 2025 scan (Chapter 8), could technically be handled differently.

◆Why Organics in the Mixed Bin Is a Double Loss

First loss: organics contaminate all other fractions in the mixed bin – plastic, cardboard and metal in the same container become wet and dirty, reducing their recyclability.

Second loss: organics landfilled decompose anaerobically and release methane. Most Israeli landfills are not equipped with full landfill gas capture systems, meaning a significant proportion of methane enters the atmosphere.

The alternative – anaerobic digestion or composting – turns organics into a resource (biogas, fertiliser) rather than an environmental threat. This is not a distant prospect but a technology operating in dozens of countries and beginning to appear in Israel.

2. Technologies: What Science Can Do

Anaerobic Digestion (עיכול אנאירובי)

Anaerobic digestion is the decomposition of organic material without oxygen, under the action of microorganisms. It is a natural process used in industrial installations called bioreactors or digesters.[2]

What goes in and what comes out:

INPUT: household organic waste, food waste, garden waste (*gezem*), sewage sludge, manure.

PROCESS: in a sealed bioreactor at controlled temperature and humidity, microorganisms decompose organics over several weeks.

OUTPUT – two valuable products:
→ **Biogas (ביוגז):** a mixture of methane (~60%) and CO_2. Used directly as fuel or for electricity and heat production.
→ **Digestate (חומר מעוכל):** nutrient-rich organic matter. When input stream quality is high (source-separated at origin) – a valuable agricultural fertiliser. From a mixed stream – quality is lower and market demand is limited.

◆Under the Hood: Why Input Stream Quality Determines Everything

MoEP explicitly prefers anaerobic digestion over simple composting – as a more climate-valuable process (captures methane rather than releasing it) and as a source of renewable energy.

But there is a fundamental constraint: if organics have been mixed with plastic, glass and metal, the digestate will contain their fragments. Such a product cannot be used as agricultural fertiliser and has no market. In Europe it is designated CLO (Compost-Like Output) – material that has technically undergone processing but is ecologically unusable.

Conclusion: anaerobic digestion works well only with source-separated organics (brown bin). From a mixed stream, even after mechanical sorting, output quality is critically reduced.

This is why Israeli MBT (mechanical-biological treatment) installations processing the mixed stream produce low-quality digestate – not due to a technology failure, but due to input material quality.[2]

Composting (קומפוסטציה)

Composting is an aerobic (oxygen-present) decomposition process for organics. The result is compost – a nutrient-rich organic fertiliser. Technologically simpler and cheaper than anaerobic digestion, but produces no energy.

Characteristic	**Anaerobic digestion vs Composting**
Product 1	Biogas (energy) | None
Product 2	Digestate (fertiliser) | Compost (fertiliser)
Input stream quality requirement	High (source separation) | Medium
Installation cost	High | Medium
Land area	Compact (closed reactor) | Large (open rows)
Odour emissions	Low (closed process) | Significant if unmanaged
MoEP Israel position	Priority | Acceptable as supplement

3. What Exists in Israel Now: An Honest Picture

The situation in Israel in 2025–2026 is a combination of considerable potential, limited existing infrastructure, and two new major facilities under construction.[3]

Current state of organic infrastructure in Israel:

OPERATIONAL CAPACITY (~530,000 tonnes/year):

→ **Tovlan** (Beka'at Hayarden): the largest facility, receiving ~90% of all organics treated in the country. Operates on primitive open composting rows. Product: CLO, not standard compost. Facility under MoEP investigation.

→ **Dudaim:** second operational facility; composting.

UNDER CONSTRUCTION (opening ~2026–2027):

→ **Zero Waste Shafdan, Rishon LeZion:** anaerobic digestion, 400,000 tonnes/year. Largest facility of this type in the country.

→ **Dia (Tzealim, Eshkol Regional Council):** GES/YSB, 250 million NIS, 200,000 tonnes/year. 25-year concession. Combination: anaerobic digestion + composting of digestate in sealed concrete chambers.[4]

THE GAP:

After the new facilities open, total capacity will reach ~1.2 million tonnes/year.
Israel's actual annual organic waste generation: ~3 million tonnes/year.
Capacity deficit for meeting 2030 targets: 7–12 additional facilities still required.

◆A Disturbing Precedent: What Happened to Organics During the Veridis Crisis

In 2022 MoEP imposed restrictions on Veridis – the largest organic processing operator. Veridis halted intake of organics.

Immediate result: capacity for organic waste treatment in the country virtually disappeared. Sorting facilities were forced to send the organic fraction directly to landfill, paying the full landfill levy rate.

This exposed a systemic problem: dependence on one or two facilities with no reserve capacity creates fragility across the entire system. A single major player and a single major facility is not a system – it is a single point of failure.[3]

4. Small-Scale Formats: What Works Now

Not all solutions require large infrastructure. A number of Israeli municipalities and clusters have implemented small-scale organic formats operating close to the point of waste generation.

Kibbutz Yagur (יגור): a small anaerobic digester converts kitchen food waste into biogas for water heating and fertiliser for kitchen gardens. This is the first documented example of "on-site" community-scale urban organic waste digestion with biogas production in Israel.[5]

Zichron Ya'akov (זכרון יעקב): food waste is transported to an anaerobic facility in the Hefer Valley; garden waste goes to a separate composting facility. A local facility for composting garden waste and high-quality organic waste at source is planned.[6]

These examples show: small scale, orientation towards source separation, and proximity to the point of waste generation are the conditions under which organics becomes a resource rather than continuing to be landfilled. This is the principle described in Chapter 24 on circularity hubs: local infrastructure close to the resident.

5. What Is Needed for This to Work Systemically

Academic literature and MoEP documents agree on one point: the key condition is source separation of organics (separation in the brown bin at the resident level), not mechanical extraction from the mixed stream.[2]

This requires the greatest investment – not in technologies but in first-mile infrastructure (Chapter 4): convenient brown bins in every courtyard, frequent collection (organics decompose quickly), and clear labelling.

Note: connection to other chapters
MoEP's 2025 data (6% of packaging in dedicated bins) shows that the first-mile problem (Chapters 4 and 8) applies to organics with the same force as to plastic. Expanding brown bin coverage is an infrastructure project, not just an educational campaign. The 2026 reform (Chapter 16) potentially opens funding through a new regulator for accelerating the construction of organic treatment capacity – currently financed primarily through the Cleanliness Fund, whose future is uncertain under the reform.

Chapter Summary

Key conclusions:

→ Organics is the largest waste fraction (~40–45%) and the most harmful when landfilled (methane) – and simultaneously the most valuable when handled correctly (biogas + fertiliser).[1]
→ Anaerobic digestion is MoEP's preferred technology: produces biogas and digestate, but requires source-separated collection for high output quality.[2]
→ Current infrastructure: ~530,000 t/year capacity against a need of ~3 million t/year. The deficit is critical even accounting for facilities under construction (Shafdan + Dia).[34]
→ Dependence on one or two facilities creates systemic fragility – demonstrated by the Veridis crisis of 2022.[3]
→ Small local formats (Yagur, Zichron Ya'akov) show that workable solutions are possible at community level.[56]
→ The systemic solution requires investment in the first mile: brown bins at source, not only large processing facilities.[2]

Chapter 23. Clothing, Toys, Furniture: Ethical Waste and Early-Stage Rescue

"A wardrobe left by a skip is not a waste problem. It is a problem of a system that has no mechanism to intercept an item before it becomes waste."

Among all the waste streams examined in this book, clothing, toys and furniture occupy a special place. They do not "become" waste in the same sense that packaging does after use. In the vast majority of cases – when a person brings them to a container – they still work, are still intact, are still needed by someone.

This is "ethical waste": a category where an item becomes waste not because its useful life is exhausted, but because there is no infrastructure to intercept it before that point.

1. Scale: How Much There Actually Is

Clothing and Textiles

The global scale of the problem is well documented. The fashion industry produces 92 million tonnes of textile waste annually. Clothing production has doubled since 2000 – while the average number of times a garment is worn has halved.[7] Recycling into new clothing accounts for less than 1% of textile waste globally.

In Israel, textile waste is not systematically tracked in national statistics – this is itself a sign of the absence of a systemic approach to this stream.[8]

Furniture and Bulky Items

Furniture and bulky items (*psholet gushit*, פסולת גושית) are a separate legal category under the Cleanliness Law. The municipality is obliged to organise their removal, but the frequency and collection system varies significantly.[9]

Per State Comptroller data (2022), bulky items left near containers – furniture, appliances – are among the most frequent complaints on the 106 line. This is a symptom: the resident has no convenient channel for properly transferring these items.[9]

2. What Happens to Items: Three Paths

Path	What it means
Early-stage rescue – the ideal scenario	The item is intercepted while still functional: given to acquaintances, deposited at a themed organisation or hub, sold via second-hand platforms. Lifecycle extended without any processing.
Repair and restoration	The item is broken but repairable. Repair cafés, cobblers, seamstresses, electricians – with accessible services the item is back in use.
Textile recycling	The item is worn out and cannot be passed on. Possible conversion into wiping material, wadding, filling. Technically possible; infrastructure is limited and volumes small.
Landfill or incineration	The item ends up in the mixed container or is left by a skip. For furniture –

Path	What it means
– the failure scenario	landfill. For clothing – either landfill, or if lucky, it ends up in a charitable organisation's container.

◆Under the Hood: Charitable Organisation Containers – Not Part of the Official System
In Israel, clothing collection containers placed by organisations such as Yad Sarah, Second Hand, Sela and others are widespread. They are NOT part of the TAMIR system and are NOT included in the official recycling statistics published by MoEP or TAMIR.

This means the real scale of textile "rescue" in Israel is systematically underestimated – the data simply is not collected.

On the other hand, part of the material from these containers is exported to Africa and Asia within the "second-hand clothing" stream – which is itself contentious: there are documented negative consequences for local textile markets in recipient countries.

3. The First Stage – "Early-Stage Rescue": The Principle

Early-stage rescue is an intervention before the item reaches a container. This is fundamentally different from recycling (which operates after): here the concern is interception at the stage when the item still has value, still requires no processing, and can be immediately transferred to the next user.

Three levels of early-stage rescue:

Level 1: P2P (person to person). Apps and platforms (Yad2, Facebook Marketplace, KSP, local neighbourhood chats). Minimal infrastructure, maximum speed. Constraint: requires time from the "donor."

Level 2: Organised drop-off points. Second-hand shops (*chanuyot yad shniya*), charitable organisations, circularity hubs (Chapter 24). Requires a physical point within walking distance and a regular operating schedule.

Level 3: Corporate buyback/extended life. Producers offering to take back their own products (example: Keter Group – collecting used plastic garden furniture for recycling into new products). Still rare in Israel.[10]

4. Re-Haifa Lab: Experience and Lessons

Re-Haifa Lab has operated since 2018 in Haifa as an organisation specialising in early-stage rescue of textiles and furniture, repair, and community swap events.[11]

Over several years of work, practical lessons have crystallised that are applicable to any similar initiative.

What Re-Haifa learned in practice – 2025 update:

Textiles and small items are the most accessible entry point. Removing a sofa requires transport; accepting a bag of clothing requires two volunteers. Starting with textiles is almost always the right choice.

Furniture requires logistics. Without an agreement with the municipality or an organisation on transport – impossible. The most realistic scenario: coordinating with the municipality around scheduled bulky waste collection days, so the hub can arrive first.

Repair is the highest-value service, the most difficult to scale. A repair café works if there are regular skilled practitioners. Enthusiastic volunteers are a necessary but not sufficient condition.

Data matters. Without tracking the number of items, categories and ultimate fate of each item – it is impossible to demonstrate real impact to the municipality or a grant-funder. Re-Haifa has maintained this tracking since 2022.

Collaboration with other NGOs is a multiplier. Transferring items that the hub cannot process to partner organisations (youth clubs, care homes, social services) extends reach without increasing costs.[11]

5. What Does Not Work: An Honest Picture

It would be dishonest to examine this topic only from the perspective of successes. A number of persistent barriers significantly limit the scale of early-stage rescue in Israel.

Barrier	Why it exists and what to do about it
Absence of data in official statistics	The textile stream is not regulated by the Packaging Law and not tracked by TAMIR. Without data – no policy. A separate category in national MoEP statistics is needed.[8]
No statutory obligation for clothing producers	The EU has EPR for textiles (from 2025). Israel does not. The producer bears no responsibility for the fate of their product.
Low quality of the "rescued" stream when collection is disorganised	Items left by a container in an open bag in the rain quickly lose value. Closed, clearly marked drop-off points are needed.
Furniture requires freight transport and storage space	For an NGO without municipal support – often an unsolvable logistical problem.
Export of second-hand clothing is not a solution	Some "rescued" items from Israel enter export streams to countries with weaker regulation – this is not closing the loop, it is shifting the problem.

Chapter Summary

Key conclusions:

→ Clothing, toys and furniture are "ethical waste": they become rubbish primarily due to the absence of interception infrastructure, not due to exhausted useful life.
→ Early-stage rescue is a higher priority than recycling: intercepting an item before the container is cheaper, more ecological and more effective than processing it afterwards.
→ The textile stream in Israel is not tracked in official statistics – this is structural invisibility,

meaning the absence of policy.[8]
→ Charitable organisation containers are not part of the official EPR system – their contribution is not included in recycling figures.
→ Re-Haifa demonstrates a working model of early-stage rescue at city level, with documented lessons about real barriers (logistics, data, scaling repair).[11]
→ A systemic solution requires: separate tracking of the textile stream, statutory producer responsibility (following the EU EPR for textiles model), and municipal support for organising drop-off locations and logistics.

◆Practical Questions for the Activist

1. Are there organised clothing and furniture drop-off points with a known schedule in your city – or only containers in inconvenient locations?
2. When is bulky waste collection scheduled in your neighbourhood – and can this be used as an opportunity for hub interception?
3. Does MoEP publish data on textile waste separately – or is it hidden in the "other waste" category?
4. Is there a repair café or workshop operating on a regular basis in your city?

The next chapter moves from working with individual streams to the infrastructure that brings them together in one place: circularity hubs and the principles of building a local circular economy.

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Chapter 24. Before It Becomes Waste: Circularity Hubs as Local Prevention Infrastructure

" Circularity is not a personal virtue. It is infrastructure. And the best moment to manage waste is before it becomes waste."

There is a point at which the formal waste system usually begins: a bag is placed in a bin, a truck arrives, a contractor records a route, and the material enters a chain of transfer stations, sorting facilities, landfills, recyclers or energy-recovery plants. By that point, however, many better options have already been lost.

A broken chair has become bulky waste instead of a repair task. Extra food has become organic waste instead of a shared resource. Old clothes have become textile waste instead of second-hand circulation or upcycling material. Kitchen scraps have become a methane problem instead of compost. A repairable appliance has entered the residual stream because there was no affordable repair option within reach.

A circularity hub is the missing layer before this happens. It is a physical place and a set of services that stand between everyday household life and the formal waste collection system. Its purpose is not to ask residents to be more conscious in the abstract, but to make prevention, repair, reuse, sharing, composting and correct downstream sorting practically available.

This chapter describes the circularity hub as a model of local prevention infrastructure: not a volunteering project, not a substitute for municipal waste services, and not a decorative community activity, but a civic mechanism for reducing the amount of material that reaches the waste system in the first place.

1. Why Personal Awareness Is Not Enough

Imagine a resident who wants to do the right thing before throwing something away. They want to repair a broken toaster, pass on a child's toy, share edible food, compost kitchen scraps, or move a usable table to someone who needs it. In many neighbourhoods, none of these actions is simple. There may be no repair service nearby, no organised exchange point, no food-sharing channel, no local composting option, and no affordable way to transport a bulky item.

The toaster, the toy, the food, the table and the kitchen scraps enter the waste stream not because residents failed morally, but because the necessary service layer was absent. This is not a problem of awareness alone. It is a problem of access, proximity, affordability and infrastructure.

The same applies to organics. A person knows that food waste should not go in the grey bin. But there is no brown bin in their courtyard. The nearest composting point is in a park a 15-minute walk away. The organics go into mixed waste.

This is a systemic pattern: we demand behaviour from people for which we have not created conditions. A circularity hub changes the equation – it creates – those conditions.

◆Under the Hood: Infrastructure vs Awareness – What the Research Base Shows

Renee Lertzman (Environmental Melancholia, 2015) shows: most people genuinely care about the environment but experience "environmental melancholia" – a sense of powerlessness before the scale of the problem. Appeals to "be more conscious" intensify this feeling rather than reducing it.

Zero Waste Europe (2020) documents: in areas where community reuse centres operate, the source-separation rate is 18–31% higher than in control areas without such centres. Infrastructure predicts behaviour more reliably than educational campaigns.[1]

OECD (2020) formulates this as the "proximity and convenience" principle: the closer and more convenient a service, the higher its uptake. This is trivial for retail and transport – but is somehow ignored in waste management.

2. What a Circularity Hub Is: Definition and Functions

A circularity hub (neighbourhood zero-waste centre) is a multifunctional place within walking distance of residents that brings several services together in one location.

The key word is "multifunctional." A single repair café or a single drop-off point is good, but insufficient. A hub works as an ecosystem of services that reinforce each other.

Seven functions of a circularity hub:

1. Early-stage rescue

Residents bring items they are "discarding but feel bad about": clothing, furniture, household appliances, books, toys. The hub checks, sorts, and prepares them for reuse. What is suitable stays in circulation. What is not goes into the correct disposal stream – not onto the pavement next to a container.

2. Repair and restoration

A repair café, workshop, or partnership with local craftspeople. Fix a toaster, bicycle, clothing. This not only reduces the waste stream – it creates jobs and skills.

3. Sharing and a library of things

A drill, stepladder, children's bicycle, tent – items needed infrequently. Borrow and return. This reduces consumption of new goods and the burden on storage space.

4. Organics: a controlled stream

A drop-off point for organic waste with on-site composting or transfer to anaerobic digestion. Critical for neighbourhoods without a brown bin.

5. Verified secondary material streams

Acceptance of clean, sorted secondary materials – HDPE, PP, paper, metal – and direct transfer to a recycler, bypassing or complementing the standard TAMIR circuit. The hub knows the downstream fate of its material.

6. Education and information

Workshops, Q&A sessions, visual materials. Not education through shame, but education through inclusion: "here is how this works, here is what happens to your bottle."

7. MRV: measurement, reporting, verification
The hub tracks everything that passes through it: tonnage by fraction, number of items repaired, CO_2 saved. This makes it a measurable partner for the municipality.

None of these functions is revolutionary in itself. What is revolutionary is their combination in one place, with permanent staff and an accounting system.

3. Jobs and the Economics of a Hub

One of the most underappreciated arguments for circularity hubs is employment. According to the European Rreuse network,[2] the reuse and repair sector creates 200–300 jobs per 10,000 tonnes of goods processed – compared to 20–40 jobs from landfilling and 10–15 from incineration.

This is not abstract statistics. A hub in a neighbourhood means:

- A **hub coordinator** – a person who organises the streams, keeps records, communicates with residents and the municipality.
- **Repair technicians** – electrician, cobbler, seamstress, bicycle mechanic.
- **Sorters and operators** – those who receive, inspect and prepare items.
- **Educators and volunteers** – those who run workshops.
- A **logistics coordinator** – if the hub organises transfer of verified streams to the recycler.

Some of these roles are full-time. Some are part-time. Some are voluntary, but with real support. Importantly, all of them are local: money stays in the neighbourhood.

◆Under the Hood: The Hub's Economic Model – Where the Money Comes From

Municipal service contract. The hub provides the municipality with a measurable service: diverting waste from landfill, reducing the burden on the sorting system, educating residents. The municipality pays for this – as it pays for waste collection. This is not a grant, it is a service contract.

Sale of sorted material. Verified streams of clean secondary material are sold directly to recyclers – at a higher price than material from the mixed stream.

Sale of repaired goods. A commission shop or market – some items are sold, with revenue partially covering operating costs.

Cleanliness Fund (קרן הניקיון). A hub with documented stream diversion from landfill can apply for grant support through the Fund.

Social employment programmes. Some roles can be financed through employment programmes for vulnerable population groups.

Important: a hub does not need to be profitable. It needs to be sustainable – meaning diversified financing without dependence on a single source.

4. Re-Haifa as One Possible Model

Re-Haifa Lab is not the only model of a circularity hub and not necessarily the best. But it is an Israeli experience that can be studied, evaluated and adapted.

Founded in 2018 in Haifa, Re-Haifa Lab works in three main areas: early-stage rescue of textiles and furniture, a repair café, and pilot organic waste projects.[3] Over several years, thousands of items passed through the hub that would otherwise have ended up at landfill or next to a container.

But more important than the specific figures is what this experience taught.

What Re-Haifa learned in practice: five lessons

Lesson 1. A physical place matters.
Events "in the air" – markets, festivals – create visibility but not a flow. A permanent point with a known address and operating schedule creates a habit. Residents begin to think: "I have an old lamp – I'll bring it to Re-Haifa."

Lesson 2. Staff is critical.
A hub does not run "by itself." You need a coordinator who opens the door every day, accepts items, and answers questions. Volunteering is necessary but not sufficient: volunteers do not replace permanent staff.

Lesson 3. The municipality is a partner, not a sponsor.
The most sustainable form of collaboration is a service contract, not a grant. A grant ends. A contract obliges both sides. This requires the hub to be able to provide the municipality with measurable KPIs.

Lesson 4. MRV is not bureaucracy – it is survival.
Without tracking tonnage, number of items and CO_2, there is no argument for a conversation with the municipality, a grant-funder or MoEP. A hub without data is invisible to the system.

Lesson 5. Do not replace municipal responsibility.
When a hub works too well, the municipality may start cutting basic services – "they have Re-Haifa after all." This is a trap. The hub complements the system, it does not substitute for it.

5. European Experience: What Works and What Does Not

Circularity hubs have been operating in Europe for 10–15 years – long enough to separate what works from what sounds good but does not scale.[14]

What works	What does not work
Municipal contract with measurable KPIs	Dependence on a single grant-funder
Permanent location with regular schedule	"Events" without a permanent point
Paid coordinator	Only volunteers without a staff core
Integration of organics and secondary materials in	Isolated monofunctional service

What works	What does not work
one place	
Transparent MRV with public reports	Hub without an accounting system
Partnership with local recyclers and craftspeople	Attempting to do everything in-house without partners
Orientation towards resident convenience	Orientation towards "correct behaviour"
Inclusion of vulnerable groups as workers	Only volunteers from the "conscious" audience

◆Under the Hood: Brussels and Amsterdam – Two Formats

Brussels (Neighbourhood Circularity Hubs, 2020–2023): six pilot hubs in different communes. Key condition: each hub must serve at least 5,000 households within walking distance. Result: 1,200–4,500 tonnes processed per year per site; contamination reduction in service zones of 18–31%.[1] *Model: partial municipal financing + material sales + social employment.*

Amsterdam (Neighbourhood Repair Hubs, 2019–2022): 14 hubs, average diversion from landfill of 340 tonnes per hub per year; 60–70% of items brought in successfully repaired or redistributed.[4] *Key success factor: integration into the city's "circular economy strategy 2020–2025" with a 5-year budget commitment.*

Spain (Puntos Limpios ampliados, Zero Waste Europe 2020): more than 40 expanded drop-off points operating as hubs. Key difference from a standard "drop-off point": on-site workshop, trained staff assess item suitability before deciding on disposal.

6. How to Build a Hub: From Pilot to Infrastructure

A hub is not built at full scale immediately. A sustainable model passes through several stages.

Four stages of hub development:

Stage 1. Pilot (6–12 months)
One or two functions. A fixed location, even a few hours per week. Simple tracking: count items and kilograms. Goal: prove to yourself and partners that demand exists.

Stage 2. Extended pilot (12–24 months)
Three or four functions. One permanent coordinator (at least half-time). First agreement with the municipality – at least on use of premises or collection of a verified stream. Start of MRV: tonnage by fraction, number of items repaired/redistributed.

Stage 3. Sustainable model (24–36 months)
All seven functions, or close to it. Municipal service contract with KPIs. Diversified financing. Annual public report with data. Replication: helping other neighbourhoods launch similar models.

Stage 4. Systemic integration
The hub becomes part of the city's waste management infrastructure – a recognised operator with a

formal role in the chain. Its data enters municipal reporting. This is not the end of the journey but a change of status: from "civic initiative" to "local infrastructure."

What is needed to launch a pilot

The minimum set for starting is not as large as it seems:

What is needed	**Comment**
A physical place	A room in a community centre, a corner in a synagogue/mosque/church, garage space in a residential complex. No separate building required.
One responsible person	Does not need to do everything. Must open the door, respond to requests and keep records.
Basic tracking	A spreadsheet: date, item type, weight (if possible), what happened to it (repair / redistribution / recycling / disposal).
One downstream partner	A recycler or charitable organisation that accepts what the hub cannot process itself.
Basic municipal sign-off	Not a contract – a letter of intent or meeting minutes. So the hub does not operate "in the shadows."
Public communication	Schedule and address on the neighbourhood website, in building chats, on notice boards.

7. MRV: How the Hub Makes Its Impact Visible

MRV (monitoring, reporting, verification) is not a bureaucratic term. It is what transforms a hub from a "good idea" into a convincing argument.

Without MRV the hub cannot:

- Obtain a municipal contract – no evidence of service delivered.
- Apply to the Cleanliness Fund – no documented stream diversion.
- Participate in city planning – no data for inclusion in statistics.
- Scale – no evidence base to persuade new partners.

Minimum MRV for a hub: what to measure

By tonnage and function:
— Items received for early-stage rescue (kg/month)
— Repaired (kg/month, number of items)
— Redistributed/sold (kg/month)
— Organics received and directed to composting/digestion (kg/month)
— Verified secondary material (kg by fraction/month)
— Directed to standard disposal stream (kg/month)

People:
— Number of visits per month
— Number of workshop participants

Derived indicators (calculable from the above):
— Tonnes diverted from landfill per year
— CO_2 equivalent (using IPCC coefficients)
— Cost of waste prevented (at the landfill levy rate)

An annual public report with this data is the best negotiating tool with the municipality. Not "we are doing a good thing," but "we diverted X tonnes from landfill, which at current gate fees saved the city Y shekels, and created Z jobs."

8. Hub and Municipality: Formats of Collaboration

The municipality's relationship with a circularity hub passes through several stages as its measurability grows.

Stage	**Relationship format**
Hub is invisible or seen as a hobby	No contract. Hub operates on its own premises; municipality does not notice.
Hub noticed but not recognised	Informal contacts. Municipality sometimes helps with collection or premises, but without obligations.
Hub recognised as a partner	Cooperation protocol. Municipality provides premises or logistical support. Hub provides data.
Hub as a service contractor	Service contract with KPIs. Municipality pays for a verified outcome: tonnes diverted, residents served, data provided.
Hub as an infrastructure element	Hub is included in the city's waste management programme. Its indicators enter municipal reporting. Formal role in the chain.

The goal is not necessarily the last stage right now. The goal is to move up this ladder sequentially, adding measurability and formalising the relationship as trust grows.

◆Under the Hood: An Important Principle – Complementarity, Not Substitution

The most serious trap in hub operations is becoming a substitute for municipal services. This happens imperceptibly: the hub works well, the municipality cuts a parallel service, residents begin to depend on the hub for what the city should provide.

Legally and politically this is a dangerous position. If the hub closes – residents are left without a service. The municipality gains the argument: "we managed without them before."

The correct frame: the hub provides services that the municipal system does not have and will not have in the near future (repair, early-stage rescue, verified streams). But basic collection, basic *sorting and basic container servicing are the municipality's responsibility – which the hub does not take on and should not take on.*

OECD (2020) formulates this as the "complementarity, not substitution" principle: NGOs in service delivery should be contracted for additional services, not for what the state operator is obliged to do.

Chapter Summary

Key conclusions:

→ Circularity is infrastructure, not personal virtue. A hub creates conditions in which the right behaviour becomes convenient.
→ Sorting remains necessary, but it is not the starting point of a prevention-oriented system. It is the last gate after repair, reuse, sharing, organics diversion and early-stage rescue have done as much as they can.
→ A circularity hub is seven functions in one place: early-stage rescue, repair, sharing, organics, verified streams, education, MRV. Multifunctionality creates synergy.
→ A hub creates jobs – significantly more per tonne processed than a landfill or WtE.2
→ Hub sustainability is built on diversified financing: municipal contract + material sales + social programmes. Not on grants.
→ MRV is not bureaucracy – it is the argument. Without data the hub is invisible to the system.
→ The hub complements the municipal system, it does not substitute for it. A clear boundary of responsibility is the condition for a long-term partnership.
→ Sorting remains necessary, but it is not the starting point of a prevention-oriented system. It is the last gate after repair, reuse, sharing, organics diversion and early-stage rescue have done as much as they can.
→ Re-Haifa is one possible path, not the only model. Each neighbourhood adapts to its own scale, resources and political context.[3]

◆Questions for Those Thinking About Launching a Hub

1. Which of the seven functions are most needed in your neighbourhood – based on what is currently absent?
2. Is there a physical space that could be used for even a few hours per week?
3. Who among residents or organisations could serve as coordinator in the initial stage?
4. Which municipal official is responsible for ecology and sanitation – and are they willing to have an informal meeting?
5. How will you measure results? What exactly will you track from day one?
6. Which downstream partner (recycler, NGO, charitable organisation) can accept what the hub cannot process itself?

The following chapters of Part VII describe practical tools – data requests, civic audits and a weekly action plan. A circularity hub is the place from which these actions unfold most naturally: it is already a point of contact between residents and the system.

Sources for Chapter 24

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10. Lertzman R. Environmental Melancholia: Psychoanalytic Dimensions of Engagement. Routledge, 2015.

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Footnotes

1. Brussels Environment Agency (see source 3). Six pilot hubs, 2020–2023: 1,200–4,500 tonnes/site/year; contamination reduction 18–31% in service zones vs control zones. ↩ ↩[2] ↩[3]

2. Rreuse (see source 1). Reuse and repair: 200–300 jobs per 10,000 tonnes processed vs 20–40 for landfilling and 10–15 for incineration. ↩ ↩[2]

3. Re-Haifa Lab (see source 2). Founded 2018, Haifa. Activities: textile and furniture rescue, repair café, community swaps, organic waste piloting. Internal operations data 2023–2025. ↩ ↩[2]

4. Amsterdam Metropolitan Area (see source 4). 14 hubs; average diversion 340 tonnes/hub/year; 60–70% of items repaired or reused. ↩ ↩[2]

Chapter 25. Public Data: What Citizens Have the Right to Ask

"Transparency is not a gift from the system to society. It is an obligation of the system to society. And in Israel there is a law that enshrines this."

One of the most common activist mistakes is writing complaints without first requesting data. A complaint says: "things are bad here." A data request says: "show me the figures, and I will tell you how bad, why – and what specifically needs to change."

The difference is fundamental. A complaint is an emotion. Data is an argument. In a conversation with a municipality, a contractor or a ministry, it is the argument that carries weight.

Israel has a Freedom of Information Law from 1998[1] that gives every citizen the right to request information from any public body – including municipalities, the Ministry of Environmental Protection and TAMIR as a recognised operator (*guf mukar*) under the Packaging Law. This chapter explains: what exactly to request, from whom, in what form – and what to do if you are refused.

1. The Data Map: What Exists and Where It Lives

Before sending a request, it is useful to understand what data exists in principle – and who is responsible for it. In Israel's waste management system, data is distributed across several levels.

Level / Body	**What data should exist**
Ministry of Environmental Protection (MoEP)	Registry of licensed facilities (landfills, sorting facilities, recyclers). National municipal solid waste statistics. EPR system data in aggregate form. Emissions standards and reports.
Municipality / local council	Collection routes and schedules by stream. Tonnage by stream (mixed, orange, blue, brown). Contracts with operators (including KPIs). Sanitation budget expenditure. 106 data (complaints, response times).
TAMIR (תאגיד מחזור יצרנים)	Packaging tonnage collected by city. Sorting facilities serving the region. Share of plastic, metal, beverage cartons – by fraction. Payments to municipalities. Downstream operators (where the system permits disclosure).
Sorting facility operators	Accepted tonnage by fraction. Rejection rate. Downstream recyclers. Share going to RDF and landfill.
Collection contractors	Routes and collection frequency. Tonnage by collection point. KPIs under contract.
Landfill operators	Accepted tonnage by source. Gate fee. Licence reports to MoEP.

◆Under the Hood: What Is Actually Published Without a Request

MoEP publishes a registry of licensed facilities (gov.il) and some aggregate MSW statistics. However, stream-level data (by fraction and downstream operator) is systematically absent from open access.

TAMIR publishes general system information and recycling figures against the declared base on its website (tmir.org.il). Data by specific city, sorting facility and recycler is not published in open access.

Tel Aviv publishes 106 data and some routing data through its open data portal (tel-aviv.gov.il). Haifa – significantly less.

The State Comptroller (Report 71C, 2021) found: many municipalities cannot provide stream-level data even for internal use – accounting systems by fraction are absent.[2]

Conclusion: most data needed for meaningful dialogue with the system requires a formal request.

2. The Freedom of Information Law: What You Need to Know

The Freedom of Information Law 1998 is the primary tool. Here are the key parameters.[1]

Freedom of Information Law: key parameters

Who can request: any Israeli citizen or resident.

From whom: from any public body – ministry, municipality, state company.

Response deadline: 30 days. Extension of a further 30 days is possible with notification.

Format of response: written. The body is obliged to provide the information or justify refusal.

Grounds for refusal: state secrecy; privacy of third parties; commercial secrecy (if disclosure would cause real harm); internal working documents without a concluded decision.

Cost: the law permits charging for large requests; a standard request is free.

Appealing refusal: the Ombudsman (ממונה על חופש המידע) at the Ministry of Justice. Administrative court.

Important: TAMIR – as a recognised operator (*guf mukar*) under public law – is de facto subject to transparency requirements. The exact scope of TAMIR's FOI obligations is a matter of legal debate; approaching via MoEP as the supervisory body is often more effective than a direct request to TAMIR.

◆Under the Hood: The Aarhus Convention – International Context

The Aarhus Convention (1998) enshrines citizens' right to access environmental information, participate in decision-making and access justice on environmental matters. It is one of the strongest international instruments in this area.

Israel signed the Convention but has not ratified it. This means the international standards have no direct legal force, but can be used as an argument in correspondence and public advocacy.

For comparison: in EU countries that have ratified the Convention, refusal to provide environmental information is subject to appeal in court with the presumption in the applicant's favour. In Israel the burden of proof is reversed: the applicant must demonstrate the illegality of the refusal.

3. The Full Request List: By Stream and By Actor

Below is a working list of requests, grouped by actor. Each can be used as a standalone request or as part of a comprehensive submission. Letter templates are in Appendix D.

Requests to the Municipality

What to request	Why it matters
Collection routes and schedules for each stream (mixed, orange, blue, brown, if applicable) with the contractor named	Basic verification: does actual collection match what is declared? Without routes a civic audit is impossible.
Tonnage by stream, monthly, for the past 2 years	Shows trends. Is source separation growing? Or static?
Contracts with waste collection operators (including KPIs)	Shows what the city is paying for: tonnes or quality. Presence of contamination penalties is a good signal.
Sanitation expenditure from the city budget broken down by stream	How much the city actually spends on each stream. Helps identify the hidden EPR subsidy.
106 data: number of complaints by type, average response time, areas with most complaints	Map of problem zones. Tool for dialogue on infrastructure priorities.
Specific destination of orange bin contents from our district: name and address of sorting facility	First step in a downstream audit. Without this it is impossible to trace the subsequent fate of material.
Payments received from TAMIR for organising source separation over the past 2 years	Allows comparison with real costs and identification of the "hidden subsidy" (see Chapter 14).

Requests to TAMIR

What to request	Why it matters
Packaging tonnage collected in our city/region, by fraction (plastic, metal, beverage cartons), for the last reporting year	Provides baseline figures for comparison with municipal data.
Name and address of the sorting facility to which packaging from our district is sent	First link in the downstream chain.
Rejection rate at the sorting facility: what percentage of accepted material is deemed unfit for recycling	Key indicator of actual sorting quality.
Which operators accept sorted plastic from the orange bin	Second link in the chain: who actually

What to request	Why it matters
for recycling in Israel	recycles.
What share of sorted plastic is exported abroad, and to which countries	Critically important for assessing real recycling (export ≠ recycling).
Payments to municipalities in our city for organising source separation, per the contract formula	Allows calculation of the gap between compensation and real costs.
List of producers and importers registered in the TAMIR system, with volumes of declared packaging (aggregated, without commercial secrecy)	Gives a sense of coverage: what percentage of the market is actually in the system.

Requests to MoEP

What to request	Why it matters
Registry of licensed plastic recycling operators in Israel, by polymer type and capacity	Official basis for Appendix B (actor directory).
Data on the national plastic waste stream: generation, collection, sorting, recycling, export, landfill and RDF – by fraction, for the past 3 years	Allows comparison of official TAMIR statistics with the real flow.
Emissions reports from landfills and sorting facilities in our region for the past 2 years	Baseline data for environmental monitoring.
Status and timeline of the Neot Hovav WtE facility: design capacity, guaranteed waste volume under contract, contract duration	Data needed to assess lock-in (Chapter 3).
TAMIR monitoring results for the last year: compliance with recycling targets	Allows assessment of the quality of state oversight of the EPR system.

Requests to Sorting Facility Operators

Sorting facility operators are private companies and are not directly obliged to disclose data under the FOI Law. But they submit reports to MoEP as a condition of their licence. Two routes: request via MoEP as the licensing body, or approach directly with a journalistic or research request.

What to request	How to request
Accepted material tonnage by fraction	Via MoEP (licence report) or directly as a research request
Rejection rate by polymer	Via MoEP or directly

What to request	How to request
Downstream recyclers: who accepts sorted material and where it goes	Via MoEP or TAMIR as counterparty
Share of material directed to RDF and landfill	Via MoEP (mandatory reporting under licence)
Sorting line technical specifications: is NIR present, number of lines, throughput capacity	Directly as a technical request; some data available on company websites

4. How to Write a Request: Structure and Wording

An effective FOI request is not just a question – it is a document with a specific structure. Vague questions get vague answers – or a justified refusal. Precise questions are harder to ignore.

◆Template: Request to Municipality on Routes and Tonnage

To: Sanitation Department [municipality name]
Copy: [Mayor or Deputy Mayor for Municipal Affairs]
Subject: Data Request under the Freedom of Information Act (חוק חופש המידע)

Dear Sir/Madam,

In accordance with Article 2 of the Freedom of Information Act (1998), I request the following data for the period [year/years]:

1. Collection routes and schedules for each waste stream:
   a) mixed waste (grey/green bin)
   b) packaging waste (orange bin)
   c) paper and cardboard (blue bin)
   d) organic waste (brown bin, if applicable)

2. Tonnage collected per stream, monthly.

3. Name and address of the facilities to which each stream is directed after collection.

4. Contracts with waste collection operators, including KPIs.

If any portion of this information is confidential, please provide the available portion with grounds for restriction stated for each item.

Yours sincerely,
[Name, address, ID number]
Date: [date]

Several practical notes on writing requests:

Break your request into specific questions rather than one general question. "Provide all information about waste" is a legitimate basis for refusal. "Provide orange bin tonnage for 2023–2024" is a specific and justified request.

State the time period. Without one, the body may respond with only "current" data.

Request the available portion even if there is a partial refusal. This is a standard right under the FOI Law.

Send by email with acknowledgement of receipt – this records the start of the 30-day period.

If no response is received by the deadline – this is itself a violation of the law. Approach the Ombudsman.

◆Under the Hood: What to Do with a Refusal

The most common ground for refusal is commercial secrecy (סוד מסחרי). This is a legitimate basis – but only if the body can demonstrate that disclosure would cause real harm to a third party. "We don't want to disclose this" is not commercial secrecy.

First step: request a written justification of the refusal citing the specific provision of the law.

Second step: approach the Freedom of Information Ombudsman at the Ministry of Justice. The procedure is free; the review period is up to 45 days.

Third step: administrative court. This is slower and more expensive, but there are precedents in applicants' favour.

Parallel route: media request. If a journalist requests the same information, a refusal acquires a public dimension. This often accelerates a response.

Re-Haifa Lab from its own experience: of several requests to Haifa Municipality, one received a partial response about the orange bin collection schedule; a request for downstream data was refused on the grounds of contractor commercial secrecy. This is the typical scenario.

5. What to Do with the Data Once Received

Getting the data is half the work. The second half is understanding it, comparing it with official statements, and turning it into an argument.

Four steps from data to argument:

Step 1. Check internal consistency.
Does the tonnage the municipality reports to TAMIR match what it reports to you? If not – that is already a question.

Step 2. Compare with official statements.
The municipality states: "we recycle 40% of packaging." What does this mean – 40% of what was collected? Of all generated? Of the TAMIR declared base? Without knowing the denominator the figure is meaningless.

Step 3. Apply the evidence scale from Chapter 8.
Mark each claim: ✓✓confirmed from an official document, ✓probable from several sources, ~ reconstructed, ? unknown.

Step 4. Write it up as a short civic report.
Not a social media post – a document with a date, sources and clear questions to the system. This is what persists in correspondence and can be cited in media.

Chapter Summary

Key conclusions:

→ The Freedom of Information Law gives citizens the right to request data from municipalities, MoEP and, through them, TAMIR. This is not a privilege – it is the system's obligation.[1]
→ Most downstream data (rejection rate, recyclers, exports) is not published by default. It must be requested.
→ A specific, structured request with a time period and a reference to the FOI Law is harder to ignore than a general question.
→ Refusal is not the end. The Ombudsman, administrative court and media request are working tools.
→ Received data only works when it is framed as an argument, not as an accusation.

Chapter 26. How to Audit a Waste Stream Without Being an Expert

"A good civic audit is not an investigation. It is documentation of the gap between what the system says and what can be observed. Sometimes that is enough to start a conversation."

Chapter 25 explained what to request. This chapter explains what to do with the responses – and what to do when responses are absent or incomplete.

A civic audit of a waste stream is not the same as a corporate environmental audit. You have no laboratories, no legal powers, no access to closed facilities. But you have something else: the right to observe public space, the right to submit official requests, the right to document and publish.

This is enough to conduct a meaningful audit – if you follow the methodology. Methodology is precisely what distinguishes a civic report that is taken seriously from a social media post that is ignored.

1. Principles of a Civic Audit

Before describing the steps, it is important to establish several principles that define how this method works.

Five principles of a civic audit:

1. One stream, one territory, one period.
Attempts to cover "everything" end in nothing. Choose: the orange bin in the Neve Sha'anan neighbourhood for January–March 2024. This is specific and verifiable.

2. Separate observation from interpretation.
"The truck arrived at 9:15 and left at 9:40, the bin was full" – observation. "The bin overflows because collection frequency is insufficient" – interpretation. Both are important but must not be mixed.

3. Apply the evidence scale (Chapter 8).
Everything you did not personally observe or receive from an official document is marked at the appropriate level. This is not a weakness – it is a strength: honest marking makes the report immune to the criticism "you made this up."

4. Document the process, not just the result.
When a request was sent, what was responded, what was not responded, what was observed on which date – all of this is part of the audit. A refusal to provide data is also a fact.

5. The goal is a question, not a verdict.
A good civic report does not accuse. It documents a gap and asks a specific question: "We observed X. Official data says Y. How is this explained?"

◆Under the Hood: The Methodological Basis – Where This Approach Comes From

Break Free From Plastic developed the brand audit methodology – a citizen count of plastic by brand in public spaces. We adapt its principles: choose an observation point, document what you see, classify, compare with official data.

OCCRP (Organized Crime and Corruption Reporting Project) uses a source hierarchy: primary (direct documents) > secondary (independent sources) > reconstruction (calculation from indirect data) > unknown. This is exactly what we call the evidence scale.

Zero Waste Europe, "Citizen Science for Waste Monitoring" (2021): successful programmes in Spain and Germany show that citizens without specialist training can produce data that municipalities then use to adjust routes and infrastructure.

2. Seven Steps of an Audit

Step 1. Choose One Stream and One Territory

Specificity is the key to success. Not "how does the orange bin work in Haifa," but "how does the orange bin work on Ha-Ored and Arlosoroff streets in the Adar neighbourhood in January 2025." This sounds narrower, but precisely this scale allows observation, documentation and comparison.

A good stream to choose is one for which you have official data to compare against. If TAMIR claims 60% packaging recycling in your city – that is a good starting point.

Step 2. Collect Official Data and Statements

Before observing anything, document what the system says about itself. This may include:

- the municipal website: what is written about source separation;
- data received via FOI request from Chapter 25;
- TAMIR statements on the website and in reports;
- a response to your direct question to the sanitation department.

Record exact wording and sources. This will be your comparison baseline.

Step 3. Observe and Document the Container Site

Choose 2–5 container sites in your territory. Visit them several times on different days and at different times. Document:

- condition of containers: how full, damage, contamination around;
- visible composition of contents through transparent walls or the open hatch;
- resident behaviour: what they place in it, what they don't, any visible sorting errors;
- truck arrival: time, route (which way it turns after your site), type of vehicle.

A photo with a timestamp and GPS is the minimum documentation standard. Video is better where possible.

Step 4. Request Route Information

Using the tools from Chapter 25, request from the municipality or contractor: where does the truck go from your container site? To which sorting facility?

If a response is received – this is valuable information. If refused – the fact of the refusal is documented in the report: "In a request about the orange bin route dated [date] the municipality refused, citing [grounds]."

Step 5. Request Downstream Information from the Sorting Facility

If you know which facility receives your stream – the next step is a request to that facility (directly or via MoEP).

- What is the rejection rate for the packaging stream?
- Who accepts the sorted plastic/metal/cardboard?
- What share goes to recycling, and what share to RDF or landfill?

Responses to these questions – even partial – allow several cells of the evidence scale to be filled. Absence of responses is also data.

Step 6. Compare the Statement with the Observation

This is the central step. Compare what the system says about itself with what you observed and received in response to your requests. Possible gaps:

Official statement	**Possible observed gap**
"The orange bin is collected twice a week"	By observation – once; the bin is full by Friday
"60% of packaging is recycled"	Rejection rate at the sorting facility is 40%; where the "tails" go – unknown
"We work with TAMIR"	No data on payments received from TAMIR in the FOI response
"The sorting facility accepts all types of plastic"	Per Chapter 11 data, black plastic and multilayer packaging are not detected by NIR
"Our residents actively sort"	By observation: food waste and non-sortable items visible in the orange bin

Step 7. Write a Short Civic Report

A document of 2–4 pages – not a post, not a petition. Structure:

Title: "Civic Audit of the Orange Bin: [Neighbourhood Name], [Period]"

Section 1. What we studied and why (1 paragraph).

Section 2. Official data and statements. List with sources and dates.

Section 3. Observation methodology. Which sites, how many visits, which tools (photos, requests).

Section 4. Findings. Each fact with evidence level: √√Confirmed / √Probable / ~ Reconstructed / ? Unknown

Section 5. Gaps between declared and observed. Specific discrepancies with source.

Section 6. Questions to the municipality / TAMIR / MoEP. Not accusations – specific questions requiring specific answers.

Section 7. Appendices. Photographs, copies of requests, responses or refusal notices.

◆Under the Hood: How a Civic Report Is Used in Practice

The report can be sent to the municipality's sanitation department with a request to comment on the gaps. A written request with the attached report is not a complaint – it is an invitation to dialogue.

The report can be passed to the media – local journalists covering the city beat. A well-documented gap between an official statement and observed reality is a story.

The report can be presented at a city council meeting or environmental committee. A public comment with data in hand is a more effective tool than a verbal complaint.

Several reports from different neighbourhoods or cities can combine into a comparative analysis – material for an NGO or academic research.

Re-Haifa Lab: the first orange bin audit in Haifa (2023) revealed a discrepancy between the declared collection schedule and what was actually observed. This document was used in negotiations with the sanitation department and led to the clarification of public information on the municipality's website.

3. Common Mistakes and How to Avoid Them

Mistake	How to avoid
Too broad a scope: "audit of the entire recycling system in Haifa"	Narrow down to one stream, one neighbourhood, one period.
Mixing observation and interpretation in one sentence	Always separate: "observed" / "from this one might infer."
Using data without citing source and date	Every fact = source + date + evidence level.
Accusatory tone	Rephrase: not "the municipality is hiding this" but "the request dated [date] received no response."
Publishing before receiving a response to the request	Wait for the response deadline (30 days). Its absence is also a fact.
Ignoring technical explanations	If the municipality explains the gap – reflect the explanation in the report and ask a follow-up question.

Mistake	How to avoid
Shaming or accusatory tone towards residents	Residents are partners, not the problem. Sorting errors are a consequence of insufficient infrastructure and information, not ill intent.

Summary of Chapters 25 and 26

Key conclusions:

→ The right to data is a legal right, not a request for a favour. The Freedom of Information Law works.[1]
→ Downstream data on waste fate is catastrophically scarce in open access in Israel's system. This is a structural deficit, not an accident.
→ A specific, structured request with a time period is more effective than a vague question. Templates in Appendix D.
→ A civic audit is a methodology, not activism. Following the principles (one stream, evidence scale, separating observation from interpretation) makes the report unassailable.
→ The goal of an audit is not a verdict – it is a question. A well-formulated question to the system is stronger than any accusation.
→ Refusal to provide information is also data. Document it.

◆Control Questions Before Starting an Audit

1. Have I chosen one specific stream, one territory and one time period?
2. Have I collected the system's official statements to compare my observations against?
3. Have I sent an FOI request and recorded the date of submission?
4. Am I keeping observation and interpretation separate in my notes?
5. Does every fact in my draft have a source and an evidence level?
6. Does my final report end with questions rather than accusations?

The next chapter – "One Week of Smart Action" – connects the tools from both these chapters into a concrete seven-day plan. This is not an abstract programme: it is something that can be done immediately after reading.

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2. State Comptroller of Israel. Report 71C: Municipal Solid Waste Management. Jerusalem, 2021. mevaker.gov.il
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10. Aarhus Convention (Convention on Access to Information, Public Participation in Decision-Making and Access to Justice in Environmental Matters). UNECE, 1998.

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11. Re-Haifa Lab. Internal correspondence with Haifa Municipality, 2023–2024. [Available on request from Re-Haifa Lab.]

12. Lertzman R. Environmental Melancholia: Psychoanalytic Dimensions of Engagement. Routledge, 2015.

Footnotes

1. Freedom of Information Act (Israel), 1998 (see source 1). Art. 2: every citizen has the right to receive information from a public authority. Art. 7: public authority must respond within 30 days; extension of 30 days is possible. Art. 8–9: grounds for refusal include state secrecy, privacy, commercial secrecy (only where disclosure causes real harm), and internal working documents. Waste management data held by municipalities and MoEP falls within the scope of the Act. ↩ ↩[2] ↩[3] ↩[4]

2. State Comptroller of Israel. Report 71C, 2021 (see source 2). Finding: many municipalities cannot provide stream-level data even for internal use – accounting systems by fraction are absent. This structural data deficit is a key finding underlying the argument of this chapter. ↩

Chapter 27. One Week of Smart Action: Working with NGOs, Schools and the Reform Process

"Knowledge of the system is a necessary but not sufficient condition. This chapter is about what to do with that knowledge in the next seven days."

This book has taken the reader through law and actors (Part I), the physical movement of waste (Part II), the fate of specific materials (Part III), packaging policy (Part IV), money and contracts (Part V), waste prevention (Part VI), and the tools of data and audit (Part VII). This is a substantial body of material – and it does not assume the reader will apply all of it at once.

This chapter offers a concrete, realistic plan for one week – separately for an activist, an NGO, a teacher, and a person who wants to participate in the discussion of the 2026 reform. The plans do not compete – they complement each other. Some steps can be taken in parallel.

Before starting: it is important to understand where this plan comes from.

What gives grounds for action: connection to previous chapters:

Chapter 25 provided tools: what to request, from whom, in what form. Request templates in Appendix D.
Chapter 26 provided methodology: how to observe, document, compare with official statements, and frame as an argument.
Chapter 16 showed: decisions about the reform are being made right now – and civic participation has a literal, not rhetorical, meaning precisely now.
Chapters 8 and 19–20 provided data: 6% of packaging in dedicated bins, the "pay twice" paradox, ILA royalties – these are specific facts documented by official sources that can be used in conversation with the system.

1. Seven Days: A Plan for the Activist

This plan assumes you are acting as an individual or small group – without NGO status and without specialist expertise.

Day 1 – Choosing a Stream and Territory
Task: identify one specific waste stream in one specific area.
Tool: the principle from Chapter 26 – "one stream, one territory, one period."
In practice: "the orange bin in neighbourhood X over the next three weeks" – specific and verifiable.
Day's result: a written choice + initial research: what does the municipality officially say about this stream on its website?

Day 2 – Collecting Official Statements
Task: record what the system says about itself.
Tool: search on the municipal website, TAMIR website (tmir.org.il/faq), official MoEP statistics (gov.il).
Record exact wording with dates and URLs – this is your comparison baseline.
Day's result: 3–5 official statements with sources. For example: "TAMIR states that X% of packaging is recycled" or "The municipality promises orange bin collection twice a week."

Day 3 – Submitting an FOI Request[1]

Task: submit a request under the Freedom of Information Act.

Tool: the template from Appendix D and the question list from Chapter 25.

What to request: routes, tonnage by stream, name of sorting facility, TAMIR payments over the past 2 years.

Where: to the municipality's sanitation department – by email with acknowledgement of receipt. This records the start of the 30-day period.

Day's result: a submitted request with confirmed date. A refusal is also data – document it.

Day 4 – Field Observation[2]

Task: visit 2–3 container sites and document reality.

Tool: the methodology from Chapter 26 – timestamped photos, notes on fill levels, contamination, resident behaviour.

What to document: did the truck arrive at the stated time? Is the bin full by mid-period? Are sorting errors visible from outside?

Remember: photographing in public space is lawful. Blur or exclude residents' faces.

Day's result: 10–20 photos with metadata + brief observations in a text file.

Day 5 – Comparison and First Draft

Task: compare official statements (Day 2) with observations (Day 4).

Tool: the evidence scale from Chapter 8 – each fact receives a level (√√/ √/ ~ / ?).

Draft structure – per the Chapter 26 template: what was studied, official data, methodology, findings, gaps, questions.

Day's result: 1–2 pages of text with specific gaps and questions. Not accusations – questions.

Day 6 – Sending the Report and Choosing the Next Step

Task: send the draft civic report to the municipal sanitation department requesting comment on the gaps.

Optional: pass to a journalist (local, covering the city beat) or publish via social networks/neighbourhood chats.

Important: publish only after the FOI response deadline has passed or a response has been received – the document becomes stronger with a response or a documented refusal.

Day's result: report sent. A new 30-day period – awaiting reaction.

Day 7 – Reflection and the Next Cycle

Task: assess what worked and what did not.

Reflection questions: what was harder than expected? What question remains unanswered? Is there someone who could help in the next cycle?

One civic audit is not an endpoint. The next cycle with the same data but a different period, or the same stream but a different neighbourhood – this is already a comparative analysis.

Day's result: notes for the next cycle and a list of potential partners (neighbours, NGOs, journalists, activists).

2. Working with NGOs: Building a Partnership

If you represent an NGO – or want to establish collaboration with an existing organisation – the logic is somewhat different. An NGO works with a longer horizon and has greater potential for influence, but requires a clear division of roles.

Type of NGO–municipality partnership	How to build it
Information partnership	The NGO provides civic monitoring data – the municipality uses it to improve services. The simplest type; requires the least trust.
Educational partnership	The NGO runs workshops for residents and schools – the municipality provides venues and communication channels. Mutually beneficial.
Service partnership	The NGO provides a measurable service (early-stage rescue of goods, source-separated organics) – the municipality pays under a contract. The most sustainable; requires KPIs and reporting. More in Chapter 24.[3]
Advocacy partnership	The NGO formulates a position on reform or tender specifications – the municipality brings it into dialogue with the state. Requires trust and political will on both sides.

◆Under the Hood: The Key Principle of NGO Partnership – You Are Not the Municipality's Volunteers

The most common trap: the NGO takes on functions the municipality should perform, without a formal contract. The municipality receives a free service, the NGO burns out, the function disappears at the first organisational crisis.

The correct frame: the NGO does what the municipal system does not do and will not do in the near future – it complements the system, does not substitute for it. For this – a formal contract with KPIs, not an informal "thank you."

Data is the basis of the conversation. Any proposal to the municipality for partnership that does not contain a measurable result ("we are prepared to intercept X tonnes per year and report quarterly") sounds like a request for help, not a business proposal.

A refusal to share data is also a response. If the municipality wants neither a contract, nor data, nor reporting – this is information about their readiness for partnership, to be recorded and taken into account in strategy.

3. Working with Schools: A Practical Format

Schools are one of the most effective entry points for civic action. Not because children are "more ecological," but because a school waste audit has a fundamentally different public dynamic: when children present data to a headteacher or mayor, it is significantly harder to ignore than when an adult activist does the same thing.[4]

Four-week school waste audit format:

Week 1: Introduction.
A lesson on where waste comes from and where it goes. Use Chapters 4–5 as a source. Photograph the contents of the school's bins at the start of the week – these are the baseline data.

Week 2: Observation and documentation.
Students divide into groups – each responsible for one stream (organics, paper, plastic, mixed). They document the composition of their fraction using the Chapter 26 methodology: photos, observations, counting.

Week 3: Analysis and requests.
Comparison with official data: how many times a week is waste collected? Where does the truck go? Does the school have a contract – and what does it say about source separation?

Week 4: Presentation.
Students prepare a short report and present it to the headteacher, a municipal representative, or parents. Three specific improvement proposals are better than one general demand.

Several practical recommendations from the experience of European Zero Waste Schools:

Start with observation, not with conclusions. Students who collect their own data are far more effective than those who have absorbed ready-made findings.

Do not make the headteacher the villain. The goal is to improve the system, not to punish a specific person. This principle from Chapter 26 works equally in a school and in a municipality.

Connect the school audit to a broader civic initiative. When students know their data will go into a city report – motivation is qualitatively higher.

One specific result is better than a large report. "This school has no brown bin – we request one be installed" is a statement that is easy to verify and to act on.

4. Participation in the 2026 Reform: What to Do Right Now

The waste system reform (Chapter 16) is not a process that happens without citizens. It is a process in which citizens can participate concretely – if they know exactly where.[5]

Participation in the 2026 reform (→ Chapter 16)

The 2026 reform provides for public consultations – both on the broader Ministry of Finance reform and on the narrower MoEP memorandum on the Packaging Law (Chapter 17). New rounds of consultations can be tracked via: tazkirim.gov.il (memoranda open for comment), knesset.gov.il/committees (Knesset committees), and infospot.co.il (waste news aggregator).[5]

Knesset Interior and Environmental Protection Committee. Open sessions on waste reform are public hearings. Civic organisations can request the opportunity to present or submit a written position.

The local council as a channel. If your municipality is a member of a regional cluster (*eshkol*) or inter-municipal association (*igud arim*), your position via the local council can reach the Union of Local Authorities (*merkaz ha-shilton ha-mekomi*) – a body that has already publicly called for reform (Chapter 16).

Participation in reform is most effective through specific demands on specific elements of the law – not through a general position "for" or "against" reform. Chapter 16 already formulated seven key questions any new law should contain: mandatory environmental targets, downstream reporting by the regulator, a coordination mechanism between EPR and the deposit system, and so on. These are precisely the points where a written comment from a citizen or NGO can change the wording.

◆Checklist for Participation in Public Consultations

1. Have I read the full text of the proposal or memorandum – not just press summaries?
2. Does the text contain measurable environmental targets – or only procedural description?
3. What specific element of the text do I want to change or add?
4. Can I formulate the proposed change in one paragraph – so it could literally be inserted into the law?
5. Have I coordinated my position with other organisations working on this topic? A coalition comment carries more weight.
6. Have I recorded the fact of submitting the comment with a date – so I can track whether it was taken into account?

5. Format Table: Who Can Do What

Audience and possibilities	Concrete steps this week
Individual resident without organisational status	— Choose a stream and territory (Day 1) – Submit an FOI request (Day 3) – Conduct field observation (Day 4) – Write a post in the neighbourhood group with a specific question to the municipality
Activist with environmental experience	— Full 7-day cycle (section 1) – Pass the civic report to a local media outlet – Register for a session of the municipal environmental committee
NGO / environmental organisation	— Conduct an audit of one stream using the Chapter 26 methodology – Submit a position on reform to the Knesset committee – Begin negotiations with the municipality on a service contract (Chapter 24)
Teacher / school group	— Launch the 4-week school audit (section 3) – Invite a sanitation department representative to answer students' questions – Publish results in the school newspaper or on social media
Local council representative	— Request data from the sanitation department across all four formats (section 1 of Chapter 25) – Initiate a discussion of tender KPIs at the next council session (Chapter 21) – Submit an official position on reform through the Union of Local Authorities

6. What Not to Do: Five Most Common Mistakes

A civic audit works better when a few typical patterns that reduce its effectiveness are avoided.

Five mistakes that weaken civic action:

1. Too broad a scope.

"We'll check the entire waste system of the city" ends in nothing. One stream, one territory, one period – always more specific and more convincing.

2. Publishing before receiving a response to the request.

Wait for the 30-day FOI deadline. The absence of a response is itself data – and a stronger argument than a rushed publication.

3. Accusatory rather than interrogative tone.

"The municipality is hiding this" – a position that is easy to challenge. "The request dated [date] received no response" – this is a fact.

4. Absence of the evidence scale.

A fact without a source and evidence level is a claim. A fact with a ✓✓source is an argument. The difference is fundamental, especially in dialogue with the system.

5. Ignoring positive examples within the same system.

If a neighbouring municipality does something better – this is the strongest argument: "this works next to us, why not here?"

Chapter Summary and Summary of Part VII

Key conclusions of Chapter 27:

→ The seven-day plan translates the tools of Chapters 25 and 26 into a concrete sequence of actions: choose a stream → collect official data → FOI request → observation → analysis → report → reflection.[12]

→ NGO work requires formalisation: a service contract with KPIs is more sustainable than informal volunteering. The NGO complements the system, does not substitute for it.[3]

→ A school audit is a high-impact format with a low entry threshold: four weeks, a specific stream, three improvement proposals.[4]

→ The 2026 reform creates new civic leverage points: public consultations, the Knesset committee, coalition comments. A specific proposal on a specific element of the law is stronger than a general position.[5]

→ Five typical mistakes: too broad a scope, premature publication, accusatory tone, absence of an evidence base, ignoring positive examples.

Concluding conclusions of Part VII: law, oversight and civic action:

→ The right to data is a legal right, not a request for a favour. The Freedom of Information Law of 1998 obliges the system to respond.

→ Data in the public domain is systematically insufficient – this is not accidental, it is a structural opacity that must be actively overcome through requests.

→ A civic audit is a methodology, not activism. Following the principles (one stream, evidence scale, questions rather than accusations) protects the report from challenge.

→ The reform moment is a rare opportunity. The laws being discussed in Israel now will shape the waste system for the next 15–20 years. The wording is not yet final.
→ The connection between the parts of the book is itself the system: knowledge of material streams (Parts II–VI) + understanding of financial incentives (Part V) + data tools (Part VII) = the ability to conduct a substantive, specific dialogue with the system.

The final section of the handbook – the Conclusion – summarises the book's arguments and answers the question with which everything began: what can one person, one neighbourhood, one organisation do – in a system that seems too large and too slow to change?

Sources for Chapter 27

Cross-references within the book

1. Chapter 25: FOI tools, actor-specific question lists, templates in Appendix D.
2. Chapter 26: seven-step methodological audit cycle.
3. Chapter 24: circularity hub model as a format for NGO–municipality service partnership.
4. Zero Waste Europe. "Community Zero Waste Centres: Lessons from Spain, Italy and France." Brussels, 2020. zerowasteeurope.eu
5. Infospot. "Government Ministries Invite Public to Comment on Waste Sector Regulation Plan." infospot.co.il; MoEP Memoranda Portal. tazkirim.gov.il; Knesset Interior and Environmental Protection Committee. knesset.gov.il/committees

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6. Freedom of Information Act (Israel), 1998. nevo.co.il/law_html/law00/5166.htm
7. State Comptroller of Israel. Follow-up Audit: Municipal Solid Waste Disposal. July 2025. library.mevaker.gov.il

Footnotes

1. Freedom of Information Act (Israel), 1998 (see source 6). Art. 2 and Art. 7: citizen right to request information from public authorities; 30-day response deadline. Full methodology for data requests, including sample letters and actor-specific question lists, is provided in Chapter 25. ↩ ↩[2]
2. Chapter 26 (this book): seven-step audit methodology – choose one stream and territory, collect official claims, observe and document, request route information, request downstream information, compare claimed vs observed, produce a short civic report. Steps 4 and 5 of the weekly plan draw directly on this methodology. ↩ ↩[2]
3. Chapter 24 (this book): circularity hub model. Documents that the most durable form of NGO–municipality collaboration is a service contract (not a grant), with measurable KPIs and public reporting. ↩ ↩[2]
4. Zero Waste Europe (see source 4). Documents successful school partnership programmes: when schools conduct their own waste audits and present findings to municipalities, the

civic impact is significantly higher than adult-led campaigns, due to both media attention and the political difficulty of ignoring child presenters. ↩ ↩[2]

5. Infospot (see source 5). Public consultation period for Ministry of Finance/MoEP joint consultation had a deadline of 15 February 2026. Subsequent rounds are expected as legislation advances. Monitor tazkirim.gov.il and knesset.gov.il/committees for timing of new consultations. ↩ ↩[2] ↩[3]

Conclusion

What One Person Can Do in a System That Seems Too Big

> *"The bin outside your home is connected to the system described across the 27 chapters of this book. And yes – it could not have been made shorter, because everything described here is a piece of one enormous machine, one mechanism: a physical chain of decisions made in the Knesset, in TAMIR's offices, in the mayor's office, and in the back of a truck you have never seen."*

What We Learned

Behind the 27 chapters of this book is not a list of environmental tips, nor a polemic against particular officials. It is an attempt to describe the system as it actually is: with built-in conflicts of interest, structural gaps between declared goals and real incentives, and documented instances where the system acknowledges its own failure – as the MoEP did in August 2025, when it published data showing that 6% of household packaging reaches the right container.

If the book is reduced to a few key conclusions, they look like this.

> **Eight Things This Book Wants the Reader to Know**
>
> 1. Israel's waste management system is not a single mechanism but a historically accumulated conglomerate of laws, actors, and incentives that often work against each other. This is not the result of ill intent – it is the architectural legacy of reactive lawmaking.
>
> 2. A 6% recycling rate is not a failure of residents. It is a failure of the system. MoEP data showed that 60% of the contents of the mixed bin is recyclable material – material that ended up there not because residents "don't care," but because the system never created conditions in which the right behaviour is convenient.
>
> 3. Financial incentives explain behaviour better than appeals to responsibility. The "pay-twice" paradox (sorting increases costs rather than reducing them) is not an anecdote but a documented structural problem that cannot be solved with education campaigns.
>
> 4. The 2026 reform is real, and unfinished. Two parallel attempts to change the system – the

Ministry of Finance's and the MoEP's – reflect a genuine disagreement over what matters more: economic efficiency or environmental goals. That dispute has not been resolved, which is exactly why civic participation carries literal, not symbolic, weight.

5. Organics is the largest and most neglected stream. Roughly 40% of household waste by weight, a potential 5.5% cut in greenhouse gas emissions if handled correctly – and almost none of the infrastructure needed to do so. This is not a technology problem: anaerobic digestion works. It is a problem of investment and political will.

6. The tender is the point of maximum civic leverage over the system. The contracts being signed now will set contractor KPIs for the next 5–10 years. Publication of tender terms is an invitation to participate, not a formality.

7. Data is a right, not a privilege. The 1998 Freedom of Information Act obliges the system to answer specific questions. A refusal is itself data that can be used.

8. Local scale is not a consolation prize. A circularity hub serving a single neighbourhood produces measurable data – and that data becomes an argument in the national conversation about reform. The local and the systemic are not opposites.

What Changed While This Book Was Being Written

Work on this book began at a time when the main argument against active civic engagement on waste was scepticism: "the system is too inert," "there is no data," "TAMIR doesn't answer to anyone anyway."

Over the course of writing it, events occurred that changed that context.

The State Comptroller recorded that landfill capacity will run out by the end of 2026. This is not a hypothesis – it is an official document with a figure (7.67 million cubic metres) and a date.

The MoEP published a scan: 6% of household packaging reaches dedicated bins. This is not an activist estimate – it is the regulator's own data, acknowledging the failure of the very system it is responsible for.

The Ministry of Finance folded sector reform into the Economic Arrangements bill. That means the debate over the future of Israel's waste system has moved out of internal ministry meetings and into public legislative space.

TAMIR filed its first lawsuits against free riders – fourteen years after the law was passed. This is a signal that the system is beginning to apply enforcement it long avoided.

The Neot Hovav tender was completed. The first major WtE plant is under construction. Two more are in progress. This is not the future – these are contracts already signed.

Each of these events creates a concrete entry point for civic action. The capacity crisis explains why participating in tender consultations is not abstract activism but influence over decisions being made right now. The MoEP's data gives an argument for a conversation with any official: this is not our claim – it is your data. The reform opens a window of public consultation that has not been open for the past 15 years.

What to Do With This Knowledge

The book's final section – Chapter 27 – contains a seven-day action plan. But here, in the conclusion, it is worth stating the broader principle.

Knowledge of the system is a necessary but not sufficient condition for change. The distance between understanding how TAMIR works and changing how it works cannot be closed alone, and cannot be closed quickly. This book does not promise that reading it will change the system.

What it does promise is different: that someone who has read it will not enter a conversation about waste armed only with moral indignation. Instead, they will have a concrete map: who is responsible for what, under which law, with which built-in conflict of interest. A scale for telling fact apart from assumption. A methodology for turning observation into argument. And an understanding that right now – specifically in 2026 – is a moment when these arguments can actually change something.

Three Concrete Actions Any of This Book's Four Audiences Can Start Right Now

1. Send a Freedom of Information request to your municipality's sanitation department: how many tonnes by stream, where the trucks go, who the downstream recycler is. A template is in Appendix D.

2. Watch tazkirim.gov.il and knesset.gov.il/committees for public hearings on waste reform – and submit a written position on a specific element of the law.

3. Run one civic audit of one stream in one neighbourhood, using the Chapter 26 methodology. Publish the results, even if modest. Data that enters the public record changes what the system is able to ignore.

A Final Word

Waste is not a marginal subject. It is a question of infrastructure, a question of justice (who lives next to the landfill), a question of climate (5.5% of emissions from untreated organics), a question of money (7% of a municipal budget that never quite balances), and a question of producers and consumers, and of who bears responsibility for what happens to packaging after the till.

The shift from cleanup to system design means changing the point at which public action begins. Cleanup starts after failure has become visible — after a wadi is littered, a bin site overflows, or usable objects have already been abandoned. System design starts earlier: when a repairable object can still be repaired, edible food can still be shared, organics can still become compost, and a bulky item can still be moved to another household instead of being left near the bin.

This is why local circularity infrastructure matters. It creates the missing layer between household life and the formal waste system — a layer where residents, NGOs and municipalities can prevent waste, generate data, build trust, and make prevention part of public management rather than private goodwill.

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This is why local circularity infrastructure matters. It creates the missing layer between household life and the formal waste system — a layer where residents, NGOs and municipalities can prevent waste, generate data, build trust, and make prevention part of public management rather than private goodwill.

In 2026, Israel does not face a choice between a “good” and a “bad” waste management system. It faces a choice between a system built under crisis pressure – quickly, without long-term safeguards, locked in for 25 years – and a system built with the participation of people who know which questions to ask.

This book is an attempt to give those questions names.

> Prevention matters more than recycling.
>
> Transparency matters more than optimism.
>
> Citizens are not peripheral actors.
>
> They are the quality control of public policy.

Bela Nikitina / Re-Haifa Movement

Haifa, July 2026

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Municipal Waste Management in Israel: A Civic Handbook

Bela Nikitina / Re-Haifa Lab, 2026

This bibliography lists all 110 sources cited across the 27 chapters of this handbook, deduplicated and organised by category. Where a source is cited in multiple chapters, the year-letter suffix (e.g. Infospot, 2025a) is used consistently throughout the main text.

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